

Question #1

An administrator accidentally deleted the `/boot/vmlinuz` file and must resolve the issue before the server is rebooted. Which of the following commands should the administrator use to identify the correct version of this file?

A) `rpm -qa | grep kernel; uname -a` (Correct Answer)

B) `yum -y update; shutdown -r now`

C) `cat /etc/centos-release; rpm -Uvh --nodeps`

D) `telinit 1; restorecon -Rv /boot`

Explanation

Correct Answer:

A

Community Discussion

A. `rpm -qa | grep kernel; uname -a` The administrator can use the command "`rpm -qa | grep kernel`" to identify the installed kernel packages on the system, and "`uname -a`" to display information about the current running kernel. The output of these commands can be used to determine the correct version of the `/boot/vmlinuz` file that needs to be restored.

A. query and grep kernel; information

A correct....see what's installed, and see what's running

Question #2

A cloud engineer needs to change the secure remote login port from 22 to 49000. Which of the following files should the engineer modify to change the port number to the desired value?

- A) /etc/host.conf
- B) /etc/hostname
- C) /etc/services
- D) /etc/ssh/sshd_config (Correct Answer)**

Explanation

Correct Answer:
D

Community Discussion

D. /etc/ssh/sshd_config The file the engineer needs to modify is "/etc/ssh/sshd_config". This is the configuration file for the SSH server and it controls various parameters for the server, including the port number used for secure remote login. To change the port number from 22 to 49000, the engineer should locate the line that starts with "Port 22" and change it to "Port 49000". After making the change, the engineer should save the file and restart the SSH service for the changes to take effect.

was on the test.

D is correct

Question #3

A new file was added to a main Git repository. An administrator wants to synchronize a local copy with the contents of the main repository. Which of the following commands should the administrator use for this task?

- A) git reflog
- B) git pull (Correct Answer)**

- C) git status
- D) git push

Explanation

Correct Answer:

B

Community Discussion

Pulling is the automated version of git fetch, it downloads a branch from a remote repository, then immediately merges it into the current branch.

B. git pull The administrator should use the "git pull" command to synchronize the local repository with the main repository. The "git pull" command fetches the latest changes from the remote repository and merges them with the local repository. This will ensure that the local repository has the latest version of the code, including the new file that was added to the main repository. Before running the "git pull" command, the administrator should make sure that they are in the correct branch and that their local repository is in a clean state (i.e., no changes have been made to the local repository since the last time it was synchronized with the main repository).

GIT PULL is a mixture of fetch and merge.. It is the only answer that is relevant here

Question #4

A Linux administrator needs to redirect all HTTP traffic temporarily to the new proxy server 192.0.2.25 on port 3128. Which of the following commands will accomplish this task?

- A) iptables -t nat -D PREROUTING -p tcp --sport 80 -j DNAT - -to-destination 192.0.2.25:3128
- B) iptables -t nat -A PREROUTING -p top --dport 81 -j DNAT --to-destination 192.0.2.25:3129
- C) iptables -t nat -I PREROUTING -p top --sport 80 -j DNAT --to-destination 192.0.2.25:3129

D) iptables -t nat -A PREROUTING -p tcp --dport 80 -j DNAT --to-destination 192.0.2.25:3128
(Correct Answer)

Explanation

Correct Answer:

D

Community Discussion

The command "iptables -t nat -A PREROUTING -p tcp --dport 80 -j DNAT --to-destination 192.0.2.25:3128" will temporarily redirect all HTTP traffic to the new proxy server at IP address 192.0.2.25 and port 3128. The options used in this command are: -t nat: Specifies the table (nat) to be used for this operation -A PREROUTING: Specifies that the rule should be added to the PREROUTING chain, which is used to manipulate incoming packets before they are processed by the routing decision -p tcp: Specifies that the rule should apply to TCP packets --dport 80: Specifies that the rule should apply to incoming packets destined for port 80 (the default HTTP port) -j DNAT: Specifies that the target of the rule should be DNAT (Destination NAT), which rewrites the destination address of a packet --to-destination 192.0.2.25:3128: Specifies the destination address and port to which incoming packets should be redirected Note: After executing this command, the administrator should make sure to save the iptables configuration, so that it will persist after a reboot.

-A --append -D --delete -t --tables nat PREROUTING(for altering packets as soon as they come in)

was on the test and i choose D.

was on the test and i choose D.

Selected Answer: D

The correct answer is D. iptables -t nat -A PREROUTING -p tcp --dport 80 -j DNAT --to-destination 192.0.2.25:3128.

Question #5

Developers have requested implementation of a persistent, static route on the application server. Packets sent over the interface eth0 to 10.0.213.5/32 should be routed via 10.0.5.1. Which of the following commands should the administrator run to achieve this goal?

- A) `route -i eth0 -p add 10.0.213.5 10.0.5.1`
- B) `route modify eth0 +ipv4.routes "10.0.213.5/32 10.0.5.1"`
- C) `echo "10.0.213.5 10.0.5.1 eth0" > /proc/net/route`
- D) `ip route add 10.0.213.5/32 via 10.0.5.1 dev eth0` (Correct Answer)**

Explanation

Correct Answer:
D

Community Discussion

D. `ip route add 10.0.213.5/32 via 10.0.5.1 dev eth0` The administrator should run the following command to implement the persistent, static route: "`ip route add 10.0.213.5/32 via 10.0.5.1 dev eth0`" This command adds a static route to the routing table, which specifies that packets sent over the interface "eth0" to the destination address "10.0.213.5/32" should be routed via the gateway "10.0.5.1". The "ip" command is used to manage the Linux IP routing table, and the "route add" option is used to add a new route. The "dev" option specifies the interface over which the packets should be sent. To make the route persistent across reboots, the administrator can add the same command to the appropriate configuration file (e.g., `/etc/rc.local`).

was on the test.

Selected Answer: D

The correct answer is D. `ip route add 10.0.213.5/32 via 10.0.5.1 dev eth0`.

D seems to be correct. I tested it locally and it worked.

Question #6

A user is asking the systems administrator for assistance with writing a script to verify whether a file exists. Given the following:

```
#!/bin/bash
filename=$1
<CONDITIONAL>
echo "File exists"
else
echo "File does not exist"
fi
```

Which of the following commands should replace the <CONDITIONAL> string?

A) if [-f "\$filename"]; then (Correct Answer)

B) if [-d "\$filename"]; then

C) if [-f "\$filename"] then

D) if [-f "\$filename"]; while

Explanation

Correct Answer:

A

Community Discussion

please correct the mistake, right after the hashtag should be a bang(!) and not a number 1

he command "if [-f "\$filename"]; then" is a shell script that tests for the existence of a file named "\$filename". The syntax of this command is as follows: "if" is a shell construct that allows you to execute a command or a series of commands only if a certain condition is met. "[-f "\$filename"]" is a test command that returns true if "\$filename" is a regular file (i.e., not a directory or a symbolic link). The "-f" option is used to test for the existence of a regular file. The "\$filename" variable is enclosed in double quotes to

allow for the possibility of spaces in the file name. "then" is a keyword that specifies the start of the commands to be executed if the test is true.

Answer must be A -f file True if file exists and is a regular file THEN

Question #7

DRAG DROP -

As a Systems Administrator, to reduce disk space, you were tasked to create a shell script that does the following:

Add relevant content to /tmp/script.sh, so that it finds and compresses related files in /var/log without recursion.

INSTRUCTIONS:

Drag and drop snippets to fill the blanks to build a script that performs the actual compression of rotated log files.

If at any time you would like to bring back the initial state of the simulation, please click the Reset All button.

Snippets

done	while	xz
pgrep	locate	for
sed	filename	until
tar	then	\$log
egrep	"\$log.[1-6]\$"	"log.[1-6]@"
"\$1"	repeat	in
/tmp/tempfile	/var/log	zip
gzip	rar	awk
"\$@"		

```
#!/bin/bash

#name: script.sh

find /var/log -type f -maxdepth 1 | grep [?] > /tmp/tempfile

[?] filename [?] $(cat [?])

do
    [?] $filename
    [?]
```

Explanation

Correct Answer:

```
"log.[1-6]@"
```

```
for
```

```
in
```

```
/tmp/tempfile
```

```
gzip
```

```
done
```

Community Discussion

First, the question does not ask for files with "log" in the name (it doesn't even ask for log files - it says "related files"). Second, some of the files in /var/log do not have log or .log anywhere in the filename (e.g. messages, secure, spooler, etc.) and would be missed using your syntax. Third, ".log[1-6]@", with a leading dot (.), is not even an option to select. Your choices are "log.[1-6]@" or "\$log.[1-6]@", without a leading dot. Your version will (likely) create an empty tempfile and compress nothing. And, yes, I've tested your solution (and mine). The correct answer is:

```
#!/bin/bash #name: script.sh find /var/log -type f -maxdepth 1 | grep "$1" > /tmp/tempfile for filename in $(cat /tmp/tempfile) do gzip $filename done
```


Question #8

A systems administrator is deploying three identical, cloud-based servers. The administrator is using the following code to complete the task:

```
resource "aws_instance" "ec2_instance" {  
    ami                        = data.aws_ami.vendor-Linux-2.id  
    associate_public_ip_address = true  
    count                     = 3  
    instance_type             = "instance_type"  
    vpc_security_group_ids    = [aws_security_group.allow_ssh.id]  
    key_name                   = aws_key_pair.key_pair.key_name  
  
    tags = {  
        Name = "${var.namespace} ${count.index}"  
    }  
}
```

Which of the following technologies is the administrator using?

- A) Ansible
- B) Puppet
- C) Chef

D) Terraform (Correct Answer)

Explanation

Correct Answer:

D

Community Discussion

In this example, Terraform is used to deploy an Amazon Web Services (AWS) EC2 instance. The first line of the code defines the AWS provider, and the region attribute is set to "us-west-2". The second section of the code defines the aws_instance resource. The ami attribute specifies the Amazon Machine Image (AMI) ID to use, and the instance_type attribute specifies the type of instance to deploy. The tags attribute is used to add a tag to the instance, with the key "Name" and the value "example-instance". This code is a simple example of how Terraform can be used to deploy a cloud-based server. In a real-

world scenario, the code would likely be more complex and include additional resources and configuration options, such as security groups, subnets, and more.

In this example, Terraform is used to deploy an Amazon Web Services (AWS) EC2 instance. The first line of the code defines the AWS provider, and the region attribute is set to "us-west-2". The second section of the code defines the aws_instance resource. The ami attribute specifies the Amazon Machine Image (AMI) ID to use, and the instance_type attribute specifies the type of instance to deploy. The tags attribute is used to add a tag to the instance, with the key "Name" and the value "example-instance". This code is a simple example of how Terraform can be used to deploy a cloud-based server. In a real-world scenario, the code would likely be more complex and include additional resources and configuration options, such as security groups, subnets, and more.

Here is an example code for deploying cloud-based servers using Terraform: provider "aws" { region = "us-west-2" } resource "aws_instance" "example" { ami = "ami-0c55b159cbf1f0" instance_type = "t2.micro" tags = { Name = "example-instance" } }

Question #9

Which of the following technologies can be used as a central repository of Linux users and groups?

A) LDAP (Correct Answer)

- B) MFA
- C) SSO
- D) PAM

Explanation

Correct Answer:

A

Community Discussion

A. LDAP (Lightweight Directory Access Protocol) can be used as a central repository of Linux users and groups. LDAP is a widely used protocol for accessing directory services and is commonly used to store user and group information in a central repository. This central repository can be used to manage user and group information for multiple

systems, making it easier to manage and maintain user and group information across an organization.

A LDAP= Lightweight Director Access Protocol MFA= Multi-factor Authentication SSO= Single Sign On PAM= Pluggable Authentication Module

Selected Answer: A

lightweight directory access protocol others are multi-factor authentication, single sign on, and pluggable authentication modules...they can use ldap for authenticating

Selected Answer: A

The answer is A. LDAP.

Question #10

A systems administrator is troubleshooting connectivity issues and trying to find out why a Linux server is not able to reach other servers on the same subnet it is connected to. When listing link parameters, the following is presented:

```
# ip link list dev eth0
2: eth0: <NO-CARRIER, BROADCAST, MULTICAST, UP> mtu 1500, qdisc
fq_codel state DOWN mode DEFAULT group default qlen 1000
link/ether ac:00:11:22:33:cd brd ff:ff:ff:ff:ff:ff
```

Based on the output above, which of following is the MOST probable cause of the issue?

- A) The address ac:00:11:22:33:cd is not a valid Ethernet address.
- B) The Ethernet broadcast address should be ac:00:11:22:33:ff instead.
- C) The network interface eth0 is using an old kernel module.
- D) The network interface cable is not connected to a switch. (Correct Answer)**

Explanation

Correct Answer:

D

Community Discussion

D. The network interface cable is not connected to a switch. The output shows that the state of the network interface eth0 is "DOWN". This means that the interface is not currently transmitting or receiving data. The "NO-CARRIER" status also suggests that the interface is not connected to a network. The most probable cause of the issue is that the network interface cable is not connected to a switch, or that the switch the cable is connected to is not functioning correctly. When a network interface is down, it usually means that there is an issue with the physical or logical connection to the network.

The fact that it says 'no carrier' should be dead giveaway to there being no cable.

Question #11

A Linux administrator was asked to run a container with the httpd server inside. This container should be exposed at port 443 of a Linux host machine while it internally listens on port 8443. Which of the following commands will accomplish this task?

A) podman run -d -p 443:8443 httpd (Correct Answer)

B) podman run -d -p 8443:443 httpd

C) podman run -d -e 443:8443 httpd

D) podman exec -p 8443:443 httpd

Explanation

Correct Answer:

A

Community Discussion

Answer is B Source: <https://www.howtoforge.com/getting-started-with-podman-manage-images-container-and-volumes/> look at section 3: Manage Containers

Selected Answer: A

<host_port>:<container_port>.

Selected Answer: A

I checked the documentation and first is host port and then the container port, so option A should be the right one. since you are exposing the port 443 on the host to the port 8443 on the container

According to the Podman documentation, the host machine port is the first parameter in the command when used with the -p option: -p=[[ip:][hostPort:]:]containerPort[/protocol]
Under this construction, answer A would be correct. I know that Podman can be seamlessly aliased to Docker in Ubuntu but I am not sure if Docker uses a different construction. I also think that the fact that Podman is a daemonless application influences how it interfaces with the O/S ports and protocols.

A. If you're using docker compose the pattern is HOST:CONTAINER, i.e. 443 on the host and 8443 on the container would be 443:8443

Question #12

A Linux administrator needs to analyze a failing application that is running inside a container. Which of the following commands allows the Linux administrator to enter the running container and analyze the logs that are stored inside?

- A) `docker run -ti app /bin/sh`
- B) `podman exec -ti app /bin/sh` (Correct Answer)**
- C) `podman run -d app /bin/bash`
- D) `docker exec -d app /bin/bash`

Explanation

Correct Answer:

B

Community Discussion

B. `podman exec -ti app /bin/sh` The `podman exec` command allows a Linux administrator to enter a running container and perform operations inside the container. The `-ti` options specify that the administrator wants to run the command interactively with a TTY and

with STDIN attached. The app in the command specifies the name or ID of the running container that the administrator wants to enter. The `/bin/sh` at the end of the command specifies the shell to use inside the container.

Selected Answer: B

B is correct... `exec` option is for running containers `-t` for TTY and `-i` interactive

In this case, the administrator wants to enter the container to analyze logs that are stored inside. To accomplish this, the administrator should run the following command: `podman exec -ti app /bin/sh` This command will enter the running container named `app` and open a shell inside the container. The administrator can then use standard Linux commands to analyze the logs and diagnose the issue with the failing application

In this case, the administrator wants to enter the container to analyze logs that are stored inside. To accomplish this, the administrator should run the following command: `podman exec -ti app /bin/sh` This command will enter the running container named `app` and open a shell inside the container. The administrator can then use standard Linux commands to analyze the logs and diagnose the issue with the failing application

Question #13

A systems administrator needs to clone the partition `/dev/sdc1` to `/dev/sdd1`. Which of the following commands will accomplish this task?

- A) `tar -cvzf /dev/sdd1 /dev/sdc1`
- B) `rsync /dev/sdc1 /dev/sdd1`
- C) `dd if=/dev/sdc1 of=/dev/sdd1` (Correct Answer)**
- D) `scp /dev/sdc1 /dev/sdd1`

Explanation

Correct Answer:

C

Community Discussion

C. `dd if=/dev/sdc1 of=/dev/sdd1` The `dd` command is a low-level utility that can be used to copy data from one location to another. In this case, the administrator wants to clone the partition `/dev/sdc1` to `/dev/sdd1`. To accomplish this, the administrator should run the following command: `dd if=/dev/sdc1 of=/dev/sdd1` The `if` option specifies the input file, which is `/dev/sdc1` in this case. The `of` option specifies the output file, which is `/dev/sdd1`. The `dd` command will read the data from `/dev/sdc1` and write it to `/dev/sdd1`, effectively cloning the partition. Note that the `dd` command is a very powerful tool and can cause data loss if used improperly. The administrator should make sure to backup important data before using the `dd` command, and be very careful when specifying the input and output files.

Selected Answer: C

C is correct

Selected Answer: C

`dd` is the only command that can do this from the above `if=input-file of=output file`

Selected Answer: C

The correct answer is C. `dd if=/dev/sdc1 of=/dev/sdd1`.

Question #14

When trying to log in remotely to a server, a user receives the following message:

```
Password:
Last failed login: Wed Sep 15 17:23:45 CEST 2021 from 10.0.4.3 on ssh:notty
There were 3 failed login attempts since the last successful login.
Connection to localhost closed.
```

The server administrator is investigating the issue on the server and receives the following outputs:

Output 1:

```
user:x:1001:7374:./home/user:/bin/false
```

Output 2:

```
dzwx-----. 2 user 62 Sep 15 17:17 /home/user
```

Output 3:

```
Sep 12 14:14:05 server sshd[22958] Failed password for user from 10.0.2.8
Sep 15 17:24:03 server sshd[8460]: Accepted keyboard-interactive/pam for user from 10.0.6.5 port 50928 ssh2
Sep 15 17:24:03 server sshd[8460]: pam_unix(sshd:session): session opened for user testuser
Sep 15 17:24:03 server sshd[8460]: pam_unix(sshd:session): session closed for user testuser
```

Which of the following is causing the issue?

- A) The wrong permissions are on the user's home directory.
- B) The account was locked out due to three failed logins.
- C) The user entered the wrong password.

D) The user has the wrong shell assigned to the account. (Correct Answer)

Explanation

Correct Answer:

D

Community Discussion

The /bin/false shell is a special shell that returns a non-zero exit status, effectively denying the user access to the system. This is often used for system accounts that do not need to interact with the system, or for temporarily disabling accounts.

D. The user has the wrong shell assigned to the account.

D. The user has the wrong shell assigned to the account.

Selected Answer: D

D correct

option 1, last column after : says /bin/false ... that usually says /bin/bash for the user's shell ... /bin/false would deny user being able to login

Question #15

A new Linux systems administrator just generated a pair of SSH keys that should allow connection to the servers. Which of the following commands can be used to copy a key file to remote servers? (Choose two.)

- A) wget
- B) ssh-keygen
- C) ssh-keyscan
- D) ssh-copy-id (Correct Answer)**
- E) ftpd
- F) scp (Correct Answer)**

Explanation

Correct Answer:
DF

Community Discussion

For example: `ssh-copy-id user@remote_server` The `scp` command is used to securely copy files between systems over an SSH connection. The command can be used to copy an SSH key file to a remote server. For example: `scp ~/.ssh/id_rsa.pub user@remote_server:/tmp/id_rsa.pub`

For example: `ssh-copy-id user@remote_server` The `scp` command is used to securely copy files between systems over an SSH connection. The command can be used to copy an SSH key file to a remote server. For example: `scp ~/.ssh/id_rsa.pub user@remote_server:/tmp/id_rsa.pub`

D. `ssh-copy-id` F. `scp` The `ssh-copy-id` and `scp` commands can be used to copy an SSH key file to remote servers. The `ssh-copy-id` command is used to install the public key of an SSH key pair to the `authorized_keys` file of a remote server. This allows the user to log into the remote server without a password, using the private key of the key pair.

In this example, the public key of the SSH key pair (`~/.ssh/id_rsa.pub`) is copied to the remote server (`remote_server`) in the `/tmp` directory with the name `id_rsa.pub`. The administrator can then use the `ssh-copy-id` command to install the public key on the remote server.

In this example, the public key of the SSH key pair (`~/.ssh/id_rsa.pub`) is copied to the remote server (`remote_server`) in the `/tmp` directory with the name `id_rsa.pub`. The administrator can then use the `ssh-copy-id` command to install the public key on the remote server.

Question #16

A systems administrator needs to reconfigure a Linux server to allow persistent IPv4 packet forwarding. Which of the following commands is the correct way to accomplish this task?

- A) `echo 1 > /proc/sys/net/ipv4/ipv_forward`
- B) `sysctl -w net.ipv4.ip_forward=1` (Correct Answer)**
- C) `firewall-cmd --enable ipv4_forwarding`
- D) `systemctl start ipv4_forwarding`

Explanation

Correct Answer:

B

Community Discussion

B. `sysctl -w net.ipv4.ip_forward=1` The `sysctl` command is used to configure the Linux kernel parameters at runtime. To enable IPv4 packet forwarding, the administrator needs to set the value of the `net.ipv4.ip_forward` parameter to 1. This can be done with the following command: `sysctl -w net.ipv4.ip_forward=1` This command sets the value of `net.ipv4.ip_forward` to 1, enabling IPv4 packet forwarding. The `-w` option specifies that the change should be written to the running system and not just displayed.

Note that this change will only persist until the next reboot. To make the change persistent across reboots, the administrator needs to add the following line to the `/etc/sysctl.conf` file: `net.ipv4.ip_forward=1` After adding this line, the administrator can run the `sysctl -p` command to reload the `/etc/sysctl.conf` file and apply the changes.

Note that this change will only persist until the next reboot. To make the change persistent across reboots, the administrator needs to add the following line to the `/etc/sysctl.conf` file: `net.ipv4.ip_forward=1` After adding this line, the administrator can run the `sysctl -p` command to reload the `/etc/sysctl.conf` file and apply the changes.

<https://linuxconfig.org/how-to-turn-on-off-ip-forwarding-in-linux>

B <https://linuxconfig.org/how-to-turn-on-off-ip-forwarding-in-linux>

Question #17

A Linux administrator would like to use `systemd` to schedule a job to run every two hours. The administrator creates timer and service definitions and restarts the server to load these new configurations. After the restart, the administrator checks the log file and notices that the job is only running daily. Which of the following is MOST likely causing the issue?

- A) The `checkdiskspace.service` is not running.
- B) The `checkdiskspace.service` needs to be enabled.
- C) The `OnCalendar` schedule is incorrect in the timer definition.

D) The system-daemon services need to be reloaded. (Correct Answer)

Explanation

Correct Answer:

D

Community Discussion

In the previous question, the answer choice D was "The system-daemon services need to be reloaded." This answer is not the most likely cause of the issue because reloading the system-daemon services will not fix an incorrect OnCalendar schedule in the timer definition. The OnCalendar directive is used in the timer definition to specify the schedule on which the timer should run. If the schedule is incorrect, the timer will not run as expected, regardless of the state of the system-daemon services. The most likely cause of the issue, as stated in answer choice C, is an incorrect OnCalendar schedule in the timer definition. To resolve the issue, the administrator should check the timer definition and make sure that the OnCalendar schedule is set correctly.

In the previous question, the answer choice D was "The system-daemon services need to be reloaded." This answer is not the most likely cause of the issue because reloading the system-daemon services will not fix an incorrect OnCalendar schedule in the timer definition. The OnCalendar directive is used in the timer definition to specify the schedule on which the timer should run. If the schedule is incorrect, the timer will not run as expected, regardless of the state of the system-daemon services. The most likely cause of the issue, as stated in answer choice C, is an incorrect OnCalendar schedule in the timer definition. To resolve the issue, the administrator should check the timer definition and make sure that the OnCalendar schedule is set correctly.

Selected Answer: C

When using systemd timers, the "OnCalendar" option is used to define when the job should run. By default, systemd timers are configured to use UTC time. If the administrator specified an incorrect OnCalendar schedule, the timer may not run as expected. To schedule a job to run every two hours, the correct OnCalendar schedule should be "0 */2 * * *". This will run the job every two hours, starting at the top of the hour (i.e., at 0 minutes past the hour). If the administrator did not specify the correct OnCalendar schedule, the timer may be set to run daily or at some other incorrect interval, which would explain the behavior observed in the log file.

Selected Answer: C

Picking C Here.

Hello all, I respectfully disagree, answer is correct, its D, there is nothing wrong with the OnCalendar, you ALWAYS need to reload system-daemon after you change or update configurations, this is how you make the system aware of the changes, This will rerun all generators, reload all unit files, and recreate the entire dependency tree.

Question #18

An administrator deployed a Linux server that is running a web application on port 6379/tcp.

SELinux is in enforcing mode based on organization policies.

The port is open on the firewall.

Users who are trying to connect to a local instance of the web application receive Error 13, Permission denied.

The administrator ran some commands that resulted in the following output:

```
# semanage port -l | egrep '(^http_port_t|6379) '
http_port_t tcp 80, 81, 443, 488, 8008, 8009, 8443, 9000

# curl http://localhost/App.php
Cannot connect to App Server.
```

Which of the following commands should be used to resolve the issue?

- A) `semanage port -d -t http_port_t -p tcp 6379`
- B) `semanage port -a -t http_port_t -p tcp 6379` (Correct Answer)**
- C) `semanage port -a http_port_t -p top 6379`
- D) `semanage port -l -t http_port_tcp 6379`

Explanation

Correct Answer:

B

Community Discussion

The `semanage` command is used to manage SELinux policy settings on Linux systems. In this case, the `semanage port -a -t http_port_t -p tcp 6379` command is adding a new SELinux port type for the TCP protocol and port number 6379. Here's what each option

does: -a: The -a option stands for "add", and it is used to add a new SELinux port type. -t http_port_t: The -t option is used to specify the SELinux type for the new port, in this case, http_port_t. -p tcp 6379: The -p option is used to specify the protocol and port number, in this case, TCP protocol and port number 6379. With this command, the SELinux policy is updated to allow the TCP protocol to listen on port number 6379 with the http_port_t type. This is useful in cases where the application requires the use of a non-standard port number for HTTP traffic.

Selected Answer: B

Correct answer is B -a means Add and -t Type

Question #19

A systems administrator created a web server for the company and is required to add a tag for the API so end users can connect. Which of the following would the administrator do to complete this requirement?

- A) hostnamectl status --no-ask-password
- B) hostnamectl set-hostname "\$(perl -le "print" "A" x 86)"
- C) hostnamectl set-hostname Comptia-WebNode -H [\[email protected\]](#)** (Correct Answer)
- D) hostnamectl set-hostname Comptia-WebNode --transient

Explanation

Correct Answer:

C

Community Discussion

The `hostnamectl set-hostname Comptia-WebNode -H root@192.168.2.14` command sets the hostname of a Linux system. Here's what each option does: `set-hostname Comptia-WebNode`: The `set-hostname` option sets the hostname of the system to "Comptia-WebNode". `-H root@192.168.2.14`: The `-H` option is used to specify the remote host to connect to. In this case, the administrator is connecting to the remote host with the IP address 192.168.2.14 as the root user. This command sets the hostname of the remote host with IP address 192.168.2.14 to "Comptia-WebNode" as the root user. The new hostname will persist across reboots unless the administrator changes it again in the future.

In the context of the question, a "tag" refers to a label or identifier that is used to distinguish or identify a specific object, in this case, the web server. The tag is used to make it easier for end users to connect to the web server, as they can use the tag instead of an IP address or a hostname. The tag is typically a short and descriptive string that represents the web server, such as "Comptia-WebNode" in the example.

In the context of the question, a "tag" refers to a label or identifier that is used to distinguish or identify a specific object, in this case, the web server. The tag is used to make it easier for end users to connect to the web server, as they can use the tag instead of an IP address or a hostname. The tag is typically a short and descriptive string that represents the web server, such as "Comptia-WebNode" in the example.

Selected Answer: C

To execute a `hostnamectl` command on a remote system, use the `-H, --host`. That is change a tag remotely in my view.

For adding a tag to the API, you typically don't use the `hostnamectl` command. Instead, you might want to consider adding a label or tag directly related to the API functionality. However, among the given options, the most relevant one would be: D. `hostnamectl set-hostname Comptia-WebNode --transient` This sets the transient (temporary) hostname to "Comptia-WebNode". However, keep in mind that this command is more related to the system's hostname configuration rather than specifically tagging an API. For tagging an API, you might want to consider using metadata or labels associated with the API service itself, which is typically done in the application or service configuration rather than at the system level.

Question #20

A systems administrator wants to back up the directory `/data` and all its contents to `/backup/data` on a remote server named `remote`. Which of the following commands will achieve the desired effect?

- A) `scp -p /data remote:/backup/data`
- B) `ssh -i /remote:/backup/ /data`
- C) `rsync -a /data remote:/backup/` (Correct Answer)**
- D) `cp -r /data /remote/backup/`

Explanation

Correct Answer:

C

Community Discussion

C. `rsync -a /data remote:/backup/` The `rsync` command is used to copy files and directories from one location to another, either locally or over a network. The `-a` option is used to preserve the original file attributes, such as timestamps, permissions, and symbolic links. In this case, the administrator is using `rsync` to copy the contents of the local directory `/data` to the remote server `remote` at the location `/backup/`. The `rsync` command will copy all the files and subdirectories in `/data` to the remote server and preserve their original attributes.

The other options are not correct or do not achieve the desired effect. The `scp` command is used to copy files securely over a network, but it does not preserve the original file attributes. The `ssh` command is used to log into a remote server, but it does not copy files. The `cp` command is used to copy files locally, but it does not copy files over a network.

Option A (`scp -p /data remote:/backup/data`) is not the correct command to achieve the desired effect of backing up the `/data` directory and its contents to the remote server. The `scp` command is used to copy files and directories between hosts over SSH, but it does not support copying directories recursively by default. Therefore, the `/data` directory will not be backed up completely using this command. To copy directories recursively, we would need to use the `-r` option with `scp`. In addition, we need to specify the destination directory as `/backup` instead of `/backup/data` since we want to copy the

contents of the /data directory to /backup. Therefore, the correct scp command would be: `scp -r /data remote:/backup`

Option A (`scp -p /data remote:/backup/data`) is not the correct command to achieve the desired effect of backing up the /data directory and its contents to the remote server. The scp command is used to copy files and directories between hosts over SSH, but it does not support copying directories recursively by default. Therefore, the /data directory will not be backed up completely using this command. To copy directories recursively, we would need to use the -r option with scp. In addition, we need to specify the destination directory as /backup instead of /backup/data since we want to copy the contents of the /data directory to /backup. Therefore, the correct scp command would be: `scp -r /data remote:/backup`

The other options are not correct or do not achieve the desired effect. The scp command is used to copy files securely over a network, but it does not preserve the original file attributes. The ssh command is used to log into a remote server, but it does not copy files. The cp command is used to copy files locally, but it does not copy files over a network.

Question #21

An administrator needs to make some changes in the IaC declaration templates. Which of the following commands would maintain version control?

- A) `git clone https://github.com/comptia/linux+-.git`
`git push origin`
- B) `git clone https://qithub.com/comptia/linux+-.git`
`git fetch New-Branch`
- C) `git clone https://github.com/comptia/linux+-.git`
`git status`
- D) `git clone https://github.com/comptia/linux+-.git`
`git checkout -b <new-branch>` (Correct Answer)**

Explanation

Correct Answer:

D

Community Discussion

D. `git clone https://github.com/comptia/linux+.git` `git checkout -b <new-branch>` The `git clone` command is used to clone a remote Git repository, which in this case is the repository located at `https://github.com/comptia/linux+.git`. This command will download the entire repository to the local machine. After cloning the repository, the administrator should create a new branch using the `git checkout -b <new-branch>` command. This will create a new branch in the Git repository and make it the current branch. The administrator can then make changes to the IaC declaration templates in this branch without affecting the main branch.

Once the changes are made, the administrator can then commit them to the new branch using `git commit`. Finally, the administrator can merge the changes back into the main branch using `git merge`. This will allow the changes to be tracked and maintain version control of the IaC declaration templates. The other options are not correct or do not achieve the desired effect. The `git push origin` command is used to push changes to a remote repository, but it should not be used before making changes. The `git fetch New-Branch` command is used to retrieve changes from a remote repository, but it does not create a new branch. The `git status` command is used to check the current status of the Git repository, but it does not create a new branch or allow changes to be made.

Once the changes are made, the administrator can then commit them to the new branch using `git commit`. Finally, the administrator can merge the changes back into the main branch using `git merge`. This will allow the changes to be tracked and maintain version control of the IaC declaration templates. The other options are not correct or do not achieve the desired effect. The `git push origin` command is used to push changes to a remote repository, but it should not be used before making changes. The `git fetch New-Branch` command is used to retrieve changes from a remote repository, but it does not create a new branch. The `git status` command is used to check the current status of the Git repository, but it does not create a new branch or allow changes to be made.

Selected Answer: D

D. `git clone https://github.com/comptia/linux+-.git` `git checkout -b <new-branch>`.

Selected Answer: D

`git checkout -b|-B <new_branch> [<start point>]` Specifying `-b` causes a new branch to be created as if `git-branch(1)` were called and then checked out.

Question #22

An administrator attempts to rename a file on a server but receives the following error.

```
mv: cannot move 'files/readme.txt' to 'files/readme.txt.orig': Operation not permitted.
```

The administrator then runs a few commands and obtains the following output:

```
$ ls -ld files/
drwxrwxrwt.1  users  users  20   Sep 10   15:15  files/
$ ls -a files/
drwxrwxrwt.1  users  users  20   Sep 10   15:15  -
drwxr-xr-x.1  users  users  32   Sep 10   15:15  ..
-rw-rw-r--.1  users  users   4   Sep 12   10:34  readme.txt
```

Which of the following commands should the administrator run NEXT to allow the file to be renamed by any user?

- A) `chgrp reet files`
- B) `chacl -R 644 files`
- C) `chown users files`

D) `chmod -t files` (Correct Answer)

Explanation

Correct Answer:

D

Community Discussion

Selected Answer: D

D is correct

Selected Answer: D

D is correct the directory has Sticky bit on. in the first ls -ld you can see the "t" instead of an "x"

Selected Answer: D

The correct answer is D. chmod -t files

You can use the chmod command with the -t option to remove the sticky bit: chmod -t /path/to/directory You cannot remove a sticky bit from a file directly. The sticky bit is a property of a directory, not a file. When a sticky bit is set on a directory, it affects the behavior of file deletion within that directory. A file within a directory with a sticky bit set cannot be deleted by a user who does not have write permissions to the directory, regardless of the permissions on the file itself. Therefore, to remove a sticky bit, you need to use the path to the directory for which you want to remove the sticky bit, not a file within the directory.

Question #23

Which of the following commands will display the operating system?

- A) uname -n
- B) uname -s
- C) uname -o (Correct Answer)**
- D) uname -m

Explanation

Correct Answer:

C

Community Discussion

Selected Answer: C

uname man page -a, --all print all information, in the following order, except omit -p and -i if unknown: -s, --kernel-name print the kernel name -n, --nodename print the network node hostname -r, --kernel-release print the kernel release -v, --kernel-version print the kernel version -m, --machine print the machine hardware name -p, --processor print the

processor type or "unknown" -i, --hardware-platform print the hardware platform or "unknown" -o, --operating-system print the operating system --help display this help and exit --version output version information and exit

Selected Answer: C

The correct command to display the operating system is C. `uname -o`. The `uname` command is used to print system information. It has several options that can be used to print different types of information. Here's what each option does in the context of the given question: A. `uname -n`: prints the hostname of the machine. B. `uname -s`: prints the kernel name of the machine. C. `uname -o`: prints the operating system of the machine. D. `uname -m`: prints the machine hardware name. Therefore, the correct option to display the operating system is C (`uname -o`).

```
[bryan@almalinux etc]$ uname -n almalinux [bryan@almalinux etc]$ uname -o GNU/Linux
[bryan@almalinux etc]$ uname -s Linux [bryan@almalinux etc]$ uname -m x86_64
```

Selected Answer: C

C. `uname -o` (OS) -s (kernel) is Linux -o (OS) is x86_64 <https://www.youtube.com/watch?v=1AoKTorlEDc> (2:28)

Selected Answer: B

The option `-o` for the `uname` command does not display the operating system. Instead, it is used to display the operating system name, which is not the same as the operating system itself. The `-s` option is used to display the operating system name, which typically includes the name, version, and architecture. For example, if you run the command `uname -s` on a Linux system, it will return the following output: Linux So, in this case, the correct option for displaying the operating system would be `-s`, not `-o`.

Question #24

A systems engineer is adding a new 1GB XFS filesystem that should be temporarily mounted under `/ops/app`. Which of the following is the correct list of commands to achieve this goal?

A)

```
pvcreate -L1G /dev/app  
mkfs.xfs /dev/app  
mount /dev/app /opt/app
```

B)

```
parted /dev/sdb --script mkpart primary xfs 1GB  
mkfs.xfs /dev/sdb  
mount /dev/sdb /opt/app
```

C)

```
lvs --create 1G --name app  
mkfs.xfs /dev/app  
mount /dev/app /opt/app
```

D)

```
lvcreate -L 1G -n app app_vg  
mkfs.xfs /dev/app_vg/app  
mount /dev/app_vg/app /opt/app
```

(Correct Answer)

Explanation

Correct Answer:

D

Community Discussion

Selected Answer: D

Question should be fixed to say: mounted under "/opt/app" not "/ops/app".

Selected Answer: D

D. `lvcreate -L 1G -n app app_vg mkfs.xfs /dev/app_vg/app mount /dev/app_vg/app /opt/app` This creates a new logical volume of size 1GB named "app" in the volume group "app_vg", formats it with the XFS filesystem, and mounts it under the `/opt/app` directory.

****Discard****

****Discard****

Selected Answer: B

i get B because the others have to do with "trying" to expand a logical volume. The question says just to simply add a 1GB partition

Question #25

A Linux administrator recently downloaded a software package that is currently in a compressed file. Which of the following commands will extract the files?

- A) `unzip -v`
- B) `bzip2 -z`
- C) `gzip` (Correct Answer)**
- D) `funzip`

Explanation

Correct Answer:

C

Community Discussion

Selected Answer: A

opposite of `gzip` is `gunzip`...`gzip` compresses, `gunzip` uncompresses ... just because `.gz` suffix on a file, it will not uncompress with `gzip`...needs `gunzip` # `gzip file.gz` `gzip: file.gz` already has `.gz` suffix -- unchanged

Selected Answer: A

Answer is A because unzip -v decompress with verbose and gzip without the -d option will compress.

Selected Answer: C

C. gzip is the best option.

Selected Answer: D

unzip -v only displays the contents with verbosity of a compressed file but not extract
bzip2 -z will compress a file not extract
gzip will only compress a file not extract
funzip is the ONLY option here that if used just like it is will extract a file. The funzip command is used to decompress files that have been compressed with the ZIP compression algorithm. It is part of the Info-ZIP suite of tools and is typically available on Linux and Unix systems.

gzip -d decompresses,, but then they aren't giving us the -d in the answer.. These questions are ridiculous. we shouldn't have to be guessing if an option can or can't be used.

Question #26

A Linux administrator is troubleshooting SSH connection issues from one of the workstations.

When users attempt to log in from the workstation to a server with the IP address 104.21.75.76, they receive the following message:

```
ssh: connect to host 104.21.75.76 port 22: Connection refused
```

The administrator reviews the information below:

Workstation output 1:

```
eth0: <BROADCAST,MULTICAST, UP, LOWER_UP> mtu 1500 qdisc mq state UP group default
link/ether 00:15:5d:e9:e9:fb brd 5.189.153.255 scope global eth0
inet 5.189.153.89/24 brd 5.189.153.255 scope global eth0
```

Workstation output 2:

```
default via 5.189.153.1 dev eth0
5.189.153.0/24 dev eth0 proto kernel scope link src 5.189.153.89
```

Server output 1:

target	prot	opt	source	destination	
REJECT	tcp	--	101.68.78.194	0.0.0.0/0	tcp dpt:22 ctstate NEW, UNTRACKED reject-with icmp-port-unreachable
REJECT	tcp	--	222.186.180.130	0.0.0.0/0	tcp dpt:22 ctstate NEW, UNTRACKED reject-with icmp-port-unreachable
REJECT	tcp	--	104.131.1.39	0.0.0.0/0	tcp dpt:22 ctstate NEW, UNTRACKED reject-with icmp-port-unreachable
REJECT	tcp	--	68.183.196.11	0.0.0.0/0	tcp dpt:22 ctstate NEW, UNTRACKED reject-with icmp-port-unreachable
REJECT	tcp	--	5.189.153.89	0.0.0.0/0	tcp dpt:22 ctstate NEW, UNTRACKED reject-with icmp-port-unreachable
REJECT	tcp	--	41.93.32.148	0.0.0.0/0	tcp dpt:22 ctstate NEW, UNTRACKED reject-with icmp-port-unreachable

Server output 2:

```
sshd.service - OpenSSH server daemon
Loaded: loaded (/usr/lib/systemd/system/sshd.service; disabled; vendor preset: enabled)
Active: active (running) since Thu 2021-08-26 18:50:19 CEST; 2 weeks 5 days ago
```

Server output 3:

```
eth0: <BROADCAST, MULTICAST, UP, LOWER_UP> mtu 1500 qdisc mg state UP group default
link/ether 52:52:00:2a:bb:98 brd 104.21.75.255 scope global eth0
inet 104.21.75.76/24 brd 104.21.75.255 scope global eth0
```

Server output 4:

```
default via 104.21.75.254 dev eth0
104.21.75.0/24 dev eth0 proto kernel scope link src 104.21.75.76
```

Which of the following is causing the connectivity issue?

- A) The workstation has the wrong IP settings.
- B) The sshd service is disabled. (Correct Answer)**
- C) The server's firewall is preventing connections from being made.
- D) The server has an incorrect default gateway configuration.

Explanation

Correct Answer:

B

Community Discussion

Selected Answer: C

It's a Firewall issue, i'm picking `C`. Please note, it's easy to misread the title of the outputs, read carefully -i made the same mistake from `TheRealManish` contribution also... it reads: Server Output#1 Server Output#2 Server Output#3 Server Output#4 *NOT* Server #1 Output Server #2 Output Server #3 Output Server #4 Output Cheers!

Selected Answer: C

Server output 2 -> sshd "active (running)", so answer not B :-(Server output 1 -> port 22 being blocked from all sources ... firewall issue Everything else looks fine

Yes the ssh service is disabled but as you can see it has been running, what that means is that the service does not start up automatically but needs the administrator to actively start it up, but the firewall drops the packets from port 22 and thus it is a firewall issue.

Yes the ssh service is disabled but as you can see it has been running, what that means is that the service does not start up automatically but needs the administrator to actively start it up, but the firewall drops the packets from port 22 and thus it is a firewall issue.

OMG thanks so much! i totally missed the wording!! i was like why the F is it telling us all of these other outputs!

Question #27

Which of the following files holds the system configuration for journal when running systemd?

A) /etc/systemd/journald.conf (Correct Answer)

B) /etc/systemd/systemd-journalctl.conf

C) /usr/lib/systemd/journalctl.conf

D) /etc/systemd/systemd-journald.conf

Explanation

Correct Answer:

A

Community Discussion

Selected Answer: A

amazingly, it's the same file on both Debian and Red Hat class systems

Question #28

A Linux administrator is tasked with creating resources using containerization. When deciding how to create this type of deployment, the administrator identifies some key features, including portability, high availability, and scalability in production. Which of the following should the Linux administrator choose for the new design?

A) Docker

B) On-premises systems

C) Cloud-based systems

D) Kubernetes (Correct Answer)

Explanation

Correct Answer:

Community Discussion

Do not understand the context of this question. Both MS Azure and AWS offer Kubernetes based container services. It seems that the cloud is far more portable, accessible, and available than an on-premises system which would rely on VPN tunnels over public internet to access the resources. What are the assumptions we are supposed to make when posed with a question like this? Also, why not Docker?

Selected Answer: D

Kubernetes is a powerful container orchestration platform that provides features such as portability, high availability, and scalability in production environments. It automates the deployment, scaling, and management of containerized applications, allowing for efficient resource utilization and seamless scaling. Kubernetes can run on both on-premises systems and cloud-based systems, providing flexibility in deployment options. While Docker is a popular containerization platform, Kubernetes extends its capabilities by providing advanced orchestration features suitable for production environments. On-premises and cloud-based systems are infrastructure choices, whereas Kubernetes is a platform for managing containerized applications regardless of the underlying infrastructure.

Selected Answer: A

"creating this type of deployment" , kubernetes is used for container orchestration

kubernetes can be used to enhance docker or cloud based services but cannot be used as a stand alone deployment, and cloud based services are not as portable as containers, i.e.docker

kubernetes can be used to enhance docker or cloud based services but cannot be used as a stand alone deployment, and cloud based services are not as portable as containers, i.e.docker

Question #29

Which of the following tools is commonly used for creating CI/CD pipelines?

A) Chef

B) Puppet

C) Jenkins (Correct Answer)

D) Ansible

Explanation

Correct Answer:

C

Community Discussion

Selected Answer: C

C. Jenkins is commonly used for creating CI/CD (Continuous Integration/Continuous Deployment) pipelines. It is an open-source automation server that can be used to automate various tasks, including building, testing, and deploying software applications. Other tools like Chef, Puppet, and Ansible are typically used for configuration management and infrastructure automation.

Selected Answer: C

C is correct, CI/CD build/Testing step.

Selected Answer: C

C is correct... other options are for config management.

Ansible is an orchestration tool that is commonly used for Red Hat Enterprise Linux deployments, is agentless, and relies on the Python programming language. Chef is an orchestration tool that utilizes agents and the Ruby programming language. Puppet is an orchestration tool that is agentless and recognizes configuration files in Bash, Python, Ruby, YAML, and PowerShell.

Ansible is an orchestration tool that is commonly used for Red Hat Enterprise Linux deployments, is agentless, and relies on the Python programming language. Chef is an orchestration tool that utilizes agents and the Ruby programming language. Puppet is an orchestration tool that is agentless and recognizes configuration files in Bash, Python, Ruby, YAML, and PowerShell.

Question #30

A systems administrator requires that all files that are created by the user named web have read-only permissions by the owner. Which of the following commands will satisfy this requirement?

A) `chown web:web /home/web`

B) `chmod -R 400 /home/web` (Correct Answer)

C) `echo "umask 377" >> /home/web/.bashrc`

D) `setfacl read /home/web`

Explanation

Correct Answer:

B

Community Discussion

why?!!! you were absolutely right with your first answer. Anyone who think that the right answer is C should think really hard about their life. "umask 377" - in binary 3 = 011 7 = 111 7 = 111 Then, we invert each bit to get the bits that should be turned off: 011 -> 100 111 -> 000 111 -> 000 And there you have 100 which in decimal is 600 (4 - read, 2 write). Question says that the owner of the file should have read permission only - 400, so that automatically excludes C as the answer.

why?!!! you were absolutely right with your first answer. Anyone who think that the right answer is C should think really hard about their life. "umask 377" - in binary 3 = 011 7 = 111 7 = 111 Then, we invert each bit to get the bits that should be turned off: 011 -> 100 111 -> 000 111 -> 000 And there you have 100 which in decimal is 600 (4 - read, 2 write). Question says that the owner of the file should have read permission only - 400, so that automatically excludes C as the answer.

why?!!! you were absolutely right with your first answer. Anyone who think that the right answer is C should think really hard about their life. "umask 377" - in binary 3 = 011 7 = 111 7 = 111 Then, we invert each bit to get the bits that should be turned off: 011 -> 100 111 -> 000 111 -> 000 And there you have 100 which in decimal is 600 (4 - read, 2 write). Question says that the owner of the file should have read permission only - 400, so that automatically excludes C as the answer.

why?!!! you were absolutely right with your first answer. Anyone who think that the right answer is C should think really hard about their life. "umask 377" - in binary 3 = 011 7 = 111 7 = 111 Then, we invert each bit to get the bits that should be turned off: 011 -> 100 111 -> 000 111 -> 000 And there you have 100 which in decimal is 600 (4 - read, 2 write). Question says that the owner of the file should have read permission only - 400, so that automatically excludes C as the answer.

Selected Answer: C

The correct answer is C. The "echo "umask 377" >> /home/web/.bashrc" command will satisfy the requirement by adding the umask setting to the user's .bashrc file. The umask setting determines the default file permissions for newly created files and directories. The value of 377 sets the default permissions to read-only for the owner of the file. The umask is specified in octal notation, and each digit represents the permissions for a different category of users. In this case, the first digit represents the permissions for the owner of the file, the second digit represents the permissions for the group owner of the file, and the third digit represents the permissions for others. A value of 3 means that the owner has read and write permissions, and a value of 7 means that the owner has read, write, and execute permissions. By setting the umask to 377, the owner of newly created files will have read-only permissions, the group owner will have no permissions, and others will have no permissions.

Question #31

A systems administrator is tasked with preventing logins from accounts other than root, while the file /etc/nologin exists. Which of the following PAM modules will accomplish this task?

- A) pam_login.so
- B) pam_access.so
- C) pam_logindef.so
- D) pam_nologin.so (Correct Answer)**

Explanation

Correct Answer:

D

Community Discussion

The correct PAM module to accomplish the task of preventing logins from accounts other than root when the file `/etc/nologin` exists is `pam_nologin.so`. This PAM module checks for the existence of the `/etc/nologin` file and denies logins if the file exists. By using `pam_nologin.so` in the appropriate PAM configuration file, the systems administrator can ensure that only root logins are allowed when the `/etc/nologin` file is present.

Question #32

A systems administrator has been tasked with disabling the `nginx` service from the environment to prevent it from being automatically and manually started. Which of the following commands will accomplish this task?

- A) `systemctl cancel nginx`
- B) `systemctl disable nginx`
- C) `systemctl mask nginx` (Correct Answer)**
- D) `systemctl stop nginx`

Explanation

Correct Answer:
C

Community Discussion

Selected Answer: C

Answer B (`systemctl disable nginx`) and answer C (`systemctl mask nginx`) both achieve the goal of disabling the `nginx` service, but they do it in different ways. The `systemctl disable nginx` command disables the `nginx` service so that it will not start automatically when the system boots up. However, the service can still be manually started if needed. The `systemctl mask nginx` command masks the `nginx` service, which means that it cannot be started, either automatically or manually. This is a more secure option compared to `systemctl disable nginx`, as it completely prevents the service from being started. In this scenario, either answer B or answer C would be correct, depending on the level of security that the systems administrator wants to enforce. If the administrator wants to prevent the `nginx` service from starting in any circumstance, then answer C (`systemctl`

mask nginx) is the better choice. If the administrator wants to prevent the nginx service from starting automatically but still allow it to be manually started if needed, then answer B (systemctl disable nginx) is the better choice.

hello, please, read the question again, you may not be understanding what they are asking, Comptia wording is brutal so be careful. You can manually start a disabled service with the systemctl start command after the system boots, to prevent this, you must use the mask subcommand, masking the service links its configuration to /dev/null. A user or process will not be able to start this service at all (whereas with a disabled service, a user or process can still start it). Use the unmask subcommand to reverse the setting: Ex: \$ sudo systemctl mask sshd

hello, please, read the question again, you may not be understanding what they are asking, Comptia wording is brutal so be careful. You can manually start a disabled service with the systemctl start command after the system boots, to prevent this, you must use the mask subcommand, masking the service links its configuration to /dev/null. A user or process will not be able to start this service at all (whereas with a disabled service, a user or process can still start it). Use the unmask subcommand to reverse the setting: Ex: \$ sudo systemctl mask sshd

Selected Answer: C

systemctl mask will prevent the service from being automatically started by other services.

Correct answer = C

Question #33

A Linux administrator is troubleshooting an issue in which an application service failed to start on a Linux server. The administrator runs a few commands and gets the following outputs:

Output 1:

```
Dec 23 23:14:15 root systemd[1] logsearch.service: Failed to start Logsearch.
```

Output 2:

```
logsearch.service - Log Search
Loaded: loaded (/etc/systemd/system/logsearch.service; enabled; vendor preset:enabled)
Active: failed (Result: timeout)
Process: 3267 ExecStart=/usr/share/logsearch/bin/logger ...
Main PID: 3267 (code=killed, signal=KILL)
```

Based on the above outputs, which of the following is the MOST likely action the administrator should take to resolve this issue?

- A) Enable the logsearch.service and restart the service.
- B) Increase the TimeoutStartUSec configuration for the logsearch.service. (Correct Answer)**
- C) Update the OnCalendar configuration to schedule the start of the logsearch.service.
- D) Update the KillSignal configuration for the logsearch.service to use TERM.

Explanation

Correct Answer:

B

Community Discussion

The TimeoutStartUSec configuration determines the amount of time that the system service manager should wait for the service to start before timing out. If the logsearch.service is taking longer than the specified timeout to start, the system service manager may consider the service to have failed to start and may terminate it. Increasing the TimeoutStartUSec configuration for the logsearch.service would give it more time to start and prevent it from being terminated prematurely. This may be necessary if the service requires additional time to start due to resource constraints or other factors.

Question #34

A Linux administrator has installed a web server, a database server, and a web application on a server. The web application should be active in order to render the web pages. After the administrator restarts the server, the website displays the following message in the browser: Error establishing a database connection. The Linux administrator reviews the following relevant output from the systemd init files:

```
[Unit]
Description=The Apache #HTTP Server
Wants=httpd-init.service
After=network.target remote-fs.target nss-lookup-target httpd-init.service mariadb.service

[Unit]
Description=MariaDB 10.5 database server
After=network.target
```

The administrator needs to ensure that the database is available before the web application is started. Which of the following should the administrator add to the HTTP server .service file to accomplish this task?

- A) TRIGGERS=mariadb.service
- B) ONFAILURE=mariadb.service
- C) WANTEDBY=mariadb.service
- D) REQUIRES=mariadb.service (Correct Answer)**

Explanation

Correct Answer:

D

Community Discussion

Selected Answer: D

The correct answer is D. REQUIRES=mariadb.service. The REQUIRES directive in systemd tells systemd that the HTTP server service depends on the mariadb.service service. This means that systemd will not start the HTTP server service until the mariadb.service service is started and running.

any comments ? please

Question #35

Several users reported that they were unable to write data to the /oracle1 directory. The following output has been provided:

Filesystem	Size	Used	Available	Use%	Mounted on
/dev/sdb1	100G	50G	50G	50%	/oracle1

Which of the following commands should the administrator use to diagnose the issue?

A) df -i /oracle1 (Correct Answer)

B) fdisk -l /dev/sdb1

C) lsblk /dev/sdb1

D) du -sh /oracle1

Explanation

Correct Answer:

A

Community Discussion

The `df -i` command displays the amount of available inodes on the file system containing each file name argument. An inode is a data structure used by the file system to store information about a file or directory, such as its permissions, ownership, timestamps, and location of its data. If you're encountering issues with writing data to the /oracle1 directory, running `df -i /oracle1` can provide information about the inode utilization on the file system containing that directory, which could be indicative of a disk space issue. The `IUse%` column shows the percentage of inodes that are currently in use. If this value is close to 100%, it could mean that there are no available inodes for new files or directories to be created in the file system, and you may need to either free up some space or increase the size of the file system to resolve the issue.

Selected Answer: A

Correct. flag -i for inodes instead of block

Question #36

After installing some RPM packages, a systems administrator discovers the last package that was installed was not needed. Which of the following commands can be used to remove the package?

A) dnf remove packagename (Correct Answer)

- B) apt-get remove packagename
- C) rpm -i packagename
- D) apt remove packagename

Explanation

Correct Answer:

A

Community Discussion

If you are using a Red Hat-based Linux distribution, such as Fedora or CentOS, the correct command to remove an RPM package that was installed is `dnf remove packagename` or `rpm -e packagename`. In this case, the command `dnf remove packagename` should be used as it is the modern package manager for Fedora and CentOS. The `rpm -e packagename` command is also correct, but `dnf` is recommended as it provides improved functionality over `rpm`. `apt-get remove packagename` and `apt remove packagename` are commands used to remove packages on Debian-based distributions such as Ubuntu, and are not relevant for Red Hat-based distributions.

Selected Answer: A

The correct answer is A. `dnf remove packagename`.

Question #37

A systems administrator is checking the system logs. The administrator wants to look at the last 20 lines of a log. Which of the following will execute the command?

- A) `tail -v 20`

B) tail -n 20 (Correct Answer)

C) tail -c 20

D) tail -l 20

Explanation

Correct Answer:

B

Community Discussion

The correct command to look at the last 20 lines of a log is tail -n 20. The tail command is used to display the last part of a file, and the -n option is used to specify the number of lines to display. In this case, the administrator wants to display the last 20 lines of the log, so the correct command is tail -n 20.

Selected Answer: B

B is the correct answer. Tail displays the last 10 lines by the default. The flag -n refers to the "number of lines"

-l is an invalid option!

... followed by the number of lines specified

... followed by the number of lines specified

Question #38

An administrator is trying to diagnose a performance issue and is reviewing the following output:

```
avg-cpu:  %user  %nice  %system  %iowait  %steal   %idle
           2.00   0.00   3.00    32.00   0.00    63.00
```

Device	tps	kB_read/s	kB_wrtn/s	kB_read	kB_wrtn
sdb	345.00	0.02	0.04	4739073123	23849523
sdb1	345.00	32102.03	12203.01	4739073123	23849523

System Properties:

CPU: 4 vCPU -

Memory: 40GB -

Disk maximum IOPS: 690 -

Disk maximum throughput: 44Mbps | 44000Kbps

Based on the above output, which of the following BEST describes the root cause?

- A) The system has reached its maximum IOPS, causing the system to be slow.
- B) The system has reached its maximum permitted throughput, therefore iowait is increasing. (Correct Answer)**
- C) The system is mostly idle, therefore the iowait is high.
- D) The system has a partitioned disk, which causes the IOPS to be doubled.

Explanation

Correct Answer:

B

Community Discussion

Selected Answer: A

Pla any comments ? A is not correct ? IOPS Input/Output Operationd per seconds ...

Selected Answer: B

Answer is B

Selected Answer: B

is more accurate

Question #39

A systems administrator wants to test the route between IP address 10.0.2.15 and IP address 192.168.1.40. Which of the following commands will accomplish this task?

- A) route -e get to 192.168.1.40 from 10.0.2.15
- B) ip route get 192.163.1.40 from 10.0.2.15 (Correct Answer)**
- C) ip route 192.169.1.40 to 10.0.2.15
- D) route -n 192.168.1.40 from 10.0.2.15

Explanation

Correct Answer:

B

Community Discussion

Can someone please explain whey the answer is ip route get 192.163.1.40 from 10.0.2.15 and not either A or D? Where id d192.163.1.40 even come from?

I dont know if the wrong IP Address was included on purpose, but neither B or C can be correct in the current format. The incorrect IP is present in the answer.

Selected Answer: B

"Test the route" is generic. Both B & D are correct, but `ip` supplants `route`. IOW, the `net-tools` suite (containing `route`) is becoming or has become deprecated by the `iproute2` suite (containing `ip`) in mainstream Linux distributions.

Selected Answer: D

Answer is D

A. `route -e get to 192.168.1.40 from 10.0.2.15` Here's why: `route -e`: This flag displays the effective routing table, including both static and dynamic routes currently in use. `get to`: This specifies the type of information to retrieve, in this case, the route to the destination IP address. `192.168.1.40`: This is the destination IP address for which you want to test the route. `from 10.0.2.15`: This specifies the source IP address from which the route test will be initiated.

Question #40

A Linux administrator was tasked with deleting all files and directories with names that are contained in the `sobelete.txt` file. Which of the following commands will accomplish this task?

- A) `xargs -f cat toDelete.txt -rm`
- B) `rm -d -r -f toDelete.txt` (Correct Answer)**
- C) `cat toDelete.txt | rm -frd`
- D) `cat toDelete.txt | xargs rm -rf`

Explanation

Correct Answer:

B

Community Discussion

Selected Answer: D

D. `cat toDelete.txt | xargs rm -rf` The "`cat toDelete.txt`" command will display the contents of the "`toDelete.txt`" file, which contains the names of the files and directories to be deleted. The "`xargs rm -rf`" command will take the standard input from the "`cat`" command and pass it as arguments to the "`rm`" command. The "`-rf`" options of the "`rm`"

command are used to forcibly delete the specified files and directories, including any files and directories within them, even if they are write-protected. The "xargs" command is used here to process the input one line at a time and execute the "rm" command for each line.

Selected Answer: D

mod correct, the question states "sobelete.txt" and should say "toDelete.txt".

Selected Answer: D

Tested, confirmed D is correct

Selected Answer: D

D is the correct answer. The selected answer delete the file. with answer D it removes the files listed inside the file. xargs - build and execute command lines from standard input.

mod correct, the answer id D.

Question #41

A Linux administrator is troubleshooting the root cause of a high CPU load and average.

```
$ uptime
07:30:43 up 20 days, 3 min, 1 user, load average: 2.98, 3.62, 5.21

$ top
PID  USER PR  NI  VIRT  RES   SHR  S  %CPU  %MEM  TIME+  COMMAND
6295 user1 30  -10  5465  56465 8254  R   86.5   1.5   7:35.25  appl

$ ps -ef | grep user1
user1 6295 1 7:42:19 tty/1    06:48:29 /usr/local/bin/appl
```

Which of the following commands will permanently resolve the issue?

A) renice -n -20 6295 (Correct Answer)

B) pstree -p 6295

C) iostat -cy 1 5

D) kill -9 6295

Explanation

Correct Answer:

A

Community Discussion

Selected Answer: D

D. kill -9 6295 The command "renice -n -20 6295" changes the priority of the process with ID 6295 to a higher priority by assigning it a nice value of -20.

Comptia wording is brutal, sometime I wonder if they are really testing our knowledge or just trolling thre hell out of everyone.....lol, maybe both

Comptia wording is brutal, sometime I wonder if they are really testing our knowledge or just trolling thre hell out of everyone.....lol, maybe both

when you renice to -20 you are actually giving higher priority to the process, the more - higher the number the more priority

when you renice to -20 you are actually giving higher priority to the process, the more - higher the number the more priority

Question #42

A Linux administrator wants to set the SUID of a file named dev_team.text with 744 access rights. Which of the following commands will achieve this goal?

A) chmod 4744 dev_team.txt (Correct Answer)

B) chmod 744 --setuid dev_team.txt

C) chmod -c 744 dev_team.txt

D) chmod -v 4744 --suid dev_team.txt

Explanation

Correct Answer:

Community Discussion

A. `chmod 4744 dev_team.txt` The "chmod" command is used to change the access rights of a file in Linux. The "4744" options specify the access rights in binary form. The first number (4) represents the setuid (SUID) bit, which causes the file to run with the effective user ID of the owner of the file, rather than with the effective user ID of the user who is running it. The other three numbers (744) represent the permissions for the owner, group, and others, respectively. In this case, the owner has read, write, and execute permissions (7), the group has read and execute permissions (4), and others have only read permission (4). So, the command "`chmod 4744 dev_team.txt`" sets the SUID bit and the access rights to 744 for the file `dev_team.txt`. The SUID bit allows the file to be executed with the owner's permissions, which can be useful for setuid scripts or other special purpose files.

Question #43

A developer has been unable to remove a particular data folder that a team no longer uses. The developer escalated the issue to the systems administrator. The following output was received:

```
# rmdir data/
rmdir: failed to remove 'data/': Operation not permitted
# rm -rf data/
rm: cannot remove 'data': Operation not permitted
# mv data/ mydata
mv: cannot move 'data/' to 'mydata': Operation not permitted
# cd data/
# cat > test.txt
bash: test.txt: Permission denied
```

Which of the following commands can be used to resolve this issue?

- A) `chgrp -R 755 data/`
- B) `chmod -R 777 data/`
- C) `chattr -R -i data/` (Correct Answer)**
- D) `chown -R data/`

Explanation

Correct Answer:

C

Community Discussion

Selected Answer: C

"operation not permitted" error (usually) means the immutable attribute is set. `chattr -i` will remove the immutable attribute

Not at all, he tried everything on the file and got denied, C is the right answer

Not at all, he tried everything on the file and got denied, C is the right answer

`cat > test.txt` is not reading a file, it is creating it. It allows you to type content in the shell and close the file with Ctrl-D. So, immutable attribute removal is appropriate. C is correct.

The command `chattr -R -i data/` is used to change the attributes of a file or directory in Linux. Here's what each part of the command does: `chattr`: This is the command used to change file or directory attributes. `-R`: This option makes the command recursive, meaning it will apply to the specified directory and its contents. `-i`: This option removes the 'immutable' attribute. When a file is immutable, it cannot be modified, deleted, or renamed, and no link can be created to it. `data/`: This is the directory that the command will be applied to. So, `chattr -R -i data/` will remove the 'immutable' attribute from the 'data/' directory and all of its contents. This means that after running this command, files in the 'data/' directory can be modified, deleted, or renamed.

Question #44

A Linux administrator needs to ensure that Java 7 and Java 8 are both locally available for developers to use when deploying containers. Currently only Java 8 is available. Which of the following commands should the administrator run to ensure both versions are available?

A) `docker image load java:7`

B) `docker image pull java:7` (Correct Answer)

- C) docker image import java:7
- D) docker image build java:7

Explanation

Correct Answer:

B

Community Discussion

The correct command to ensure that both Java 7 and Java 8 are locally available for developers to use when deploying containers is: B. docker image pull java:7 The "docker image pull" command is used to pull images from a Docker registry, such as Docker Hub, to the local system. The "java:7" option specifies the image to be pulled, in this case the Java 7 image. So, in this example, the command "docker image pull java:7" pulls the Java 7 image to the local system, making it available for use when deploying containers.

The correct command to ensure that Java 7 is locally available for developers to use when deploying containers is: B. docker image pull java:7 Here's why: The docker image pull command is used to pull an image or a repository from a registry. This command will download the Java 7 image from the Docker Hub, making it locally available on the system.

Question #45

A cloud engineer is installing packages during VM provisioning. Which of the following should the engineer use to accomplish this task?

A) Cloud-init (Correct Answer)

- B) Bash
- C) Docker
- D) Sidecar

Explanation

Correct Answer:

A

Community Discussion

A. Cloud-init Cloud-init is a popular tool used for provisioning virtual machines (VMs) in the cloud. It is commonly used to automate the initial configuration of cloud instances, such as setting up SSH keys, creating users, installing packages, and so on. When installing packages during VM provisioning, cloud-init can be configured to run a script that installs the required packages, or to use a package management tool such as apt-get or yum to install the packages. So, in this scenario, the cloud engineer should use cloud-init to accomplish the task of installing packages during VM provisioning. This will ensure that the packages are installed consistently and correctly, and that the VMs are properly configured for use.

The correct tool for a cloud engineer to use when installing packages during VM provisioning is: A. Cloud-init Here's why: Cloud-init is a widely used approach to initialize cloud instances. It is a package that contains utilities for early boot configuration of a cloud instance. It can handle tasks such as setting up users, SSH keys, and installing packages, making it an ideal choice for this task.

Question #46

A systems administrator is tasked with creating a cloud-based server with a public IP address.

The code is as follows:

```
---
-name: start an instance with a public IP address
  community.abc.ec2_instance:
    name: "public-compute-instance"
    key_name: "comptia-ssh-key"
    vpc_subnet_id: subnet-5cjssh1
    instance_type: instance.type
    security_group: comptia
    network:
      assign_public_ip: true
    image_id: ami-1234568
    tags:
      Environment: Comptia-Items-Writing-Workshop
...
```

Which of the following technologies did the systems administrator use to complete this task?

- A) Puppet
- B) Git
- C) Ansible
- D) Terraform (Correct Answer)**

Explanation

Correct Answer:

D

Community Discussion

Selected Answer: C

This looks more like ansible than terraform? I was googling ansible and the syntax looks like an ansible playbook and not a terraform config file.

The correct answer is C. Ansible

Selected Answer: C

The technology used by the systems administrator in this task is Ansible. The code is written in YAML and is using an Ansible module, specifically the "community.abc.ec2_instance" module, to create a cloud-based server with a public IP address. The "community.abc.ec2_instance" module is part of the Ansible community collection and provides the ability to manage Amazon Web Services (AWS) Elastic Compute Cloud (EC2) instances. The code sets various parameters, such as the instance name, the SSH key, the subnet ID, the instance type, the security group, and the image ID. The "assign_public_ip" option is set to "true", which specifies that the created instance should have a public IP address. So, in this scenario, the systems administrator used Ansible to automate the creation of a cloud-based server with a public IP address.

Selected Answer: C

as i use ansible , it looks like an ansible playbook

Selected Answer: C

Terraform works with JSON, not YAML ... Ansible is YAML. JSON uses { } YAML uses - and :

Question #47

A Linux systems administrator is setting up a new web server and getting 404 - NOT FOUND errors while trying to access the web server pages from the browser. While working on the diagnosis of this issue, the Linux systems administrator executes the following commands:

```
# getenforce
Enforcing

# matchpathcon -V /var/www/html/*
/var/www/html/index.html has context unconfined_u:object_r:user_home_t:s0, should be system_u:object_r:httpd_sys_content_t:s0
/var/www/html/page1.html has context unconfined_u:object_r:user_home_t:s0, should be system_u:object_r:httpd_sys_content_t:s0
```

Which of the following commands will BEST resolve this issue?

- A) `sed -i 's/SELINUX=enforcing/SELINUX=disabled/' /etc/selinux/config`
- B) `restorecon -R -v /var/www/html` (Correct Answer)**
- C) `setenforce 0`

D) `setsebool -P httpd_can_network_connect_db on`

Explanation

Correct Answer:

B

Community Discussion

Selected Answer: B

The correct answer is B. `restorecon -R -v /var/www/html`.

Option B is the best answer to resolve the issue, as the output of the `matchpathcon` command indicates that the context of the files under `/var/www/html` is incorrect. `restorecon -R -v /var/www/html` will restore the default SELinux context to the files under `/var/www/html`, which will allow the web server to access them properly.

Question #48

To harden one of the servers, an administrator needs to remove the possibility of remote administrative login via the SSH service. Which of the following should the administrator do?

A) Add the line `DenyUsers root` to the `/etc/hosts.deny` file.

B) Set `PermitRootLogin to no` in the `/etc/ssh/sshd_config` file. (Correct Answer)

C) Add the line `account required pam_nologin.so` to the `/etc/pam.d/sshd` file.

D) Set `PubKeyAuthentication to no` in the `/etc/ssh/ssh_config` file.

Explanation

Correct Answer:

B

Community Discussion

Selected Answer: B

The correct answer is B. Set PermitRootLogin to no in the `/etc/ssh/sshd_config` file.

The administrator should set the PermitRootLogin option to no in the `/etc/ssh/sshd_config` file. This option determines whether the root user is allowed to log in to the SSH service remotely. Setting this option to no disables remote administrative login via SSH for the root user, providing additional security to the server.

Question #49

Which of the following is a function of a bootloader?

A) It initializes all the devices that are required to load the OS. (Correct Answer)

- B) It mounts the root filesystem that is required to load the OS.
- C) It helps to load the different kernels to initiate the OS startup process.
- D) It triggers the start of all the system services.

Explanation

Correct Answer:

A

Community Discussion

I hope i never see this question..

Selected Answer: A

Option A is correct. A bootloader is a software that initializes the system and starts the operating system. The bootloader is responsible for initializing the system, setting up the environment, and loading the operating system into memory. Option B is incorrect. The bootloader does not mount the root filesystem. Mounting the root filesystem is usually done by the operating system after it has been loaded into memory by the bootloader. Option C is correct to some extent. The bootloader does help to load the different kernels to initiate the OS startup process, but it's not limited to only kernels. The bootloader also has the ability to load other operating systems or other software. Option D is incorrect. The bootloader does not trigger the start of all system services. Starting

system services is usually done by the operating system after it has been loaded into memory by the bootloader.

Selected Answer: A

A. It initializes all the devices that are required to load the OS. The bootloader plays a crucial role in the boot process of an operating system. One of its primary functions is to initialize the hardware devices that are necessary to load the operating system. This includes tasks such as identifying and configuring the CPU, memory, storage devices, and other essential hardware components to ensure that the system is in a suitable state to load the OS.

Selected Answer: C

The correct answer is "C. It helps to load the different kernels to initiate the OS startup process." A bootloader is a program that runs before the operating system (OS) starts. Its main function is to load the OS into memory and initiate the OS startup process. This involves loading the kernel, which is the core part of the OS that manages hardware resources and provides a platform for running user applications. The bootloader helps to load different kernels, which can be useful for testing and recovery purposes. The other options listed are not functions of the bootloader: Option A: Initializing devices is performed by the BIOS or UEFI (depending on the system), not the bootloader. Option B: Mounting the root filesystem is performed by the kernel, not the bootloader. Option D: Triggering the start of system services is performed by the OS, not the bootloader.

I would go with C. It is the responsibility of the boot loader to load the kernel, but the responsibility of the kernel to load the root file system and initialize devices.

Question #50

A systems administrator configured firewall rules using firewalld. However, after the system is rebooted, the firewall rules are not present:

```
Chain INPUT (policy ACCEPT)                destination
target          prot opt source
Chain FORWARD (policy ACCEPT)              destination
target          prot opt source
Chain OUTPUT (policy ACCEPT)               destination
target          prot opt source
```

The systems administrator makes additional checks:

```
- dynamic firewall daemon
  Loaded: loaded (/usr/lib/systemd/system/firewalld.service: disabled; vendor preset: enabled)
  Active: inactive (dead)
  Docs: man: firewalld (1)

firewalld is not running
```

Which of the following is the reason the firewall rules are not active?

- A) iptables is conflicting with firewalld.
- B) The wrong system target is activated.
- C) FIREWALL_ARGS has no value assigned.
- D) The firewalld service is not enabled. (Correct Answer)**

Explanation

Correct Answer:

D

Community Discussion

Selected Answer: D

firewalld needs to be enabled so that it starts up on reboot

Selected Answer: D

Firewalld is disabled. Tested on my machine. when i create a rule in firewalld it does not show up on iptables output. I have both services running so no conflict.

Question #51

A newly created container has been unable to start properly, and a Linux administrator is analyzing the cause of the failure. Which of the following will allow the administrator to determine the FIRST command that is executed inside the container right after it starts?

- A) `docker export <container_id>`
- B) `docker info <container_id>`
- C) `docker start <container_id>`
- D) `docker inspect <container_id>` (Correct Answer)**

Explanation

Correct Answer:

D

Community Discussion

Selected Answer: D

The correct answer is D. `docker inspect <container_id>`.

Selected Answer: D

D. `docker inspect <container_id>` The `docker inspect` command provides detailed information about a Docker container, including its configuration and state. Among the information available from this command is the `entrypoint`, which is the first command that is executed inside the container when it starts. By inspecting the container with `docker inspect`, the administrator can determine the first command that is executed inside the container and potentially diagnose why it is unable to start properly.

Selected Answer: D

D is correct here, you use the inspect and default template which will output the docker run command.

Is it really docket inspect? when i run docker inspect i just get a bunch of info about the container, but nothing about commands executed.

Question #52

A Linux administrator is scheduling a system job that runs a script to check available disk space every hour. The Linux administrator does not want users to be able to start the job. Given the following:

```
[Unit]
Description=Check available disk space
RefuseManualstart=yes
RefuseManualStop=yes

[Timer]
Persistent=true
OnCalendar=*-*-* *:00:00
Unit=checkdiskspace.service

[Install]
WantedBy=timers.target
```

The Linux administrator attempts to start the timer service but receives the following error message:

```
Failed to start checkdiskspace.timer: Operation refused ...
```

Which of the following is MOST likely the reason the timer will not start?

- A) The checkdiskspace.timer unit should be enabled via systemctl.
- B) The timers.target should be reloaded to get the new configuration.
- C) The checkdiskspace.timer should be configured to allow manual starts. (Correct Answer)**

D) The checkdiskspace.timer should be started using the sudo command.

Explanation

Correct Answer:

C

Community Discussion

C is correct. Even root cannot start the service if the config file prevents it.

Question #53

A Linux administrator wants to find out whether files from the wget package have been altered since they were installed. Which of the following commands will provide the correct information?

- A) rpm -i wget
- B) rpm -qf wget
- C) rpm -F wget
- D) rpm -V wget (Correct Answer)**

Explanation

Correct Answer:
D

Community Discussion

D. rpm -V wget The rpm -V (or "verify") command can be used to check the integrity of installed RPM packages. The wget package, as well as all its files, can be verified using this command. If any files have been altered since the package was installed, rpm -V will report the differences. This can help the administrator determine whether any unauthorized changes have been made to the package, or if the files have been damaged in some way.

Question #54

A Linux engineer set up two local DNS servers (10.10.10.10 and 10.10.10.20) and was testing email connectivity to the local mail server using the mail command on a local machine when the following error appeared:

```
Send-mail: Cannot open mail:25
```

The local machine DNS settings are:

```
$ cat /etc/resolv.conf
nameserver 10.10.10.10 #web records
nameserver 10.10.10.20 #email records

Mail server: mail.example.com
```

Which of the following commands could the engineer use to query the DNS server to get mail server information?

- A) dig @example.com 10.10.10.20 a
- B) dig @10.10.10.20 example.com mx (Correct Answer)**
- C) dig @example.com 10.10.10.20 ptr
- D) dig @10.10.10.20 example.com ns

Explanation

Correct Answer:

B

Community Discussion

B. dig @10.10.10.20 example.com mx The dig command is a useful tool for querying DNS servers to retrieve information about domain names.

Answer B: Syntax to specify DNS query to mail domain server: dig @10.10.10.20 example.com mx "mx" is mail exchange - directs to mail server

Question #55

A Linux engineer has been notified about the possible deletion of logs from the file `/opt/app/logs`. The engineer needs to ensure the log file can only be written into without removing previous entries.

```
# lsattr /opt/app/logs
-----e--- logs
```

Which of the following commands would be BEST to use to accomplish this task?

A) `chattr +a /opt/app/logs` (Correct Answer)

B) `chattr +d /opt/app/logs`

C) `chattr +i /opt/app/logs`

D) `chattr +c /opt/app/logs`

Explanation

Correct Answer:

A

Community Discussion

Selected Answer: A

The `chattr +a` command is used to set the "append only" attribute for a file or directory. This attribute ensures that once a file has been created, it cannot be deleted or modified. Any attempts to do so will result in a "permission denied" error. In this case, the command sets the "append only" attribute for the `/opt/app/logs` directory, which will prevent any changes to existing files or the deletion of files in that directory. New files can still be added to the directory, but their contents cannot be modified. This is useful in cases where the logs stored in the `/opt/app/logs` directory are critical for debugging or auditing purposes, and need to be kept for an extended period of time. The "append only" attribute helps to ensure the integrity of the logs, by preventing any accidental or malicious changes to the log files.

should be 'lsattr' not 'lsattz'

Question #56

A systems administrator needs to check if the service `systemd-resolved.service` is running without any errors. Which of the following commands will show this information?

A) `systemctl status systemd-resolved.service` (Correct Answer)

B) `systemctl enable systemd-resolved.service`

C) `systemctl mask systemd-resolved.service`

D) `systemctl show systemd-resolved.service`

Explanation

Correct Answer:

A

Community Discussion

Selected Answer: A

The correct answer is A. `systemctl status systemd-resolved.service`.

A. `systemctl status systemd-resolved.service` The `systemctl status` command is used to show the current status of a system service in a Linux system that uses `systemd`. The status information includes the name of the service, its state (e.g. running, stopped), and any related error messages or warnings.

Question #57

SIMULATION -

Junior system administrator had trouble installing and running an Apache web server on a Linux server. You have been tasked with installing the Apache web server on the Linux server and resolving the issue that prevented the junior administrator from running Apache.

INSTRUCTIONS -

Install Apache and start the service. Verify that the Apache service is running with the defaults.

Typing “help” in the terminal will show a list of relevant event commands.

If at any time you would like to bring back the initial state of the simulation, please click the Reset All button.

CentOS Command Prompt

```
[root@centos7]#
```

Explanation

Correct Answer:

```
yum install httpd
```

```
systemctl --now enable httpd
```

```
systemctl status httpd
```

```
netstat -tunlp | grep 80  
pkill <processname>  
systemctl restart httpd  
systemctl status httpd
```

Community Discussion

```
sudo yum install httpd systemctl start httpd systemctl enable httpd systemctl status  
httpd
```

Not sure why we need all those commands to install Apache and confirm it is running with the defaults. I would just do: yum install httpd systemctl enable httpd systemctl start httpd systemctl status httpd

Actually after looking again you have the right commands but wrong order: yum install httpd systemctl start httpd systemctl enable httpd systemctl status httpd

Actually after looking again you have the right commands but wrong order: yum install httpd systemctl start httpd systemctl enable httpd systemctl status httpd

correct, but no need to sudo since its already on root

Question #58

A Linux administrator needs to remove software from the server. Which of the following RPM options should be used?

- A) rpm -s
- B) rpm -d
- C) rpm -q
- D) rpm -e (Correct Answer)**

Explanation

Correct Answer:

D

Community Discussion

Selected Answer: D

The correct answer is D. rpm -e.

-e flag is for erase

A Linux administrator needs to remove software from the server. Which of the following RPM options should be used? A. rpm -s B. rpm -d C. rpm -q D. rpm -e

D. rpm -e The rpm -e option is used to remove software packages from a Linux system that uses the RPM Package Manager. This option allows the administrator to remove one or more installed packages, along with all their dependencies.

D. rpm -e The rpm -e option is used to remove software packages from a Linux system that uses the RPM Package Manager. This option allows the administrator to remove one or more installed packages, along with all their dependencies.

Question #59

A Linux system fails to start and delivers the following error message:

```
Checking all file systems.  
/dev/sda1 contains a file system with errors, check forced.  
/dev/sda1: Inodes that were part of a corrupted orphan linked list found.  
/dev/sda1: UNEXPECTED INCONSISTENCY;
```

Which of the following commands can be used to address this issue?

A) fsck.ext4 /dev/sda1 (Correct Answer)

B) partprobe /dev/sda1

C) fdisk /dev/sda1

D) mkfs.ext4 /dev/sda1

Explanation

Correct Answer:

A

Community Discussion

The `fsck.ext4` command is used to check and repair file system errors on an ext4 file system. In this case, the command is checking the file system located on `/dev/sda1`. The `fsck.ext4` utility is used to detect and correct file system inconsistencies, such as corrupt or lost inodes, missing block groups, and other issues. It's typically run automatically by the operating system during system boot, or manually by the administrator when file system problems are suspected. By running the `fsck.ext4 /dev/sda1` command, the administrator is checking the ext4 file system on the first partition of the first SATA disk (`/dev/sda1`) for any errors. If any problems are found, `fsck.ext4` will attempt to repair them. It's important to note that running `fsck.ext4` on a mounted file system can cause data loss, so it's recommended to run it on an unmounted file system or in a maintenance mode.

Question #60

Based on an organization's new cybersecurity policies, an administrator has been instructed to ensure that, by default, all new users and groups that are created fall within the specified values below.

```
# Min/max values for automatic uid selection in useradd
#
UID_MIN 1000
UID_MAX 60000
# Min/max values for automatic gid selection in groupadd
#
GID_MIN 1000
GID_MAX 60000
```

To which of the following configuration files will the required changes need to be made?

- A) `/etc/login.defs`
- B) `/etc/security/limits.conf`
- C) `/etc/default/useradd` (Correct Answer)**
- D) `/etc/profile`

Explanation

Correct Answer:

C

Community Discussion

Selected Answer: A

This answer is wrong, cat out your /etc/login.defs file. those values are in there

Sorry, I agree with answer A, /etc/login.defs.

I dont understand how this can be wrong, correct answer is A, please fix it!

linux_admin has it right: A is definitely correct, it's all right here, a quick look into the file shows below: cat /etc/login.defs yields: ... # Min/max values for automatic uid selection in useradd # UID_MIN 1000 UID_MAX 60000 # System accounts SYS_UID_MIN 201 SYS_UID_MAX 999 # # Min/max values for automatic gid selection in groupadd # GID_MIN 1000 GID_MAX 60000 # System accounts SYS_GID_MIN 201 SYS_GID_MAX 999

I can definitely understand why etc/default/useradd would be the answer. The question specifically stated that the users and groups were added "by default".

Question #61

A Linux administrator is trying to remove the ACL from the file `/home/user/data.txt` but receives the following error message:

```
setfacl: data.txt: operation not permitted
```

Given the following analysis:

```
/dev/mapper/linux-home on /home type xfs (rw,relatime,seclabel,attr2,inode64,usrquota)

-rw-rw-r--+ 1 user staff 2354 Sep 15 16:33 data.txt
-rw-rw-r--+ user staff unconfined_u:object_r:user_home_t:s0 data.txt

# file: data.txt
# owner: user
# group: staff
user::rw-
user:accounting:rw-
group::r-
mask::rw-
other::r-

Attributes:
-----a-----
```

Which of the following is causing the error message?

- A) The administrator is not using a highly privileged account.
- B) The filesystem is mounted with the wrong options.
- C) SELinux file context is denying the ACL changes.
- D) File attributes are preventing file modification. (Correct Answer)**

Explanation

Correct Answer:

D

Community Discussion

I think it is D. With the append only attribute activated, you can only add information to the file; you cannot change it. This includes deleting or changing the ACL.

Selected Answer: D

D. "File attributes are preventing file modification." It is not C: "SELinux file context is denying the ACL changes." SELinux does not discriminate: if an attribute can be set, then

it can be unset. It's true that SELinux could be configured to totally disallow or selectively allow access to 'chattr', but that is not its default configuration and the question has no hint in regard to a change in SELinux policy. This can be easily demonstrated on any system using SELinux. touch file sudo chattr +a file lsattr file rm file sudo chattr -a file lsattr file

Selected Answer: C

The correct option is C. SELinux file context is denying the ACL changes. The error message "setfacl: data.txt: operation not permitted" is caused by the SELinux file context denying the ACL changes.

Selected Answer: D

it has the -----a-- (append attr) hence you cant remove nor modify the file

C. SELinux file context is denying the ACL changes. The file context of the file is set to user_home_t and since the file is located under the /home directory, it is most likely that the file is located in a user's home directory which is restricted by SELinux. SELinux provides Mandatory Access Control (MAC) to restrict the access of the files and processes in a Linux system.

Question #62

A Linux administrator needs to create a new cloud.cpio archive containing all the files from the current directory. Which of the following commands can help to accomplish this task?

- A) ls | cpio -iv > cloud.epio
- B) ls | cpio -iv < cloud.epio
- C) ls | cpio -ov > cloud.cpio (Correct Answer)**
- D) ls cpio -ov < cloud.cpio

Explanation

Correct Answer:

C

Community Discussion

C. `ls | cpio -ov > cloud.cpio` The `cpio` command is used to create and extract archive files in the `cpio` format. The `-o` option is used to create an archive file, and the `-v` option provides verbose output during the creation process. This will pipe the output of the `ls` command (which lists the files in the current directory) into the `cpio` command, which will create a new archive file `cloud.cpio` containing all the listed files. The `>` symbol is used to redirect the output of the `cpio` command to the `cloud.cpio` file, overwriting any existing file with the same name.

C is correct, STDOUT from `ls` (current directory) into the `-ov` (create archive and display files processed by `cpio`) and redirecting into the filename name `cloud.cpio`

Question #63

A systems administrator made some changes in the `~/.bashrc` file and added an alias command. When the administrator tried to use the alias command, it did not work. Which of the following should be executed FIRST?

A) `source ~/.bashrc` (Correct Answer)

B) `read ~/.bashrc`

C) `touch ~/.bashrc`

D) `echo ~/.bashrc`

Explanation

Correct Answer:

A

Community Discussion

A. `source ~/.bashrc` The `source` command is used to re-read a shell configuration file and make its changes effective immediately in the current shell session. In this case, the administrator made changes to the `~/.bashrc` file and added an alias command, but the alias is not working. To make the changes in the `~/.bashrc` file effective, the administrator should run the following command: `source ~/.bashrc` this will cause the shell to re-read the `~/.bashrc` file and apply the changes, including the new alias. After running this

command, the administrator should be able to use the alias as expected. Option B is incorrect because the read command is used to read input from the user, not to re-read a configuration file. Option C is incorrect because the touch command is used to update the modification time of a file, not to re-read a configuration file. Option D is incorrect because the echo command is used to display text on the screen, not to re-read a configuration file.

Question #64

A junior systems administrator has just generated public and private authentication keys for passwordless login. Which of the following files will be moved to the remote servers?

- A) id_dsa.pem
- B) id_rsa
- C) id_ecdsa
- D) id_rsa.pub (Correct Answer)**

Explanation

Correct Answer:
D

Community Discussion

D. id_rsa.pub The public authentication key, id_rsa.pub, is typically used to set up passwordless login, also known as SSH key-based authentication. In this scenario, the junior administrator has generated public and private authentication keys, and they need to be moved to the remote servers to set up passwordless login. The public key, id_rsa.pub, is usually copied to the remote server and added to the ~/.ssh/authorized_keys file on the remote server. This allows the local system to authenticate with the remote server using the private key, id_rsa, without requiring a password.

Question #65

A Linux administrator cloned an existing Linux server and built a new server from that clone. The administrator encountered the following error after booting the cloned server:

```
Device mismatch detected
```

The administrator performed the commands listed below to further troubleshoot and mount the missing filesystem:

```
#ls -al /dev/disk/by-uuid/  
total 0  
drwxr-xr-x 2 root 220 Jul 08:59 .  
drwxr-xr-x 2 root 160 Jul 08:59 ..  
lrwxrwxrwx 1 root 26 Jul 11:10 2251a54-6c14-9187-df8629373 -> ../../sdb  
lrwxrwxrwx 1 root 26 Jul 11:10 4211c54-2a13-7291-bd8629373 -> ../../sdc  
lrwxrwxrwx 1 root 26 Jul 11:10 3451b54-6d10-3561-ad8629373 -> ../../sdd
```

Which of the following should administrator use to resolve the device mismatch issue and mount the disk?

- A) mount disk by device-id
- B) fsck -A
- C) mount disk by-label
- D) mount disk by-blkid (Correct Answer)**

Explanation

Correct Answer:

D

Community Discussion

Selected Answer: C

Changed yet again, but i'm serious this time. IT's C!!

Selected Answer: C

Here ya' go: <https://unix.stackexchange.com/questions/644708/how-can-i-prevent-disk-uuid-mismatch-when-cloning-a-machine>

Discard

Discard

Selected Answer: C

C. By lable. Why? (A) There is no "device-id." (B) The 'fsck' command repairs file systems. (D) The 'blkid' command shows block device attributes. (The sane choice is UUID because the UUID is embedded in the file system of the block device, but that's not an option to choose.)

Question #66

A systems administrator installed a new software program on a Linux server. When the systems administrator tries to run the program, the following message appears on the screen.

```
Hardware virtualization support is not available on this system.
Either is not present or disabled in the system's BIOS
```

Which of the following commands will allow the systems administrator to check whether the system supports virtualization?

- A) dmidecode -s system-version
- B) lscpu (Correct Answer)**
- C) sysctl -a
- D) cat /sys/device/system/cpu/possible

Explanation

Correct Answer:

B

Community Discussion

Selected Answer: B

lscpu will output following line for virtualization support. ... Virtualization: VT-x ...

B. lscpu The lscpu command is used to display information about the CPU architecture and the system topology on Linux systems. It provides information about the number of CPUs, cores, and threads, the architecture, clock speed, and cache size. If the system supports virtualization, it should be reflected in the output of the lscpu command. The other options are not relevant for checking virtualization support: dmidecode -s system-version is used to retrieve the version number of the system's BIOS or UEFI firmware. sysctl -a is used to display all system control parameters, including both kernel parameters and sysctl settings. cat /sys/device/system/cpu/possible does not exist, it should be /sys/devices/system/cpu/possible. This file provides information about the maximum number of CPUs that can be configured for the system, but does not indicate whether the system supports virtualization.

Question #67

A Linux administrator created the directory /project/access2all. By creating this directory, the administrator is trying to avoid the deletion or modification of files from non-owners. Which of the following will accomplish this goal?

A) chmod +t /project/access2all (Correct Answer)

- B) chmod +rws /project/access2all
- C) chmod 2770 /project/access2all
- D) chmod ugo+rwx /project/access2all

Explanation

Correct Answer:

A

Community Discussion

You are wrong. The sticky bit does prevent users, other than the owner, to delete files. The question is asking for a solution that will prevent non-owner(groups and others)

from deleting files. `chmod +t` will do the job. Answer C is giving `rwx` to groups (non-owner)

You are wrong. The sticky bit does prevent users, other than the owner, to delete files. The question is asking for a solution that will prevent non-owner(groups and others) from deleting files. `chmod +t` will do the job. Answer C is giving `rwx` to groups (non-owner)

You are wrong. The sticky bit does prevent users, other than the owner, to delete files. The question is asking for a solution that will prevent non-owner(groups and others) from deleting files. `chmod +t` will do the job. Answer C is giving `rwx` to groups (non-owner)

You are wrong. The sticky bit does prevent users, other than the owner, to delete files. The question is asking for a solution that will prevent non-owner(groups and others) from deleting files. `chmod +t` will do the job. Answer C is giving `rwx` to groups (non-owner)

Selected Answer: A

Option A, `chmod +t /project/access2all`, is better than option C, `chmod 2770 /project/access2all`, for preventing the deletion or modification of files from non-owners because of the following reasons: Sticky Bit: Option A sets the sticky bit on the directory. The sticky bit restricts deletion or renaming of files within the directory to the file owners, the directory owner, or the root user. This means even if a user has write permission to the directory, they cannot delete or rename files owned by others. Setgid and Permissions: Option C sets the setgid bit and gives the group read, write, and execute permissions. The setgid bit causes new files and directories created in the directory to inherit the group ownership of the directory, but it does not prevent file deletion or modification by non-owners. The permissions 770 give the owner and the group full permissions (read, write, execute), but they do not prevent file deletion or modification by non-owners who are in the group. Therefore, to specifically prevent deletion or modification of files from non-owners, option A is the better choice.

Question #68

A Linux systems administrator needs to persistently enable IPv4 forwarding in one of the Linux systems. Which of the following commands can be used together to accomplish this task? (Choose two.)

A) `sysctl net.ipv4.ip_forward`

B) sysctl -w net.ipv4.ip_forward=1 (Correct Answer)

C) echo "net.ipv4.ip_forward=1" >> /etc/sysctl.conf (Correct Answer)

D) echo 1 > /proc/sys/net/ipv4/ip_forward

E) sysctl -p

F) echo "net.ipv6.conf.all.forwarding=1" >> /etc/sysctl.conf

Explanation

Correct Answer:

BC

Community Discussion

Selected Answer: CE

The commands that can be used together to persistently enable IPv4 forwarding in a Linux system are: C. echo "net.ipv4.ip_forward=1" >> /etc/sysctl.conf E. sysctl -p The first command echo "net.ipv4.ip_forward=1" >> /etc/sysctl.conf is used to add the net.ipv4.ip_forward=1 line to the end of the /etc/sysctl.conf file. This file is used to configure kernel parameters at runtime. The change made in this file will persist across reboots. The second command sysctl -p is used to apply the changes made to the /etc/sysctl.conf file. It reloads the configuration file and sets the new values for the specified parameters. The other options sysctl net.ipv4.ip_forward, sysctl -w net.ipv4.ip_forward=1, echo 1 > /proc/sys/net/ipv4/ip_forward, and echo "net.ipv6.conf.all.forwarding=1" >> /etc/sysctl.conf are not suitable for persistently enabling IPv4 forwarding, as the changes made with these commands will not persist after a reboot.

Correct answer = BC

Correct answer = BC

Selected Answer: BC

1.to allow persistent IPv4 packet forwarding: sysctl -w net.ipv4.ip_forward=1 2. Once the system is satisfactorily tuned, make the new values permanent by modifying /etc/sysctl.conf or the /etc/sysctl.d/ directory.

Selected Answer: BC

BC are the only ones that enable IPv4 forwarding persistently

Question #69

Due to low disk space, a Linux administrator finding and removing all log files that were modified more than 180 days ago. Which of the following commands will accomplish this task?

A) `find /var/log -type d -mtime +180 -print -exec rm {} \;` (Correct Answer)

B) `find /var/log -type f -modified +180 -rm`

C) `find /var/log -type f -mtime +180 -exec rm {} \`

D) `find /var/log -type c -atime +180 -remove`

Explanation

Correct Answer:

A

Community Discussion

Selected Answer: C

C is Correct.

Disagree. Correct answer should be C. The question refers to removing log "files". The type option for find needs -f for files.

Selected Answer: C

The correct answer is C, "`find /var/log -type f -mtime +180 -exec rm {} \;`". This command will accomplish the task of finding and removing all log files that were modified more than 180 days ago. Here's a breakdown of the command: The find command is used to search for files and directories in a specific location, in this case, "/var/log". The -type f option is used to search only for regular files, excluding directories and other file types. The -mtime +180 option is used to select only files that have a modification time more than 180 days ago. The -exec rm {} \; option is used to execute the rm command on each file that matches the criteria specified in the previous options. The curly braces {} are replaced with the name of each file found and the backslash \; is used to end the command.

Selected Answer: C

The answer is C

THE CORRECT OPTION IS C. I've tested this, but used 1600 days so found only one file that old, used option C and it deleted it only, results below: # find /var/log -type f -mtime +1600 /var/log/anaconda/syslog <<<<<<<<<< found this file over 1600 days old-didn't want to remove anything important # find /var/log -type f -mtime +1600 -exec rm {} \; <<<<<<<<<< Executed option C # find /var/log -type f -mtime +1600 <<<<<<<<<< Ran command again and /var/log/anaconda/syslog is no longer present

Question #70

A junior administrator is setting up a new Linux server that is intended to be used as a router at a remote site. Which of the following parameters will accomplish this goal?

A)

```
echo 1 > /proc/sys/net/ipv4/ip_forward
iptables -t nat -A PREROUTING -i eth0 -j MASQUERADE
```

(Correct Answer)

B)

```
echo 1 > /proc/sys/net/ipv4/ip_forward
iptables -t nat -D POSTROUTING -o eth0 -j MASQUERADE
```

C)

```
echo 1 > /proc/sys/net/ipv4/ip_forward
iptables -t nat -A POSTROUTING -o eth0 -j MASQUERADE
```

D)

```
echo 1 > /proc/sys/net/ipv4/ip_forward
iptables -t nat -A PREROUTING -o eth0 -j MASQUERADE
```

Explanation

Correct Answer:

A

Community Discussion

Selected Answer: C

The more i research this one, the more the answer is C. Several links agree.

I don't know the answer for sure, but this link makes it seem like it could be C? <https://bobcares.com/blog/iptables-nat-masquerade/>

Option C. did execute without errors: `# echo 1 > /proc/sys/net/ipv4/ip_forward # iptables -t nat -A POSTROUTING -o eth0 -j MASQUERADE` # So I'm with LRISB, C is the correct answer

The iptables command "`-t nat -A POSTROUTING -o eth0 -j MASQUERADE`" configures NAT on the server. It adds a rule to the nat table in iptables that performs source NAT (SNAT) on outgoing packets. The "`-o eth0`" option specifies the outgoing interface (eth0 in this case), and "`-j MASQUERADE`" instructs iptables to modify the source IP address of outgoing packets to match the IP address of the interface, effectively masquerading the internal IP addresses. Options a and d are incorrect because they use the PREROUTING chain in iptables, which is used for modifying packets as they enter the system. In the context of setting up a router, we need to modify packets as they leave the system, so we should use the POSTROUTING chain. Option b is incorrect because it uses the "-D" flag, which stands for "delete," to remove a rule from iptables. However, in this case, we need to add a rule to configure NAT, not delete an existing rule. Therefore, the correct answer is

Selected Answer: A

I think the answer is A, as with IP forwarding and PREROUTING, both modifying the packet as it arrives from outside the private LAN, and altering the destination address to a preconfigured mapping

Question #71

Some servers in an organization have been compromised. Users are unable to access to the organization's web page and other services. While reviewing the system log, a systems administrator notices messages from the kernel regarding firewall rules:

```
Oct 20 03:45:50 hostname kernel: iptables denied: IN=eth0 OUT=
MAC=xx:xx:xx:xx:xx:xx:xx:xx:xx:xx:xx:xx:xx:xx:xx:xx SRC=x.x.x.x DST=x.x.x.x LEN=1059 TOS=0x00
PREC=0x00 TTL=115 ID=31368 DF PROTO=TCP
SPT=17992 DPT=80 WINDOW=16477 RES=0x00 ACK PSH URG=0
Oct 20 03:46:02 hostname kernel: iptables denied: IN=eth0 OUT=
MAC=xx:xx:xx:xx:xx:xx:xx:xx:xx:xx:xx:xx:xx:xx:xx:xx SRC=x.x.x.x DST=x.x.x.x LEN=52 TOS=0x00
PREC=0x00 TTL=52 ID=763 DF PROTO=TCP SPT=20229 DPT=22 WINDOW=15598 RES=0x00 ACK URG=0
Oct 20 03:46:14 hostname kernel: iptables denied: IN=eth0 OUT=
MAC=xx:xx:xx:xx:xx:xx:xx:xx:xx:xx:xx:xx:xx:xx:xx:xx SRC=x.x.x.x DST=x.x.x.x LEN=324 TOS=0x00
PREC=0x00 TTL=49 ID=64245 PROTO=TCP SPT=47237 DPT=80 WINDOW=470 RES=0x00 ACK PSH URG=0
Oct 20 03:46:26 hostname kernel: iptables denied: IN=eth0 OUT=
MAC=xx:xx:xx:xx:xx:xx:xx:xx:xx:xx:xx:xx:xx:xx:xx:xx SRC=x.x.x.x DST=x.x.x.x LEN=52 TOS=0x00
PREC=0x00 TTL=45 ID=2010 PROTO=TCP SPT=48322 DPT=80 WINDOW=380 RES=0x00 ACK URG=0
```

Which of the following commands will remediate and help resolve the issue?

A)

```
Iptables -A FORWARD -i eth0 -p tcp --dport 80 -j ACCEPT
Iptables -A FORWARD -i eth0 -p tcp --dport 22 -j ACCEPT
```

B)

```
Iptables -A INPUT -i eth0 -p tcp --dport 80 -j ACCEPT
Iptables -A INPUT -i eth0 -p tcp --dport 22 -j ACCEPT
```

(Correct Answer)

C)

```
Iptables -A INPUT -i eth0 -p tcp --sport 80 -j ACCEPT
Iptables -A INPUT -i eth0 -p tcp --sport 22 -j ACCEPT
```

D)

```
Iptables -A INPUT -i eth0 -p tcp --dport :80 -j ACCEPT
Iptables -A INPUT -i eth0 -p tcp --dport :22 -j ACCEPT
```

Explanation

Correct Answer:

B

Community Discussion

This iptables command is adding a new rule to the INPUT chain in the iptables firewall. The rule allows incoming traffic on interface "eth0" using the TCP protocol, destined for port 80, to be accepted. The "-A" option is used to append the rule to the end of the chain, and the "-j" option specifies the target action for the rule, in this case ACCEPT. This command is typically used to configure a firewall to allow specific types of traffic to enter the system.

Question #72

A junior administrator is trying to set up a passwordless SSH connection to one of the servers. The administrator follows the instructions and puts the key in the `authorized_key` file at the server, but the administrator is still asked to provide a password during the connection.

Given the following output:

```
junior@server:~$ ls -lh .ssh/auth*  
-rw----- 1 junior junior 566 sep 13 20:56 .ssh/authorized_key
```

Which of the following commands would resolve the issue and allow an SSH connection to be established without a password?

- A) `restorecon -rv .ssh/authorized_key`
- B) `mv .ssh/authorized_key .ssh/authorized_keys` (Correct Answer)**
- C) `systemctl restart sshd.service`
- D) `chmod 600 mv .ssh/authorized_key`

Explanation

Correct Answer:

B

Community Discussion

B. `mv .ssh/authorized_key .ssh/authorized_keys` This changes the name of the file from `authorized_key` to `authorized_keys`, which is the conventional name for the file that contains authorized public keys for passwordless SSH connections. By convention, the `ssh-keygen` tool generates the `authorized_keys` file, and some SSH servers expect this exact name. Changing the name of the file to `authorized_keys` may resolve the issue and allow the passwordless connection to be established.

Selected Answer: B

The correct answer is B. `mv .ssh/authorized_key .ssh/authorized_keys`

Question #73

A Linux administrator needs to resolve a service that has failed to start. The administrator runs the following command:

```
ls -l startup file
```

The following output is returned

```
-----. root root 81k Sep 13 19:01 startupfile
```

Which of the following is MOST likely the issue?

A) The service does not have permissions to read write the startupfile. (Correct Answer)

B) The service startupfile size cannot be 81k.

C) The service startupfile cannot be owned by root.

D) The service startupfile should not be owned by the root group.

Explanation

Correct Answer:

A

Community Discussion

Selected Answer: A

Selecting A - The output from the `ls -l` command shows the permissions of the startup file as a series of dashes (-----), this means that there are no permissions set for the

owner, group, or others. This would prevent any user, including the root user, from reading, writing, or executing the file.

Selected Answer: A

A. RB34561 explains it. (Tangentially, the file can be deleted by its owner and the root user. The file's permissions can be changed by its owner and the root user.)

Selected Answer: C

The file is owned by the "root" user (the owner) and belongs to the "root" group (the group). In many cases, system files should not be owned by the "root" user, especially if they are related to a specific service or application. It's generally recommended to have service-specific users and groups to manage permissions and access to files.

-l flag display file permissions owner (root) and group (root) have no permissions

Question #74

A Linux engineer is setting the sticky bit on a directory called devops with 755 file permission. Which of the following commands will accomplish this task?

A) chown -s 755 devops

B) chown 1755 devops

C) chmod -s 755 devops

D) chmod 1755 devops (Correct Answer)

Explanation

Correct Answer:

D

Community Discussion

D. `chmod 1755 devops` The sticky bit is a special permission that can be set on directories, and it is represented by the number "1" in the binary representation of the file permission mode. To set the sticky bit and keep the 755 file permission, you would set the permission mode to 1755. The "`chmod 1755 devops`" command sets the file permission mode to 1755, which includes the sticky bit and the 755 file permissions.

Also can be accomplished with `chmod +t directory_name`.

Question #75

A Linux administrator booted up the server and was presented with a non-GUI terminal. The administrator ran the command `systemctl isolate graphical.target` and rebooted the system by running `systemctl reboot`, which fixed the issue. However, the next day the administrator was presented again with a non-GUI terminal. Which of the following is the issue?

- A) The administrator did not reboot the server properly.
- B) The administrator did not set the default target to `basic.target`.
- C) The administrator did not set the default target to `graphical.target`. (Correct Answer)**
- D) The administrator did not shut down the server properly.

Explanation

Correct Answer:

C

Community Discussion

C. The administrator did not set the default target to `graphical.target`. When a Linux system is booted, it starts a target, which is a group of units that define the system state. The administrator ran the command "`systemctl isolate graphical.target`" which switches the system to the graphical target and provides a GUI interface. However, if the default target is not set to `graphical.target`, the system will revert back to the non-GUI terminal after a reboot.

Question #76

Users report that connections to a MariaDB service are being closed unexpectedly. A systems administrator troubleshoots the issue and finds the following message in `/var/log/messages`:

```
dbserver kernel: out of Memory: Killed process 1234 (mysqld).
```

Which of the following is causing the connection issue?

A) The process mysqld is using too many semaphores. (Correct Answer)

- B) The server is running out of file descriptors.
- C) Something is starving the server resources.
- D) The amount of RAM allocated to the server is too high.

Explanation

Correct Answer:

A

Community Discussion

Selected Answer: C

I'm picking C, this isn't a MySQL database test so knowing what a semaphores seems outside of the scope of linux+.

Selected Answer: C

Maybe I'm wrong, but semaphore is just a non-negative shared variable? This just seems like the kernel was starved for memory and killed the process.. C, something is starving the server resources.

Option A "The process mysqld is using too many semaphores" is not relevant to the issue described in the log message. Semaphores are a type of synchronization mechanism that are used to control access to shared resources in a multithreaded environment. They are implemented as a type of lock that can be acquired and released by multiple processes. While semaphores can be a source of performance issues and can cause problems in a system, they do not have a direct impact on memory usage. If a process is using too many semaphores, it may cause other issues such as deadlocks or performance degradation, but it will not cause the system to run out of memory. In this case, the log

message indicates that the process mysqld was terminated by the Linux kernel due to a lack of available memory, not due to an issue with semaphores. The message specifically mentions "out of memory", which suggests that the root cause of the issue is related to memory, not semaphores.

Option A "The process mysqld is using too many semaphores" is not relevant to the issue described in the log message. Semaphores are a type of synchronization mechanism that are used to control access to shared resources in a multithreaded environment. They are implemented as a type of lock that can be acquired and released by multiple processes. While semaphores can be a source of performance issues and can cause problems in a system, they do not have a direct impact on memory usage. If a process is using too many semaphores, it may cause other issues such as deadlocks or performance degradation, but it will not cause the system to run out of memory. In this case, the log message indicates that the process mysqld was terminated by the Linux kernel due to a lack of available memory, not due to an issue with semaphores. The message specifically mentions "out of memory", which suggests that the root cause of the issue is related to memory, not semaphores.

I'm with you, answer is C

Question #77

A developer is trying to install an application remotely that requires a graphical interface for installation. The developer requested assistance to set up the necessary environment variables along with X11 forwarding in SSH. Which of the following environment variables must be set in remote shell in order to launch the graphical interface?

- A) \$RHOST
- B) SETENV
- C) \$SHELL
- D) \$DISPLAY (Correct Answer)**

Explanation

Correct Answer:

D

Community Discussion

Selected Answer: D

I pick D. the only hesitation I have is that installing X11 should set that automatically.

D. \$DISPLAY The DISPLAY environment variable tells the X Window System which display to use for any graphical applications that are launched. When using X11 forwarding over SSH, the DISPLAY variable must be set in the remote shell to the value of the display on the local system. This allows graphical applications running on the remote system to display their windows on the local system.

It's a weird question these days. Imagine you have the workstation running X11. You use SSH to connect to a server, and you find software that you want to run. It's graphical software that won't run because it cannot find a display. On the remote server, you would 'export DISPLAY=hostname_or_ip_address:display_number' to cause the software to send its drawing commands to your local display.

It's a weird question these days. Imagine you have the workstation running X11. You use SSH to connect to a server, and you find software that you want to run. It's graphical software that won't run because it cannot find a display. On the remote server, you would 'export DISPLAY=hostname_or_ip_address:display_number' to cause the software to send its drawing commands to your local display.

Question #78

A systems administrator is implementing a new service task with systems at startup and needs to execute a script entitled test.sh with the following content:

```
TIMESTAMP=$(date '+%Y-%m-%d %H:%M:%S')
echo "helpme.service: timestamp $(Timestamp)" | systemd-cat -p info
sleep 60
done
```

The administrator tries to run the script after making it executable with `chmod +x`; however, the script will not run. Which of the following should the administrator do to address this issue? (Choose two.)

A) Add `#!/bin/bash` to the bottom of the script.

B) Create a unit file for the new service in `/etc/systemd/system/` with the name

helpme.service in the location. (Correct Answer)

C) Add #!/bin/bash to the top of the script. (Correct Answer)

D) Restart the computer to enable the new service.

E) Create a unit file for the new service in /etc/init.d with the name helpme.service in the location.

F) Shut down the computer to enable the new service.

Explanation

Correct Answer:

BC

Community Discussion

NO!, its B and C, #!/bin/bash is the very first line at the beginning on the script not the bottom

NO!, its B and C, #!/bin/bash is the very first line at the beginning on the script not the bottom

Selected Answer: BC

B: Create unit file in /etc/systemd/system/ C: Add #!/bin/bash to the top of the script.

Selected Answer: BC

C is correct because every bash script should start with #!/bin/bash B is correct because /etc/systemd/system is the correct folder for a unit file. E is INCORRECT because /etc/init.d is an invalid folder path. E appears to be there to through people off the track of the correct answer.

Selected Answer: AB

A. Add #!/bin/bash to the bottom of the script. B. Create a unit file for the new service in /etc/systemd/system/ with the name

Question #79

A Linux administrator needs to correct the permissions of a log file on the server. Which of the following commands should be used to set filename.log permissions to -rwxr--. ?

- A) `chmod 755 filename.log`
- B) `chmod 640 filename.log`
- C) `chmod 740 filename.log`
- D) `chmod 744 filename.log` (Correct Answer)**

Explanation

Correct Answer:

D

Community Discussion

D. `chmod 744 filename.log` The `chmod` command is used to change the permissions of files in Linux. The number passed to `chmod` specifies the binary representation of the desired permissions. The binary representation of -rwxr-- is 111 100 100, which corresponds to the decimal value 744. So, the command to set the permissions to -rwxr-- is "`chmod 744 filename.log`".

Question #80

After listing the properties of a system account, a systems administrator wants to remove the expiration date of a user account. Which of the following commands will accomplish this task?

- A) `chgrp system accountname`
- B) `passwd -s accountname`
- C) `chmod -G system account name`
- D) `chage -E -1 accountname` (Correct Answer)**

Explanation

Correct Answer:

D

Community Discussion

D. chage -E -1 accountname The chage command is used to modify the aging information of a user account, including the expiration date. The -E option is used to specify the expiration date of an account, and the -1 option sets the expiration date to a value of "never". So, the command "chage -E -1 accountname" will remove the expiration date of the account "accountname".

Selected Answer: D

-E is for expiration

Selected Answer: D

D is the correct answer

Question #81

A systems administrator wants to be sure the sudo rules just added to /etc/sudoers are valid. Which of the following commands can be used for this task?

A) visudo -c (Correct Answer)

B) test -f /etc/sudoers

C) sudo vi check

D) cat /etc/sudoers | tee test

Explanation

Correct Answer:

A

Community Discussion

Selected Answer: A

-c flag is to check-only mode

Selected Answer: A

The correct answer is A. visudo -c.

Selected Answer: A

A. visudo -c The visudo command is used to edit the /etc/sudoers file, and the -c option is used to check the syntax of the file without actually making any changes to it. If there are any syntax errors in the file, visudo will return an error message and prevent the changes from being saved. So, the command "visudo -c" can be used to validate the sudo rules in /etc/sudoers. Option B "test -f /etc/sudoers" checks if the file /etc/sudoers exists, but it does not validate the syntax of the file. Option C "sudo vi check" opens the file "check" with the vi editor, but it does not relate to the /etc/sudoers file. Option D "cat /etc/sudoers | tee test" displays the contents of the /etc/sudoers file and saves it to a file named "test", but it does not validate the syntax of the file.

Aamm033 is right, its A.

Aamm033 is right, its A.

Question #82

A user generated a pair of private-public keys on a workstation. Which of the following commands will allow the user to upload the public key to a remote server and enable passwordless login?

A) `scp ~/.ssh/id_rsa user@server:~/`

B) `rsync ~ /.ssh/ user@server:~/`

C) `ssh-add user server`

D) `ssh-copy-id user@server` (Correct Answer)

Explanation

Correct Answer:

Community Discussion

Selected Answer: D

D is the correct command but is missing the path to the public key that is to be copied over \$ ssh-copy-id -i ~/.ssh/id_rsa.pub user@host_address

Selected Answer: D

The correct answer is D. ssh-copy-id user@server.

Question #83

A Linux administrator created a new file system. Which of the following files must be updated to ensure the filesystem mounts at boot time?

- A) /etc/sysctl
- B) /etc/filesystems
- C) /etc/fstab (Correct Answer)**
- D) /etc/nfsmount.conf

Explanation

Correct Answer:
C

Community Discussion

C. /etc/fstab The /etc/fstab file is used to specify the file systems that are mounted at boot time in Linux. The administrator must add an entry for the new file system to this file, which includes information such as the file system device, mount point, file system type, and mount options. This information is used by the mount command to mount the file system at boot time.

Question #84

A Linux administrator is troubleshooting a memory-related issue. Based on the output of the commands:

```
$ vmstat -s --unit M

968 M total memory
331 M used memory
482 M active memory
279 M inactive memory
99 M free memory

$ free -h

              total        used        free      shared  buff/cache   available
Mem:           968M        331M        95M         13M         540M        458M
Swap:           0          0          0

$ ps -aux | grep script.sh
USER  PID   %CPU  %MEM  VSZ   RSS    TTY  STAT  START  TIME  COMMAND
user  8321  2.8   40.5  3224846  371687  7    SN    16:49  2:09  /home/user/script.sh
```

Which of the following commands would address the issue?

- A) `top -p 8321`
- B) `kill -9 8321`
- C) `renice -10 8321` (Correct Answer)**
- D) `free 8321`

Explanation

Correct Answer:

C

Community Discussion

Answer B is correct. The `renice` command speeds up or slows down processes in regards to CPU access. Killing the process would free up the memory.

A Linux administrator is troubleshooting a memory-related issue- but what is the issue that the process needs more memory or that we need to free memory ?

A Linux administrator is troubleshooting a memory-related issue- but what is the issue that the process needs more memory or that we need to free memory ?

A Linux administrator is troubleshooting a memory-related issue- but what is the issue that the process needs more memory or that we need to free memory ?

A Linux administrator is troubleshooting a memory-related issue- but what is the issue that the process needs more memory or that we need to free memory ?

Question #85

A systems administrator made some unapproved changes prior to leaving the company. The newly hired administrator has been tasked with revealing the system to a compliant state. Which of the following commands will list and remove the correspondent packages?

- A) dnf list and dnf remove last
- B) dnf remove and dnf check
- C) dnf info and dnf upgrade

D) dnf history and dnf history undo last (Correct Answer)

Explanation

Correct Answer:

D

Community Discussion

D is correct. check the following link: https://access.redhat.com/documentation/en-us/red_hat_enterprise_linux/9/html/managing_software_with_the_dnf_tool/assembly_handling-package-management-history_managing-software-with-the-dnf-tool#:~:text=Reverting%20transactions%20To%20revert%20a%20particular%20transaction%2C,last%20

I tested it on my VM and D worked for me.. as for a when do - dnf remove last, it says "no mach for argument last"

I tested it on my VM and D worked for me.. as for a when do - dnf remove last, it says "no mach for argument last"

Selected Answer: D

D. dnf history and dnf history undo last The "dnf history" command displays the history of all transactions performed using the DNF package manager, including package installs, upgrades, and removals. The "dnf history undo last" command undoes the last transaction, effectively removing the package that was installed. By using these two commands, the systems administrator can reveal the system to a compliant state by removing the unapproved packages. Option A "dnf list and dnf remove last" lists the packages installed on the system, but it does not undo the last transaction. The "dnf remove last" command is not a valid command.

Selected Answer: D

Yeah D is the correct answer. If there were multiple unwanted changes, the admin could also run the "rollback" option instead of "last"

Question #86

An administrator transferred a key for SSH authentication to a home directory on a remote server. The key file was moved to `.ssh/authorized_keys` location in order to establish SSH connection without a password. However, the SSH command still asked for the password. Given the following output:

```
[admin@linux ~]$ -ls -lhZ .ssh/auth*
-rw-r--r--. admin unconfined_u:object_r:user_home_t:s0 .ssh/authorized_keys
```

Which of the following commands would resolve the issue?

A) restorecon .ssh/authorized_keys (Correct Answer)

- B) `ssh_keygen -t rsa -o .ssh/authorized_keys`
- C) `chown root:root .ssh/authorized_keys`
- D) `chmod 600 .ssh/authorized_keys`

Explanation

Correct Answer:

A

Community Discussion

Selected Answer: A

"restorecon" is a command in SELinux (Security-Enhanced Linux) that is used to reset the security context of a file to its default SELinux security context. The "restorecon" command can be useful in cases where the SELinux security context of a file has been altered or changed, causing issues with the file's behavior or access. The "restorecon .ssh/authorized_keys" command specifically resets the security context of the "authorized_keys" file in the ".ssh" directory to its default SELinux security context. This can be useful in cases where the SELinux security context of the "authorized_keys" file has been altered, causing issues with SSH authentication.

D. `chmod 600 .ssh/authorized_keys`. It must have 600 permissions to work properly.

D. `chmod 600 .ssh/authorized_keys` This command sets the file permissions of `.ssh/authorized_keys` to 600, which means that only the owner (in this case, the user) will have read and write permissions, and no other users will have any permissions. By setting the correct permissions on the `authorized_keys` file, SSH will be able to use the key for authentication without asking for a password.

Selected Answer: A

(A) is most plausibly correct. A typical SELinux label for `authorized_keys` looks like this: `'unconfined_u:object_r:ssh_home_t:s0'`. (B) There is no command, `'ssh_keygen'`. (It's `'ssh-keygen'`. (C) The file is owned by its user, not root. (D) The file, `authorized_keys`, works fine in 0600, 0640, and 0644. (But it's true that the best practice is 0600.)

Selected Answer: A

A does appear to be the correct method to restore the default context of a file

Question #87

A cloud engineer needs to remove all dangling images and delete all the images that do not have an associated container. Which of the following commands will help to accomplish this task?

A) `docker images prune -a` (Correct Answer)

- B) `docker push images -a`
- C) `docker rmi -a images`
- D) `docker images rmi --all`

Explanation

Correct Answer:

A

Community Discussion

Selected Answer: A

A. docker images prune -a The "docker images prune -a" command will remove all dangling images, which are images that are not associated with any existing containers, as well as all images that do not have an associated container. The "-a" option specifies that all images should be removed, regardless of whether they are in use or not.

Selected Answer: A

A is the correct answer: \$ docker image prune [OPTIONS] --all , -a Remove all unused images, not just dangling ones

Question #88

A Linux system is failing to boot with the following error:

```
error: no such partitions
Entering rescue mode...
grub rescue>
```

Which of the following actions will resolve this issue? (Choose two.)

- A) Execute `grub-install --root-directory=/mnt` and reboot.
- B) Execute `grub-install /dev/sdX` and reboot. (Correct Answer)**
- C) Interrupt the boot process in the GRUB menu and add `rescue` to the kernel line.
- D) Fix the partition modifying /etc/default/grub and reboot. (Correct Answer)**
- E) Interrupt the boot process in the GRUB menu and add `single` to the kernel line.
- F) Boot the system on a LiveCD/ISO.

Explanation

Correct Answer:

BD

Community Discussion

Selected Answer: BF

Option F "Boot the system on a LiveCD/ISO" would help along with option B, as booting the system on a LiveCD/ISO would allow the administrator to access the file system and make necessary changes to fix the issue. Additionally, the administrator could use the bootable LiveCD/ISO to run the grub-install command with the appropriate arguments, such as "/dev/sdX" (Option B), to install GRUB on the appropriate disk. This would help resolve the issue with the missing partition and allow the system to boot properly.

Selected Answer: BF

B. Execute grub-install /dev/sdX and reboot. This command reinstalls the GRUB bootloader on the specified disk (/dev/sdX). Replace /dev/sdX with the appropriate device identifier for the system's boot disk (e.g., /dev/sda, /dev/nvme0n1). After executing the command, reboot the system and check if it successfully boots. F. Boot the system on a LiveCD/ISO. Booting the system from a LiveCD/ISO allows you to access the system's filesystem and repair the GRUB bootloader. Once booted into the LiveCD/ISO environment, you can mount the system's root partition, chroot into it, and then reinstall GRUB using the appropriate commands (e.g., grub-install and update-grub). After performing the necessary repairs, restart the system and check if it boots correctly.

B. Execute grub-install /dev/sdX and reboot. This action involves reinstalling the GRUB bootloader on the specified device (/dev/sdX). By reinstalling GRUB, it can help resolve any issues related to the bootloader and allow the system to boot successfully. F. Boot the system on a LiveCD/ISO. Booting the system using a LiveCD/ISO allows accessing the system's filesystem from an external environment. Once booted into the LiveCD/ISO, you can perform various troubleshooting steps such as checking the filesystem, repairing any errors, or modifying configuration files if necessary.

Selected Answer: BF

B)Execute grub-install /dev/sdX and reboot. F)Boot the system on a LiveCD/ISO.

Selected Answer: BF

I think its BF, but most resources state to: 1. Boot from live CD or ISO 2. Mount root partition 3. Reinstall GRUB

Question #89

A Linux administrator needs to create an image named sda.img from the sda disk and store it in the /tmp directory. Which of the following commands should be used to accomplish this task?

- A) `dd of=/dev/sda if=/tmp/sda.img`
- B) `dd if=/dev/sda of=/tmp/sda.img` (Correct Answer)**
- C) `dd --if=/dev/sda --of=/tmp/sda.img`
- D) `dd --of=/dev/sda --if=/tmp/sda.img`

Explanation

Correct Answer:

B

Community Discussion

The correct command is B: `dd if=/dev/sda of=/tmp/sda.img`. The `dd` command is used to create an image of a disk or partition, and the syntax is as follows: `dd if=input-file of=output-file`. In this case, the input file is the sda disk `/dev/sda` and the output file is the image file in the /tmp directory `/tmp/sda.img`.

Selected Answer: B

B is the only one that looks like the correct command syntax

I-input O-output

Question #90

A Linux administrator is creating a primary partition on the replacement hard drive for an application server. Which of the following commands should the administrator issue to verify the device name of this partition?

- A) `sudo fdisk /dev/sda`
- B) `sudo fdisk -s /dev/sda`
- C) `sudo fdisk -l` (Correct Answer)**
- D) `sudo fdisk -h`

Explanation

Correct Answer:

C

Community Discussion

Selected Answer: C

The correct answer is C. `sudo fdisk -l`.

The correct command is C. `sudo fdisk -l` The `sudo fdisk -l` command is used to list information about all the partitions on the system. The output will include the device name, file system type, size, and other relevant information for each partition. In this case, the administrator is looking for the device name of the primary partition on the replacement hard drive for an application server. By using the `sudo fdisk -l` command, the administrator will be able to see a list of all the partitions on the system and identify the device name for the new partition.

Selected Answer: C

"-l" = "--list" List the partition tables for the specified devices and then exit

Question #91

A systems administrator is investigating why one of the servers has stopped connecting to the internet.

```
#curl http://google.com
curl: (6) Could not resolve host: google.com

#cat /etc/resolv.conf
search user.company.com company.com
#nameserver 10.10.10.10

#ip route
0.0.0.0/0 via 10.0.5.1 dev eth0 proto static metric 100
10.0.0.0/16 dev eth0 proto kernel scope link src 10.0.3.60 metric 101

#nmcli connection show
NAME                                UUID                                TYPE      DEVICE
eth0                                ba4a3d30-efdc-4fa5-83d3-3721fd4aff75  ethernet  eth0
Wired connection 1                  8d569d5a-22a2-356d-8532-9a2638f11b5a5  ethernet  --
```

Which of the following is causing the issue?

A) The DNS address has been commented out in the configuration file. (Correct Answer)

B) The search entry in the /etc/resolv.conf file is incorrect.

C) Wired connection 1 is offline.

D) No default route is defined.

Explanation

Correct Answer:

A

Community Discussion

Selected Answer: A

The correct answer is A. The DNS address has been commented out in the configuration file.

Selected Answer: A

A. The DNS address has been commented out in the configuration file. #nameserver 10.10.10.10

Selected Answer: A

Whenever you want to comment a line, put a # in an appropriate place in a file. Anything beginning after # and ending at the end of the line won't get executed. This comments out the complete line. This comments out only the last part of the line starting at #

Question #92

A systems administrator is tasked with installing GRUB on the legacy MBR of the SATA hard drive. Which of the following commands will help the administrator accomplish this task?

- A) `grub-install /dev/hda`
- B) `grub-install /dev/sda` (Correct Answer)**
- C) `grub-install /dev/sr0`
- D) `grub-install /dev/hd0,0`

Explanation

Correct Answer:
B

Community Discussion

Selected Answer: B

The GRUB boot loader is a crucial component of a Linux system. It is responsible for loading the operating system into memory and allowing the user to select the desired operating system to boot into. In order to install GRUB on a legacy MBR (Master Boot Record) of a SATA hard drive, the administrator needs to specify the correct device name. The device name is used to identify the hard drive in the system, and it must be specified correctly in order for GRUB to be installed and function properly. The correct device name to use in this scenario would be `/dev/sda`. The `/dev/sda` device name is the standard name used to identify the first SATA hard drive on a system. This is the device name that should be specified in the `grub-install` command

Selected Answer: B

<https://www.thegeekdiary.com/grub-install-command-options/>

Question #93

A junior Linux administrator is tasked with installing an application. The installation guide states the application should only be installed in a run level 5 environment.

```
$ systemctl get-default  
getty.target
```

Which of the following commands would ensure the server is set to runlevel 5?

- A) systemctl isolate multi-user.target
- B) systemctl isolate graphical.target (Correct Answer)**
- C) systemctl isolate network.target
- D) systemctl isolate basic.target

Explanation

Correct Answer:

B

Community Discussion

System V runlevel (Purpose) runlevel 0 - (for System shutdown) runlevel1 (for Single-user mode) runlevel2 (for Local multiuser without remote network) runlevel3 (for Full multiuser with network) runlevel4 (for Unused/User-defined) runlevel5 (for Full multiuser with network and display manager) runlevel6 (for System reboot)

Is it suppose to be A?

A runlevel is an operating state on a Unix and Unix-based operating system that is preset on the Linux-based system. Runlevels are numbered from zero to six. Runlevels determine which programs can execute after the OS boots up. The runlevel defines the state of the machine after boot. Systems administrators set the default runlevel of a system according to their needs, or use the runlevel command to find out the machine's current runlevel to assess a system. Runlevel 0 shuts down the system Runlevel 1 single-user mode Runlevel 2 multi-user mode without networking Runlevel 3 multi-user mode with networking Runlevel 4 user-definable Runlevel 5 multi-user mode with networking Runlevel 6 reboots the system to restart it

Question #94

A Linux administrator is tasked with adding users to the system. However, the administrator wants to ensure the users' access will be disabled once the project is over. The expiration date should be 2021-09-30. Which of the following commands will accomplish this task?

A) `sudo useradd -e 2021-09-30 Project_user` (Correct Answer)

B) `sudo useradd -c 2021-09-30 Project_user`

C) `sudo modinfo -F 2021-09-30 Project_uses`

D) `sudo useradd -m -d 2021-09-30 Project_user`

Explanation

Correct Answer:

A

Community Discussion

Selected Answer: A

The correct answer is A. `sudo useradd -e 2021-09-30 Project_user`.

Selected Answer: A

The correct command to accomplish the task is A: `sudo useradd -e 2021-09-30 Project_user`. The `-e` option is used to specify the expiration date for the user account.

Question #95

A DevOps engineer needs to download a Git repository from `https://git.company.com/admin/project.git`. Which of the following commands will achieve this goal?

A) `git clone https://git.company.com/admin/project.git` (Correct Answer)

B) `git checkout https://git.company.com/admin/project.git`

C) `git pull https://git.company.com/admin/project.git`

D) `git branch https://git.company.com/admin/project.git`

Explanation

Correct Answer:

A

Community Discussion

Selected Answer: A

The correct command to download a Git repository from `https://git.company.com/admin/project.git` is: A. `git clone https://git.company.com/admin/project.git`

Selected Answer: A

`git clone` is the best option if its a new project. `git pull` would work best if it was an ongoing project that had multiple team members that had pushed their updates to the main repository

Question #96

An administrator installed an application from source into `/opt/operations1/` and has received numerous reports that users are not able to access the application without having to use the full path `/opt/operations1/bin/*`. Which of the following commands should be used to resolve this issue?

A) `echo 'export PATH=$PATH:/opt/operations1/bin' >> /etc/profile` (Correct Answer)

B) `echo 'export PATH=/opt/operations1/bin' >> /etc/profile`

- C) echo 'export PATH=\$PATH/opt/operations1/bin' >> /etc/profile
 D) echo 'export \$PATH:/opt/operations1/bin' >> /etc/profile

Explanation

Correct Answer:

A

Community Discussion

Selected Answer: A

This command adds the path /opt/operations1/bin to the \$PATH environment variable in the /etc/profile file. This will allow users to access the application without having to use the full path /opt/operations1/bin/*.

Selected Answer: A

A looks like the best answer here. It would be better to put the directory before \$PATH, so that way the directory is searched before the rest of the pathing options in \$PATH.

Question #97

A Linux system is getting an error indicating the root filesystem is full. Which of the following commands should be used by the systems administrator to resolve this issue? (Choose three.)

- A) df -h /
- B) fdisk -l /dev/sdb
- C) growpart /dev/mapper/rootvg-rootlv
- D) pvcreate /dev/sdb (Correct Answer)**
- E) lvresize -L +10G -r /dev/mapper/rootvg-rootlv (Correct Answer)**
- F) lsblk /dev/sda
- G) parted -l /dev/mapper/rootvg-rootlv
- H) vgextend /dev/rootvg /dev/sdb (Correct Answer)**

Explanation

Correct Answer:

Community Discussion

Selected Answer: DEH

```
pvcreate /dev/sdb vgextend /dev/rootvg /dev/sdb lvresize -L +10G -r /dev/mapper/  
rootvg-rootlv
```

Discard Im going with DEH.

Discard Im going with DEH.

Correct is pvcreate /dev/sdb vgextend /dev/rootvg /dev/sdb lvresize -L +10G -r /dev/
mapper/rootvg-rootlv

Selected Answer: DEH

DEH. But in this order: DHE. D. 'pvcreate /dev/sdb' (Create a physical volume for LVM on the new disk.) H. 'vgextend /dev/rootvg /dev/sdb' (Extend volume group to include new disk.) E. 'lvresize -L +10G -r /dev/mapper/rootvg-rootlv' (Add space to logical volume.)

Question #98

A cloud engineer is asked to copy the file deployment.yaml from a container to the host where the container is running. Which of the following commands can accomplish this task?

- A) `docker cp container_id/deployment.yaml deployment.yaml`
- B) `docker cp container_id:/deployment.yaml deployment.yaml` (Correct Answer)**
- C) `docker cp deployment.yaml local://deployment.yaml`
- D) `docker cp container_id/deployment.yaml local://deployment.yaml`

Explanation

Correct Answer:

B

Community Discussion

Selected Answer: B

Given answer appears correct(B) The file is being copied from the container (SRC) to the host (DST) `docker cp [OPTIONS] CONTAINER:SRC_PATH DEST_PATH|- docker cp [OPTIONS] SRC_PATH|- CONTAINER:DEST_PATH`

B. `docker cp container_id:/deployment.yaml deployment.yam`

Question #99

A Linux system is failing to start due to issues with several critical system processes. Which of the following options can be used to boot the system into the single user mode? (Choose two.)

- A) Execute the following command from the GRUB rescue shell: `mount -o remount, ro/sysroot`.
- B) Interrupt the boot process in the GRUB menu and add `systemd.unit=single` in the kernel line.
- C) Interrupt the boot process in the GRUB menu and add `systemd.unit=rescue.target` in the kernel line. (Correct Answer)**
- D) Interrupt the boot process in the GRUB menu and add `single=user` in the kernel line.
- E) Interrupt the boot process in the GRUB menu and add `init=/bin/bash` in the kernel line. (Correct Answer)**
- F) Interrupt the boot process in the GRUB menu and add `systemd.unit=single.target` in the kernel line.

Explanation

Correct Answer:
CE

Community Discussion

why isnt it BE?

Discard I'm going with CE.

Discard I'm going with CE.

Selected Answer: CE

I think the given answer (CE) is correct - C. "systems.unit=rescue.target" will put you in single-user mode - E. 1. In GRUB, press E to edit your boot entry (the Ubuntu entry). 2. Look for the line that starts with linux, and then look for ro. 3. Replace ro with rw init=/bin/bash. This action mounts your file system as read-write and uses /bin/bash as the init process. 4. Press Ctrl+X to reboot with these settings

Selected Answer: DE

Option D (adding "single=user" in the kernel line) and option E (adding "init=/bin/bash" in the kernel line) can be used to boot the Linux system into single user mode. In single user mode, the system will only run a minimal set of services and the root user will be given a shell, allowing the administrator to troubleshoot and repair any issues with the system. Option C, adding systemd.unit=rescue.target to the kernel line, is intended to boot the system into a rescue environment, not a single user mode. In the rescue environment, the system will attempt to repair the system and restore the system to a bootable state, rather than providing direct access to the system as single user mode does.

Question #100

A DevOps engineer needs to allow incoming traffic to ports in the range of 4000 to 5000 on a Linux server. Which of the following commands will enforce this rule?

A) `iptables -f filter -I INPUT -p tcp --dport 4000:5000 -A ACCEPT`

B) `iptables -t filter -A INPUT -p tcp --dport 4000:5000 -j ACCEPT` (Correct Answer)

C) `iptables filter -A INPUT -p tcp --dport 4000:5000 -D ACCEPT`

D) `iptables filter -S INPUT -p tcp --dport 4000:5000 -A ACCEPT`

Explanation

Correct Answer:

B

Community Discussion

Selected Answer: B

The command iptables is used to manage the rules in the Linux kernel's firewall. The options used in the command determine how the rule will be enforced. In the case of option B, -t filter specifies that the rules should be applied to the filter table, which is used for packet filtering. The -A INPUT option specifies that the rule should be appended to the INPUT chain, which is used for incoming traffic. The -p tcp option specifies that the rule should only apply to TCP traffic, and the --dport 4000:5000 option specifies that the rule should only apply to incoming traffic to ports in the range of 4000 to 5000. The -j ACCEPT option specifies that the matching traffic should be accepted, allowing it to enter the system.

Question #101

A Linux administrator needs to determine whether a hostname is in the DNS. Which of the following would supply the information that is needed?

A) nslookup (Correct Answer)

- B) rsync
- C) netstat
- D) host

Explanation

Correct Answer:

A

Community Discussion

Both 'nslookup' and 'host' can resolve hostname, even against specific name servers. This is bad question.

Selected Answer: A

nslookup

Selected Answer: D

nslookup - query Internet name servers (deprecated) host - newer DNS lookup utility

host also works, what will be the correct answer ?

Selected Answer: A

The correct command is "nslookup." The "nslookup" command allows a user to query a DNS server to determine the IP address of a hostname, or to determine the hostname associated with a particular IP address. It is commonly used to troubleshoot DNS resolution issues. The other options "rsync," "netstat," and "host" are not used for this purpose.

Question #102

A server is experiencing intermittent connection issues. Some connections to the Internet work as intended, but some fail as if there is no connectivity. The systems administrator inspects the server configuration:

Routing table:

```
default via 89.107.157.129 dev ens3 proto static metric 100
default via 10.0.5.1 dev ens11 proto dhcp metric 101
10.0.0.0/16 dev sn11 proto kernel scope link src 10.0.6.225 metric 101
89.107.157.128/26 via 89.107.157.129 dev ens3 proto static metric 100
89.107.157.129 dev ens3 proto static scope link metric 100
89.107.157.160/29 dev ens3 proto kernel scope link src 89.107.157.161 metric 100
```

IP configuration:

```
ens3:
    inet 89.107.157.161/29 brd 89.107.157.167 scope global noprefixroute ens3
ens11:
    inet 10.0.6.225/16 brd 10.0.255.255 scope global noprefixroute dynamic ens11
```

ARP table:

Address	Hwtype	Hwaddress	Flags	Mask	Iface
10.0.5.1	ether	64:d1:54:c4:75:cb	C		ens11
89.107.157.129	ether	5c:5e:ab:01:85:cf	C		ens3
89.107.157.162	ether	52:54:00:e1:44:0a	C		ens3
10.0.255.1	ether	00:50:7f:e3:aa:1c	C		ens11

```
/etc/resolv.conf:
Generated by NetworkManager
search company.com
nameserver 10.0.5.1
```

Which of the following is MOST likely the cause of the issue?

- A) An internal-only DNS server is configured.
- B) The IP netmask is wrong for ens3.
- C) Two default routes are configured. (Correct Answer)**
- D) The ARP table contains incorrect entries.

Explanation

Correct Answer:

C

Community Discussion

Selected Answer: B

well the 89 networks definitely have a subnet ID problem, but I'm not sure if that's the reason there is a problem here.

oh shit!!! good call Manish!! i just realized that the real answer is D, looks at `ens3` interface in the arp table!! it's showing twice and it's the main default gateway, you're a genius!! i'm glad a looked at this again!

Answer is D!!

Answer is D!!

Selected Answer: B

B. The IP netmask is wrong for ens3. - the IP of the GW is certainly wrong C. Two default routes are configured. - actually having a double default can use problems D. The ARP table contains incorrect entries. , you see two IP and MAC associated with the same network card...

Question #103

A cloud engineer needs to block the IP address 192.168.10.50 from accessing a Linux server. Which of the following commands will achieve this goal?

- A) `iptables -F INPUT -j 192.168.10.50 -m DROP`
- B) `iptables -A INPUT -s 192.168.10.30 -j DROP` (Correct Answer)**
- C) `iptables -i INPUT --ipv4 192.168.10.50 -z DROP`
- D) `iptables -j INPUT 192.168.10.50 -p DROP`

Explanation

Correct Answer:

B

Community Discussion

B is best, but the IP address is wrong.

Selected Answer: B

The correct answer is B. `iptables -A INPUT -s 192.168.10.50 -j DROP`.

B. `iptables -A INPUT -s 192.168.10.30 -j DROP`

MissAllen is right, B is the answer here. "-A" appends the rule, "-s" is source address, and "-j" is the action if the rule is matched, so DROP

Question #104

A Linux systems administrator is configuring a new filesystem that needs the capability to be mounted persistently across reboots. Which of the following commands will accomplish this task? (Choose two.)

A) `df -h /data`

B) `mkfs.ext4 /dev/sdc1` (Correct Answer)

C) `fsck /dev/sdc1`

D) `fdisk -l /dev/sdc1`

E) `echo "/data /dev/sdc1 ext4 defaults 0 0" >> /etc/fstab`

F) `echo "/dev/sdc1 /data ext4 defaults 0 0" >> /etc/fstab` (Correct Answer)

Explanation

Correct Answer:

BF

Community Discussion

Selected Answer: BF

`mkfs.ext4 /dev/sdc1` - This command creates the ext4 filesystem on the device `/dev/sdc1`.
`echo "/dev/sdc1 /data ext4 defaults 0 0" >> /etc/fstab` - This command appends a line to the `/etc/fstab` file that specifies the device to be mounted at a specific mount point

persistently across reboots. After running these commands, the new filesystem will be created and automatically mounted at the specified mount point every time the system reboots.

Selected Answer: BF

Answer here is BF "modify the /etc/fstab text file to automatically mount the new partition by opening it in an editor and adding the following line: /dev/ xxx 1 /data ext4 defaults 1 2 where xxx is the device name of the storage device" <https://learning.oreilly.com/library/view/mastering-linux-system/978119794455/b01.xhtml>

Question #105

A Linux administrator is alerted to a storage capacity issue on a server without a specific mount point or directory. Which of the following commands would be MOST helpful for troubleshooting? (Choose two.)

- A) parted
- B) df (Correct Answer)**
- C) mount
- D) du (Correct Answer)**
- E) fdisk
- F) dd
- G) ls

Explanation

Correct Answer:
BD

Community Discussion

Selected Answer: BD

B. df - This command shows the disk space usage of all mounted filesystems on the system. It will help identify the filesystems that are consuming the most space and the available space left. D. du - This command shows the disk usage of files and directories, including hidden files. It can help identify which directories or files are using the most space on the filesystem.

Selected Answer: BD

BD is the answer here. du is a standard Unix program used to estimate file space usage
df is a standard Unix command used to display the amount of available disk space for file systems

Question #106

A systems administrator pressed Ctrl+Z after starting a program using the command line, and the shell prompt was presented. In order to go back to the program, which of the following commands can the administrator use?

A) fg (Correct Answer)

- B) su
- C) bg
- D) ed

Explanation

Correct Answer:

A

Community Discussion

Selected Answer: A

The command the administrator can use to go back to the program is fg. A. fg - This command brings a job to the foreground. In this case, the job is the program that was suspended using Ctrl+Z. B. su - This command is used to switch to another user account. C. bg - This command starts a suspended job in the background. It is not useful in this case because the administrator wants to bring the job to the foreground. D. ed - This command starts the ed text editor. It is not useful in this case because the administrator wants to resume the program that was suspended.

Selected Answer: A

Answer is A. Ctrl+Z suspended the process, and "fg" will bring it back into the foreground of the shell

Question #107

A systems administrator received a notification that a system is performing slowly. When running the top command, the systems administrator can see the following values:

```
%Cpu(s): 2.7 us, 1.9 sy, 0.0 ni, 0.4 id, 95 wa, 0.0 hi, 0.0 si 0.0 st
```

Which of the following commands will the administrator most likely run NEXT?

A) vmstat (Correct Answer)

- B) strace
- C) htop
- D) lsof

Explanation

Correct Answer:

A

Community Discussion

Selected Answer: A

Based on the output of the top command, which shows a high percentage of CPU time spent on I/O wait (%wa), the command that the administrator would most likely run next is vmstat. A. vmstat - This command provides a summary of system memory, CPU usage, and I/O statistics. It can be used to identify I/O bottlenecks and other performance issues. B. strace - This command is used to trace system calls and signals made by a process. It is not directly relevant to identifying I/O bottlenecks. C. htop - This command is an interactive process viewer that is similar to top but provides more features and capabilities. It may be useful for more detailed analysis of system performance, but is not necessary in this case. D. lsof - This command lists open files and can be used to identify processes that are holding open file handles. It is not directly relevant to identifying I/O bottlenecks.

Selected Answer: A

A is the answer here, as the I/O wait time is 95%. vmstat will show the resource usage for each process

Question #108

Which of the following technologies provides load balancing, encryption, and observability in containerized environments?

- A) Virtual private network
- B) Sidecar pod
- C) Overlay network

D) Service mesh (Correct Answer)

Explanation

Correct Answer:

D

Community Discussion

Selected Answer: D

The answer is D. Service mesh.

Selected Answer: D

A service mesh is a dedicated infrastructure layer that provides communication management between microservices in a containerized environment. Service meshes provide a range of features including load balancing, service discovery, traffic routing, encryption, and observability. The most popular service mesh frameworks are Istio, Linkerd, and Consul. The other options provided are: A. Virtual private network (VPN) - A VPN is a secure connection between two networks over the internet, which is primarily used to ensure secure remote access to a private network. B. Sidecar pod - A sidecar pod is a design pattern in Kubernetes that allows for two containers to run within the same pod. C. Overlay network - An overlay network is a virtual network that is built on top of an existing physical network, providing additional network management features, but does not inherently provide the features mentioned in the question.

Selected Answer: D

Answer is D "A service mesh controls the delivery of service requests in an application. Common features provided by a service mesh include service discovery, load balancing, encryption and failure recovery." <https://www.techtarget.com/searchitoperations/definition/service-mesh>

Question #109

A development team asks an engineer to guarantee the persistency of journal log files across system reboots. Which of the following commands would accomplish this task?

A) `grep -i auto /etc/systemd/journald.conf && systemctl restart systemd-journald.service`
(Correct Answer)

B) `cat /etc/systemd/journald.conf | awk '(print $1,$3)'`

C) `sed -i 's/auto/persistent/g' /etc/systemd/journald.conf && sed -i 'persistent/s/^#//q' /etc/systemd/journald.conf`

D) `journalctl --list-boots && systemctl restart systemd-journald.service`

Explanation

Correct Answer:

A

Community Discussion

I don't see any correct answer, but C is the closest. A is not correct because command one searches for the pattern "auto" in the file /etc/systemd/journald.conf, ignoring case. It will find it, thereby causing command two to execute. Restarting the journald service doesn't help if you made no changes to the .conf file!. Answer C searches for the pattern "auto", globally in the same file, but replaces it with the pattern "persistent", in the file (that is the -i option). That's great, but I tested it with command two and get an error. I suggest command two be the restart of the journald service.

Selected Answer: C

C for real

C. `sed -i 's/auto/persistent/g' /etc/systemd/journald.conf && sed -i 'persistent/s/^#//q' /etc/systemd/journald.conf`

Good work MissAllen, (C) seems correct 1st command & (A) 2nd Command. Heya wanna turn that Miss into Mrs.Allen? email: mrs.allen@blackhat.io <3 you can come live on this paradise island wit me :)

Good work MissAllen, (C) seems correct 1st command & (A) 2nd Command. Heya wanna turn that Miss into Mrs.Allen? email: mrs.allen@blackhat.io <3 you can come live on this paradise island wit me :)

Question #110

A systems administrator is receiving tickets from users who cannot reach the application app that should be listening on port 9443/tcp on a Linux server. To troubleshoot the issue, the systems administrator runs netstat and receives the following output:

```
# netstat -anp | grep appd | grep -w LISTEN
tcp 0 0 127.0.0.1:9443 0.0.0.0:* LISTEN 1234/appd
```

Based on the information above, which of the following is causing the issue?

- A) The IP address 0.0.0.0 is not valid.
- B) The application is listening on the loopback interface. (Correct Answer)**
- C) The application is listening on port 1234.
- D) The application is not running.

Explanation

Correct Answer:
B

Community Discussion

B because longest answer

lol...

lol...

Selected Answer: B

Based on the output of netstat, the issue is that the application is listening on the loopback interface (127.0.0.1) rather than on a publicly accessible IP address. The netstat output shows that the application is bound to the IP address 127.0.0.1 on port 9443, which means that it is only accessible from the same system (localhost). The 0.0.0.0:* field in the output indicates that the application is listening on all available network interfaces, but the IP address 0.0.0.0 does not actually represent a valid IP address. To resolve the issue, the application needs to be reconfigured to listen on a publicly accessible IP address, such as the IP address of the system's network interface that is reachable from the users' network.

Selected Answer: B

Answer is B. The server is in a "Listen" state on port 9943 using its loopback address. The "1234" is a process-id

Question #111

A systems administrator is troubleshooting a connectivity issue pertaining to access to a system named db.example.com. The system IP address should be 192.168.20.88. The administrator issues the dig command and receives the following output:

```
;; ANSWER SECTION:
db.example.com.      15 IN A 192.168.20.89
```

The administrator runs `grep db.example.com /etc/hosts` and receives the following output:

```
192.168.20.89 db.example.com
```

Given this scenario, which of the following should the administrator do to address this issue?

A) Modify the `/etc/hosts` file and change the db.example.com entry to 192.168.20.89. (Correct Answer)

B) Modify the `/etc/network` file and change the db.example.com entry to 192.168.20.88.

- C) Modify the /etc/network file and change the db.example.com entry to 192.168.20.89.
D) Modify the /etc/hosts file and change the db.example.com entry to 192.168.20.88.

Explanation

Correct Answer:

A

Community Discussion

Selected Answer: D

Why not D?

D is the right answer

D is the right answer

Selected Answer: D

D is correct.

Selected Answer: D

D obviously

Question #112

Users have been unable to reach www.comptia.org from a Linux server. A systems administrator is troubleshooting the issue and does the following:

Output 1:

```
2: eth0: <BROADCAST,MULTICAST,UP, LOWER_UP> mtu 1500 qdisc noqueue state UP group default qlen 1000
    link/ether ac:11:22:33:44:cd brd ff:ff:ff:ff:ff:ff
    inet 192.168.168.10/24 brd 192.168.169.255 scope global dynamic noprefixroute eth0
        valid_lft 8097sec preferred_lft 8097sec
    inet fe80::4daf:8c7c:a6ff:2771/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
```

Output 2:

```
nameserver 192.168.168.53
```

Output 3:

```
FING 192.168.168.53 (192.168.168.53) 56(84) bytes of data.
64 bytes from 192.168.168.53: icmp_seq=1 ttl=64 time=2.85 ms

--- 192.168.168.53 ping statistics ---
1 packets transmitted, 1 received, 0% packet loss, time 0ms
rtt min/avg/max/mdev = 2.847/2.847/2.847/0.000 ms
```

Output 4:

```
192.168.168.0/24 dev eth0 proto kernel scope link src 192.168.168.10 metric 600
```

Output 5:

```
...
;; QUESTION SECTION:
;www.comptia.org. IN A

;; ANSWER SECTION:
. 0 CLASS4096 OPT 10 8 LgmNvk0AazU=

;; ADDITIONAL SECTION:
www.comptia.org. 3385 IN A 23.96.239.26
...
```

Based on the information above, which of the following is causing the issue?

- A) The name www.comptia.org does not point to a valid IP address.
- B) The server 192.168.168.53 is unreachable.
- C) No default route is set on the server. (Correct Answer)**
- D) The network interface eth0 is disconnected.

Explanation

Correct Answer:

C

Community Discussion

Selected Answer: C

The answer should be C. I can't see any default routes on this configuration. But the name server on the same subnet is reachable, and does appear to have a valid IP address. It looks like there is another typo on this question though. The broadcast address 192.168.169.255 is outside the subnet of 192.168.168.10/24. So any broadcasts at that IP would be useless when attempting to talk to 192.168.168.0/24 devices.

process of elimination, C is left.

process of elimination, C is left.

Selected Answer: C

going with C.

Selected Answer: C

I'm picking C because i don't see any default route. I have no idea if the static name resolution is correct or not..

Question #113

A systems technician is working on deploying several microservices to various RPM-based systems, some of which could run up to two hours. Which of the following commands will allow the technician to execute those services and continue deploying other microservices within the same terminal session?

A) gedit & disown (Correct Answer)

B) kill 9 %1

C) fg %1

D) bg %1 job name

Explanation

Correct Answer:

A

Community Discussion

Selected Answer: D

Answer D is the closest. A doesn't make any sense as what gedit will help in all this.

disown is fine, but what does gedit have to do with anything?

disown is fine, but what does gedit have to do with anything?

Answer A: bg puts existing running processes in the background; while "& disown" sends the process to the bg upon execution which is what the question is implying. "...execute those services and continue deploying other microservices within the same terminal session? gedit - has nothing to do with the question or answer it's just the command used in the answer (bad choice); imagine answer A as: job name & disown

Selected Answer: D

D. bg %1 job name: This command puts the job associated with job ID %1 in the background and allows the technician to continue deploying other microservices in the same terminal session. The job name argument is optional and allows the technician to specify a custom name for the job. This option is suitable for the scenario described in the question. A. gedit & disown: This command opens the gedit text editor in the background and disowns it so that it is no longer associated with the terminal. This option is not relevant to deploying microservices and would not allow the technician to continue deploying other microservices in the same terminal session.

Question #114

A Linux administrator was notified that a virtual server has an I/O bottleneck. The Linux administrator analyzes the following output:

```
root@linux:~# uptime
18:43:47 up 1 day, 19:58, 1 user, load average: 9.90, 5.83, 2.49
root@linux:~# vmstat 10 10
procs -----memory----- --swap----- ----io---- -system- -----cpu-----

 r b swpd   free   buff   cache  si   so bi   bo    in    cs us  sy  id  wa st
13 0 5520 141228 98932 2325312 0    2 10   28    192   167  1  0 99  0  0
10 0 5608 131280 98932 2325324 0 26211 0 26211  342   393 91  9  0  0  0
10 0 5528   1096 98932 2325324 0  5242 0  5242  333   402 96  4  0  0  0

root@linux:~# free -m
              total    used      free   shared  buff/cache   available
Mem:           3933    1454        110        33        2368        2202
Swap:           1497         5        1491
```

Given there is a single CPU in the sever, which of the following is causing the slowness?

- A) The system is running out of swap space.
- B) The CPU is overloaded. (Correct Answer)**
- C) The memory is exhausted.
- D) The processes are paging.

Explanation

Correct Answer:

B

Community Discussion

Selected Answer: B

im not seeing and swap or memory issues, what we do see is the CPU ID value at 99, so i pick B.

Selected Answer: B

Based on the output provided, the issue causing the slowness is B. The CPU is overloaded. The load average of 9.90 indicates that there are, on average, 9.90 processes in the running or runnable state. Given that there is only a single CPU, this means that

there are far more processes needing CPU time than the CPU can handle, leading to an overloaded CPU. This is likely the cause of the slowness. The other options (A, C, and D) do not seem to be supported by the provided output. The output does not indicate that the system is running out of swap space (almost all free), that the memory is exhausted (still have a lot of free mem), or that the processes are paging (only 5 for pages used). Therefore, these are not likely to be the cause of the slowness.

Selected Answer: D

While the load average is high, it doesn't directly imply that the CPU is overloaded. Instead, the "si" and "so" values in the vmstat output indicate that processes are paging in and out of swap, which typically points to I/O bottlenecks and disk activity as the primary cause of slowness. So, based on the provided information, the most likely cause of slowness is related to I/O and processes paging, rather than CPU overload. Therefore, option B (The CPU is overloaded) is less likely to be the primary cause.

Selected Answer: D

B or D ? please help As you can see we have bo values -> blocks sent to a block device. paged memory , and so value -> Amount of memory swapped to disk (/s). and id -> idle elapsed time. = 0 (normal value is 99 and 100)

B definitive id – The percentage of idle CPU.

Question #115

Employees in the finance department are having trouble accessing the file /opt/work/file. All IT employees can read and write the file. Systems administrator reviews the following output:

```
admin@server:/opt/work$ ls -al file
-rw-rw----+ 1 root it 4 Sep 5 17:29 file
```

Which of the following commands would permanently fix the access issue while limiting access to IT and finance department employees?

- A) `chattr +i file`
- B) `chown it:finance file`
- C) `chmod 666 file`
- D) `setfacl -m g:finance:rw file` (Correct Answer)**

Explanation

Correct Answer:

D

Community Discussion

Selected Answer: D

if we change the owner to finance, the IT group would then be considered "other" in permissions and have no rights. currently IT has rights,, so using setfacl to give finance RW rights would keep the IT groups, and give finance rights. D

Selected Answer: B

chown [owner]:[group] file This would allow rw for both IT and Finance.

Debian12 / -bash: setfacl: command not found

Selected Answer: D

The correct answer is D. setfacl -m g:finance:rw file.

Selected Answer: D

D. setfacl -m g:finance:rw file: This command sets a new ACL (access control list) on the file that grants read and write access to the finance group. This option would allow the finance department to access the file while also maintaining the existing permissions for IT employees. The + in the rw option means to add read and write permissions to the existing permissions. The g: specifies that the permission is being set for a group, and the finance is the name of the group.

Question #116

A Linux engineer needs to create a custom script, cleanup.sh, to run at boot as part of the system services. Which of the following processes would accomplish this task?

A) Create a unit file in the /etc/default/ directory.
systemctl enable cleanup

systemctl is-enabled cleanup

B) Create a unit file in the /etc/skel/ directory.

systemctl enable cleanup

systemctl is-enabled cleanup

C) Create a unit file in the /etc/systemd/system/ directory.

systemctl enable cleanup

systemctl is-enabled cleanup (Correct Answer)

D) Create a unit file in the /etc/sysctl.d/ directory.

systemctl enable cleanup

systemctl is-enabled cleanup

Explanation

Correct Answer:

C

Community Discussion

Selected Answer: C

To create a custom script, cleanup.sh, to run at boot as part of the system services, the process that would accomplish this task is option C, "Create a unit file in the /etc/systemd/system/ directory". Here's how this can be done: Create the cleanup.sh script and place it in a suitable location. For example, in the /usr/local/bin directory. Create a new systemd unit file with the .service extension in the /etc/systemd/system directory. For example, create a file called cleanup.service. In the cleanup.service file, define the service by setting the service name, description, and the command to execute the script. For example: [Unit] Description=Cleanup script [Service] ExecStart=/usr/local/bin/cleanup.sh [Install] WantedBy=multi-user.target

Selected Answer: C

The correct answer is C. Create a unit file in the /etc/systemd/system/ directory.

Save and close the cleanup.service file. Use the systemctl command to enable the service to run at boot time: systemctl enable cleanup.service this will create the necessary symbolic links to start the cleanup.service unit automatically at boot time. Use the systemctl command to check the status of the service: systemctl is-enabled cleanup.service This command will show whether the service is enabled to start at boot time or not. After completing these steps, the cleanup.sh script will run automatically at boot time as part of the system services.

Save and close the cleanup.service file. Use the systemctl command to enable the service to run at boot time: `systemctl enable cleanup.service` this will create the necessary symbolic links to start the cleanup.service unit automatically at boot time. Use the systemctl command to check the status of the service: `systemctl is-enabled cleanup.service` This command will show whether the service is enabled to start at boot time or not. After completing these steps, the cleanup.sh script will run automatically at boot time as part of the system services.

Question #117

A Linux system is failing to boot. The following error is displayed in the serial console:

```
[[1;33mDEPEND[Om] Dependency failed for /data.
```

```
[[1;33mDEPEND[Om] Dependency failed for Local File Systems
```

...

Welcome to emergency mode! After logging in, type "journalctl -xb" to view system logs, "systemctl reboot" to reboot, "systemctl default" to try again to boot into default mode.

Give root password for maintenance

(or type Control-D to continue)

Which of the following files will need to be modified for this server to be able to boot again?

A) /etc/mtab

B) /dev/sda

C) /etc/fstab (Correct Answer)

D) /etc/grub.conf

Explanation

Correct Answer:

C

Community Discussion

Selected Answer: C

Based on the error message and symptoms described, the file that needs to be modified for this server to be able to boot again is option C, /etc/fstab. The error message indicates that the system is failing to start the /data partition, which is causing the system to go into emergency mode and preventing it from booting. The /data partition is most likely defined in the /etc/fstab file, which contains information about the file systems that are mounted at boot time.

Selected Answer: C

I pick C: fstab.. but i'm not really sure. . anyone else more sure?

Selected Answer: C

The correct answer is C. /etc/fstab.

This does not look like output that can be properly analyzed. Should we just assume what it says?: [[1;33mDEPEND[Om] Dependency failed for /data. [[1;33mDEPEND[Om] Dependency failed for Local File Systems

Selected Answer: C

The file /etc/fstab is used by the Linux system to mount file systems during boot. It contains information about file systems to be mounted, including the device name, mount point, file system type, and mount options. The error message indicates that a dependency has failed for "/data" and for local file systems, which suggests that there is an issue with one of the file systems listed in the /etc/fstab file. It is likely that the device name, mount point, file system type, or mount options for "/data" or another file system listed in /etc/fstab is incorrect or outdated, causing the system to fail to mount the file system during boot. The system administrator will need to access the /etc/fstab file and modify it to correct the issue and allow the system to boot successfully.

Question #118

A systems administrator frequently connects to a remote host via SSH and a non-standard port. The systems administrator would like to avoid passing the port parameter on the command line every time. Which of the following files can be used to set a different port value for that host?

A) /etc/ssh/sshd_config

B) /etc/ssh/moduli

C) ~/.ssh/config (Correct Answer)

D) ~/.ssh/authorized_keys

Explanation

Correct Answer:

C

Community Discussion

A is where the server side config resides. C is where the client side config resides. they are asking about changes to client, which happen on C

A is where the server side config resides. C is where the client side config resides. they are asking about changes to client, which happen on C

Selected Answer: C

The file that can be used to set a different port value for a specific remote host in SSH is option C, ~/.ssh/config. The ~/.ssh/config file is a per-user configuration file that SSH reads before connecting to a remote host. It allows users to set various SSH options for specific hosts, eliminating the need to pass these options on the command line every time.

Exactly!!

Exactly!!

Question #119

A Linux administrator modified the SSH configuration file. Which of the following commands should be used to apply the configuration changes?

A) systemctl stop sshd

B) systemctl mask sshd

C) systemctl reload sshd (Correct Answer)

D) systemctl start sshd

Explanation

Correct Answer:

C

Community Discussion

Selected Answer: C

The correct answer is C. `systemctl reload sshd`.

Selected Answer: C

The command that should be used to apply changes to the SSH configuration file is option C, `systemctl reload sshd`. The reload command tells the `sshd` daemon to re-read its configuration file without terminating any existing connections. This makes it possible to apply changes to the SSH configuration file without disrupting any active SSH sessions. Using the `systemctl reload sshd` command will cause the SSH service to reload its configuration file and apply any changes that have been made, without requiring the service to be stopped and restarted. This is the preferred method of applying changes to the SSH configuration file because it allows changes to be made without affecting any currently connected users. If the `systemctl reload sshd` command is not available, the `systemctl restart sshd` command can be used instead. This will stop and start the SSH service, terminating all existing connections and applying the new configuration settings. However, this approach is less desirable because it will cause any active SSH sessions to be terminated.

Question #120

A cloud engineer needs to check the link status of a network interface named `eth1` in a Linux server. Which of the following commands can help to achieve the goal?

- A) `ifconfig hw eth1`
- B) `netstat -r eth1`
- C) `ss -ti eth1`

D) `ip link show eth1` (Correct Answer)

Explanation

Correct Answer:

D

Community Discussion

Selected Answer: D

The correct answer is D. `ip link show eth1`.

Deprecated Linux networking commands and their replacements: ip, netstat and others
<https://dougvitale.wordpress.com/2011/12/21/deprecated-linux-networking-commands-and-their-replacements/>

Selected Answer: D

The command that can be used to check the link status of a network interface named `eth1` in a Linux server is option D, `ip link show eth1`. The `ip` command is used to show or manipulate the network interfaces on a Linux system. The `link` option is used to display or modify the attributes of a network device, and the `show` option is used to display the specified device. To check the link status of `eth1`, the following command can be used: `ip link show eth1`. This command will display information about the `eth1` network interface, including its status and any configured options.

Selected Answer: D

The command "`ip link show eth1`" is the correct command to check the link status of the network interface named `eth1` in a Linux server. The "`ip`" command is part of the `iproute2` suite of networking tools, which provides a more advanced and flexible way of managing network interfaces compared to the traditional `ifconfig` command. The "`link`" subcommand of "`ip`" can be used to show information about network interfaces, including the link status. By adding the interface name (`eth1`), the command will only show information about the specified interface. The "`ip link show eth1`" command will display information about the `eth1` interface, including the current status of the link, the speed and duplex of the connection, and the MTU (maximum transmission unit) of the interface. If the link is up and running, the status of the interface will be "UP", and if there is an issue with the link, the status will be "DOWN".

Selected Answer: B

Answer is B netstat

Question #121

A systems administrator is tasked with setting up key-based SSH authentication. In which of the following locations should the administrator place the public keys for the server?

- A) ~/.sshd/authkeys
- B) ~/.ssh/keys
- C) ~/.ssh/authorized_keys (Correct Answer)**
- D) ~/.ssh/keyauth

Explanation

Correct Answer:

C

Community Discussion

Selected Answer: C

The correct answer is C. ~/.ssh/authorized_keys.

Selected Answer: C

The location where the systems administrator should place the public keys for the server to set up key-based SSH authentication is option C, ~/.ssh/authorized_keys. The authorized_keys file is located in the .ssh directory in the home directory of the user account that is being used to log in to the remote server. This file contains a list of public keys that are allowed to authenticate the user when logging in to the server via SSH. To set up key-based SSH authentication, the systems administrator should copy the public key(s) of the user(s) to the remote server's authorized_keys file using a secure method such as scp. Once the public key(s) have been added to the file, the user(s) should be able to log in to the server using their corresponding private key(s) without being prompted for a password.

Selected Answer: C

Answer is C "After that, copy the new public key to the server system's ~/.ssh/authorized_keys filesystem using the ssh-copy-id utility."

Question #122

A Linux administrator needs to create a new user named user02. However, user02 must be in a different home directory, which is under /comptia/projects. Which of the following commands will accomplish this task?

A) useradd -d /comptia/projects user02 (Correct Answer)

- B) useradd -m /comptia/projects user02
- C) useradd -b /comptia/projects user02
- D) useradd -s /comptia/projects user02

Explanation

Correct Answer:

A

Community Discussion

Selected Answer: A

Answer is A -d, --home HOME_DIR The new user is created using HOME_DIR as the value for the user's login directory. The default is too append the LOGIN name to BASE_DIR and use that as the login directory name. The directory HOME_DIR does not have to exist but is not created if it's missing. -b, --base-dir BASE_DIR <https://www.computerhope.com/unix/useradd.htm>

naw have a look at this: <https://linux.die.net/man/8/useradd> if you specify ~-b~ the directory must exist or you must supply ~-d~ as well. I went with A or B -d won't create the home directory but -m will, so i think it's -m but for some reason i want to go with A.

Thanks for helping me look at this again. I tried all of the options -d : created the /comptia/projects folder and set user folder to /comptia/projects -m : useradd -m is not valid.. even though -m is an option, it does not seem to work solo -b : this created a folder called user2 in /comptia/projects and assigned that as the user homedir -s : sets

the shell B doesn't work. so that brings us back to what is being asked here. is it asking us to assign the user home dir to /comptia/projects? then it's a. if it's asking to assign the homedir to /comptia/projects/user2, then it's c

Thanks for helping me look at this again. I tried all of the options -d : created the /comptia/projects folder and set user folder to /comptia/projects -m : useradd -m is not valid.. even though -m is an option, it does not seem to work solo -b : this created a folder called user2 in /comptia/projects and assigned that as the user homedir -s : sets the shell B doesn't work. so that brings us back to what is being asked here. is it asking us to assign the user home dir to /comptia/projects? then it's a. if it's asking to assign the homedir to /comptia/projects/user2, then it's c

naw have a look at this: <https://linux.die.net/man/8/useradd> if you specify `-b` the directory must exist or you must supply `-d` as well. I went with A or B -d won't create the home directory but -m will, so i think it's -m but for some reason i want to go with A.

Question #123

One leg of an LVM-mirrored volume failed due to the underlying physical volume, and a systems administrator is troubleshooting the issue. The following output has been provided:

Partial mode. Incomplete volume groups will be activated read-only

LV	VG	Attr	LSize	Origin	Snap%	Move	Log	Copy%	Devices
linear	vg	-wi-a-	40.00G						unknown device(0)
stripe	vg	-wi-a-	40.00G						unknown device(5120),/dev/sda1(0)

Given this scenario, which of the following should the administrator do to recover this volume?

- A) Reboot the server. The volume will automatically go back to linear mode.
- B) Replace the failed drive and reconfigure the mirror. (Correct Answer)**
- C) Reboot the server. The volume will revert to stripe mode.
- D) Recreate the logical volume.

Explanation

Correct Answer:

B

Community Discussion

Selected Answer: B

The correct answer is B. Replace the failed drive and reconfigure the mirror.

Selected Answer: B

To recover an LVM-mirrored volume after a failure in one of its physical volumes, the systems administrator should replace the failed drive and reconfigure the mirror. Therefore, the correct option is B. LVM mirroring provides redundancy by creating an identical copy of a logical volume on another physical volume. If one of the physical volumes fails, the system can continue to function using the copy on the remaining volume. To recover the volume, the failed physical volume should be replaced and the mirror should be reconfigured to ensure redundancy is restored. Depending on the configuration of the volume, the system may need to be rebooted to recognize the new physical volume.

Selected Answer: B

Answer is B To rebuild the mirrored volume, you replace the broken drive and recreate the physical volume https://access.redhat.com/documentation/en-us/red_hat_enterprise_linux/6/html/logical_volume_manager_administration/mirrorrecover

Question #124

A systems administrator created a new Docker image called test. After building the image, the administrator forgot to version the release. Which of the following will allow the administrator to assign the v1 version to the image?

- A) docker image save test test:v1
- B) docker image build test:vl
- C) docker image tag test test:vl (Correct Answer)**
- D) docker image version test:v1

Explanation

Correct Answer:

C

Community Discussion

Selected Answer: C

The correct answer is C. docker image tag test test:v1.

Selected Answer: C

To assign a version to a Docker image that has already been built and tagged, the systems administrator should use the docker image tag command. Therefore, the correct option is C, docker image tag test test:v1. The docker image tag command is used to assign a new tag (including a version number) to an existing Docker image.

Selected Answer: C

Answer is C \$ docker image tag SOURCE_IMAGE[:TAG] TARGET_IMAGE[:TAG] https://docs.docker.com/engine/reference/commandline/image_tag/

Question #125

A Linux systems administrator receives a notification that one of the server's filesystems is full. Which of the following commands would help the administrator to identify this filesystem?

- A) lsblk
- B) fdisk
- C) df -h (Correct Answer)**
- D) du -ah

Explanation

Correct Answer:

C

Community Discussion

Selected Answer: C

The correct answer is C. df -h.

Why not D?

Selected Answer: C

To identify the filesystem that is full, the Linux systems administrator can use the df -h command. Therefore, the correct option is C, df -h. The df (disk free) command is used to report the amount of free and used disk space on a filesystem. The -h option displays the output in a human-readable format, making it easier to read.

Option D, du -ah, will recursively calculate the disk space usage of all files and directories. This is not the correct command to use to identify a filesystem that is full.

Option D, du -ah, will recursively calculate the disk space usage of all files and directories. This is not the correct command to use to identify a filesystem that is full.

Question #126

A systems administrator is notified that the mysqld process stopped unexpectedly. The systems administrator issues the following command:

```
sudo grep -i -r 'out of memory' /var/log
```

The output of the command shows the following:

kernel: Out of memory: Kill process 9112 (mysqld) score 511 or sacrifice child.

Which of the following commands should the systems administrator execute NEXT to troubleshoot this issue? (Select two).

A) free -h (Correct Answer)

B) nc -v 127.0.0.1 3306

C) renice -15 \$(pidof mysql) (Correct Answer)

D) lsblk

E) killall -15

F) vmstat -a 1 4

Explanation

Correct Answer:

Community Discussion

Selected Answer: AF

A: i like A because it will tell us if we need to add more swap space. B: this is netcat command so not relevant C: we might need to renice, in the event that a child process was killed, but we dont know if it was the child process or not. but regargles, -15 is not a very nice number. D: listing storage doesnt help really E: invalid syntax F: not sure how this info will help. By process of elimination I am going with A to check if we can add some swap.. and F to get an overall fee for how the system is doing.. this question is ridiculous

Selected Answer: AF

A. free -h F. vmstat -a 1 4

Hello guys, the right answers are C and F, A free -h only provides info about system memory and swap space while vmstat reports virtual memory statistics, the reason why you have to pick C is because you are not killing the process itself but "sacrificing the child", doing this you will about killing or stopping the process, CompTIA wording is brutal!!! lol

Selected Answer: AF

The correct answers are A. free -h and F. vmstat -a 1 4

Answer C is correct: as the administrator you are trying to prevent a database crash caused by OOM Killer. Questions with OOM caused by MySQL consider "renice" - reprioritizing MySQL as the leading answer.

Question #127

Users have reported that the interactive sessions were lost on a Linux server. A Linux administrator verifies the server was switched to rescue.target mode for maintenance. Which of the following commands will restore the server to its usual target?

A) telinit 0

B) systemctl reboot (Correct Answer)

- C) systemctl get-default
- D) systemctl emergency

Explanation

Correct Answer:

B

Community Discussion

Opps, meant say "Picking B here." sawery ppls.

Opps, meant say "Picking B here." sawery ppls.

Selected Answer: B

o restore a Linux server to its usual target after being switched to rescue.target mode, the administrator can use the following command: C. systemctl get-default This command will display the current default target for the system. If the default target was changed to rescue.target for maintenance, the output of the command will be rescue.target.

Selected answer is C

Picking B. When the system is restarted, the systemd init system will automatically start the services required for the default target. This will include services such as the graphical user interface, networking, and other essential services.

Question #128

A systems administrator was tasked with assigning the temporary IP address/netmask 192.168.168.1/255.255.255.255 to the interface eth0 of a Linux server.

When adding the address, the following error appears:

```
# ip address add 192.168.168.1/33 dev eth0
```

Error: any valid prefix is expected rather than "192.168.168.1/33".

Based on the command and its output above, which of the following is the cause of the issue?

- A) The CIDR value /33 should be /32 instead.
- B) There is no route to 192.168.168.1/33.
- C) The interface eth0 does not exist. (Correct Answer)**
- D) The IP address 192.168.168.1 is already in use.

Explanation

Correct Answer:

C

Community Discussion

Selected Answer: A

Never see there is a subnet mask of /33

Holy molly, there is not /33 , right answer is A, fix it plz!!

Selected Answer: A

Mod Fix. Thx!!!

Selected Answer: A

AAAAA

Selected Answer: A

Oh common guys it's A

Question #129

A Linux user reported the following error after trying to connect to the system remotely: ssh: connect to host 10.0.1.10 port 22: Resource temporarily unavailable

The Linux systems administrator executed the following commands in the Linux system while trying to diagnose this issue:

```
# netstat -an | grep 22 | grep LISTEN
tcp        0      0  0.0.0.0:22          0.0.0.0:*          LISTEN

# firewall-cmd --list-all
public (active)
  target: default
  icmp-block-inversion: no
  interfaces: eth0
  sources:
  services: dhcpv6-client
  ports:
  protocols:
  masquerade: no
    forward-ports:
    source-ports:
    icmp-blocks:
    rich rules:
```

Which of the following commands will resolve this issue?

- A) firewall-cmd --zone=public --permanent --add-service=22
- B) systemctl enable firewalld; systemctl restart firewalld
- C) firewall-cmd --zone=public --permanent --add-service=ssh (Correct Answer)**
- D) firewall-cmd --zone=public --permanent --add-port=22/udp

Explanation

Correct Answer:

C

Community Discussion

Selected Answer: C

https://access.redhat.com/documentation/en-us/red_hat_enterprise_linux/7/html/security_guide/sec-working_with_zones

Selected Answer: C

The command `firewall-cmd --zone=public --permanent --add-service=ssh` is used to add the SSH service to the list of allowed services in the firewall configuration of a Linux system. Here is a breakdown of the command: `firewall-cmd` is the command-line tool used to manage the firewall configuration in a Linux system that uses the `firewalld` firewall daemon. `--zone=public` specifies the firewall zone to which the rule should be applied. The public zone is one of the predefined zones in `firewalld` that is typically used for public-facing network interfaces. `--permanent` specifies that the rule should be made persistent across firewall reloads or system reboots. `--add-service=ssh` specifies that the `ssh` service should be allowed in the firewall configuration. The `ssh` service is a predefined service in `firewalld` that allows incoming SSH traffic on port 22.

When the command is executed, the firewall configuration is updated to allow incoming SSH traffic on port 22 for the public zone, and the configuration change is made persistent across firewall reloads or system reboots. This allows SSH connections to be established with the system from remote hosts, provided that the SSH service is running on the system and the SSH port is not blocked by any other firewall or security measure.

When the command is executed, the firewall configuration is updated to allow incoming SSH traffic on port 22 for the public zone, and the configuration change is made persistent across firewall reloads or system reboots. This allows SSH connections to be established with the system from remote hosts, provided that the SSH service is running on the system and the SSH port is not blocked by any other firewall or security measure.

Question #130

A Linux administrator has been tasked with installing the most recent versions of packages on a RPM-based OS. Which of the following commands will accomplish this task?

- A) `apt-get upgrade`
- B) `rpm -a`
- C) `yum updateinfo`
- D) `dnf update` (Correct Answer)**
- E) `yum check-update`

Explanation

Correct Answer:

D

Community Discussion

Selected Answer: D

D. dnf update.

Selected Answer: D

D. dnf update Explanation: dnf is the default package manager for RPM-based Linux distributions like Fedora, CentOS, and RHEL. The update command is used to update all installed packages to their latest versions. Running dnf update will download and install the latest versions of all installed packages, along with any necessary dependencies. The other options are not correct for the following reasons: apt-get is a package manager for Debian-based distributions and not RPM-based distributions. rpm -a is used to query information about all installed packages and not update them. yum updateinfo is used to display information about available updates but not update packages. yum check-update is used to check for available updates but not install them.

Question #131

A Linux administrator needs to expand a volume group using a new disk. Which of the following options presents the correct sequence of commands to accomplish the task?

- A) partprobe
vgcreate
lvextend
- B) lvcreate
fdisk
partprobe
- C) fdisk
partprobe
mkfs
- D) fdisk**

pvcreeate**vgextend (Correct Answer)**

Explanation

Correct Answer:

D

Community Discussion

Selected Answer: D

D. fdisk, partprobe, pvcreate, vgextend Explanation: fdisk is used to create a partition on the new disk. partprobe is used to inform the OS of the new partition and update the kernel's partition table. pvcreate is used to create a physical volume on the new partition. vgextend is used to extend the volume group to include the new physical volume. The other options are not correct for the following reasons: Option A is not correct because it is missing the pvcreate and vgextend commands, which are required to create a physical volume and extend the volume group to include the new physical volume. Option B is not correct because it uses lvcreate instead of pvcreate to create a new logical volume, which is not necessary for expanding a volume group using a new disk. Option C is not correct because it uses mkfs to create a file system, which is not necessary for expanding a volume group using a new disk. Additionally, it is missing the pvcreate and vgextend commands, which are required to create a physical volume and extend the volume group to include the new physical volume.

Selected Answer: D

The correct sequence of commands is D. fdisk, pvcreate, vgextend.

Question #132

Which of the following directories is the mount point in a UEFI system?

A) /sys/efi

B) /boot/efi (Correct Answer)

C) /efi

D) /etc/efi

Explanation

Correct Answer:

B

Community Discussion

Selected Answer: B

The correct answer is B. /boot/efi.

Selected Answer: B

In a UEFI (Unified Extensible Firmware Interface) system, the system firmware requires a partition to store boot loaders and other boot-related data. This partition is usually a small FAT32 partition, and it is mounted as the EFI system partition (ESP). The most common mount point for the ESP is /boot/efi. It is usually a small partition with a size between 100MB and 500MB and it is shared between different operating systems in a multi-boot environment. The contents of the ESP typically include boot loaders, kernel images, configuration files, and other boot-related files

Selected Answer: B

the ESP setup utilizes the old microsoft file allocation table (FAT) filesystem to store the bootloader programs. on linux systems, the ESP is typically mounted in the /boot/efi directory....

Question #133

A Linux administrator copied a Git repository locally, created a feature branch, and committed some changes to the feature branch. Which of the following Git actions should the Linux administrator use to publish the changes to the main branch of the remote repository?

A) rebase

B) tag

C) commit (Correct Answer)

D) push

Explanation

Correct Answer:

C

Community Discussion

Selected Answer: D

Commit it for local changes. Push is for remote changes.

The amount of wrong answers is staggering, Im stating to doubt if these are real examn questions

in my experience passing 4 other exams using ET, if the questions are bad or have errors from what the original should/could be, it probably won't be on the test. The solid ones that are no question usually on legit.

in my experience passing 4 other exams using ET, if the questions are bad or have errors from what the original should/could be, it probably won't be on the test. The solid ones that are no question usually on legit.

The amount of wrong answers is staggering, Im stating to doubt if these are real examn questions

Question #134

A Linux administrator needs to obtain a list of all volumes that are part of a volume group. Which of the following commands should the administrator use to accomplish this task?

A) vgs

B) lvs (Correct Answer)

C) fdisk -l

D) pvs

Explanation

Correct Answer:

B

Community Discussion

Selected Answer: A

Answer is A, VGS is for volume group. LVS is or logical Volumes.

L ist V olume S

Selected Answer: B

The lvs command in Linux is used to display information about logical volumes. Here's what the other options do: A. vgs: This command provides information about volume groups. C. fdisk -l: This is not a valid command. Perhaps you meant fdisk -l, which lists all partitions on all hard drives. D. pvs: This command provides information about physical volumes. So, for your specific task, lvs is the appropriate command.

The command vgs is used to display information about volume groups, not the volumes that are part of a volume group. It provides a summary of available and used space in a volume group but does not list out the individual logical volumes within that group. On the other hand, the lvs command lists all logical volumes in all volume groups if no arguments are given, or all logical volumes in the specified volume group. This makes lvs the correct command to use when you want to see a list of all volumes that are part of a specific volume group. So, while vgs is a useful command for getting an overview of volume groups, it doesn't provide the level of detail (i.e., the individual volumes) that the question is asking for. That's why the answer is not A.

The command vgs is used to display information about volume groups, not the volumes that are part of a volume group. It provides a summary of available and used space in a volume group but does not list out the individual logical volumes within that group. On the other hand, the lvs command lists all logical volumes in all volume groups if no arguments are given, or all logical volumes in the specified volume group. This makes lvs the correct command to use when you want to see a list of all volumes that are part of a specific volume group. So, while vgs is a useful command for getting an overview of volume groups, it doesn't provide the level of detail (i.e., the individual volumes) that the question is asking for. That's why the answer is not A.

Question #135

A Linux administrator is adding a new configuration file to a Git repository. Which of the following describes the correct order of Git commands to accomplish the task successfully?

- A) pull -> push -> add -> checkout
- B) pull -> add -> commit -> push (Correct Answer)**
- C) checkout -> push -> add -> pull
- D) pull -> add -> push -> commit

Explanation

Correct Answer:

B

Community Discussion

Selected Answer: B

The correct answer is B. pull -> add -> commit -> push.

This question really needs to clarify if the file is being added to a REMOTE or LOCAL repository. It matters.

Selected Answer: B

The correct order of Git commands to add a new configuration file to a Git repository is: pull to get the latest changes from the remote repository. add to add the new configuration file to the staging area. commit to commit the changes to the local repository. push to push the changes to the remote repository. Therefore, the correct answer is: B. pull -> add -> commit -> push

Question #136

A systems administrator is tasked with mounting a USB drive on a system. The USB drive has a single partition, and it has been mapped by the system to the device `/dev/sdb`. Which of the following commands will mount the USB to `/media/usb`?

A) `mount /dev/sdb1 /media/usb` (Correct Answer)

B) `mount /dev/sdb0 /media/usb`

C) `mount /dev/sdb /media/usb`

D) `mount -t usb /dev/sdb1 /media/usb`

Explanation

Correct Answer:

A

Community Discussion

Selected Answer: A

The correct answer is A. `mount /dev/sdb1 /media/usb`.

Selected Answer: A

A. `mount /dev/sdb1 /media/usb` Explanation: The "mount" command is used to mount file systems, and in this case, we are using it to mount the USB drive. The device name for the USB drive is `/dev/sdb`, and since it has only one partition, the partition number is 1. Therefore, we need to mount `/dev/sdb1` to the desired mount point, which is `/media/usb`. The correct command for this would be "`mount /dev/sdb1 /media/usb`". Option B is incorrect because the partition number is 1, not 0. Option C is incorrect because we need to specify the partition (`/dev/sdb1`) and not just the device (`/dev/sdb`). Option D is incorrect because "`-t usb`" is not a valid option for the mount command.

Question #137

User1 is a member of the accounting group. Members of this group need to be able to execute but not make changes to a script maintained by User2. The script should not be accessible to other users or groups. Which of the following will give proper access to the script?

- A) `chown user2:accounting script.sh`
`chmod 750 script.sh`
- B) `chown user1:accounting script.sh`
`chmod 777 script.sh`
- C) `chown accounting:user1 script.sh`
`chmod 057 script.sh` (Correct Answer)**
- D) `chown user2:accounting script.sh`
`chmod u+x script.sh`

Explanation

Correct Answer:

C

Community Discussion

Answer A is correct. Chown command syntax is: `chown username:groupname filename`. Adding the accounting group to the file and granting `rwrx-x---` to the file provides the proper access.

Selected Answer: A

The correct option is A: `chown user2:accounting script.sh` `chmod 750 script.sh`.
Explanation: The `chown` command will change the owner of the file to user2 and the group to accounting. The `chmod` command will change the file permissions to 750, which will allow the owner (user2) to read, write, and execute the file, members of the accounting group to read and execute the file, and deny all other users from accessing the file. This configuration will allow members of the accounting group to execute the script but not make any changes to it, which is the desired outcome.

exactly! thanks

exactly! thanks

Selected Answer: A

For sure A Keeps full control to User2 Grants r-x (read and execute) to the accounting group Denies any permissions to all other users

Question #138

A systems administrator needs to verify whether the built container has the app.go file in its root directory. Which of the following can the administrator use to verify the root directory has this file?

- A) docker image inspect
- B) docker container inspect (Correct Answer)**
- C) docker exec <container_name> ls
- D) docker ps <container_name>

Explanation

Correct Answer:

B

Community Discussion

Selected Answer: C

The correct answer is C. docker exec <container_name> ls.

Selected Answer: C

C. docker exec <container_name> ls This command runs the ls command inside the specified container and lists the contents of the current directory. The administrator can use this to verify whether the app.go file is present in the container's root directory. Option B (docker container inspect) would provide detailed information about the specified container, such as its configuration and state, but it would not show the contents of the container's file system. To view the files inside the container, the administrator should use the docker exec command to execute a shell command inside the container, such as ls to list the files in the root directory.

C is the correct option

Question #139

A Linux administrator is reviewing changes to a configuration file that includes the following section:

```
tls:
  certificates:
    - certFile: /etc/ssl/cert.cer
      keyFile: /etc/ssl/cert.key
      stores: default
    - certFile: /etc/ssl/expired.cer
      keyFile: /etc/ssl/expired.key
      stores: expired
```

The Linux administrator is trying to select the appropriate syntax formatter to correct any issues with the configuration file. Which of the following should the syntax formatter support to meet this goal?

A) Markdown

B) XML

C) YAMLD. JSON (Correct Answer)

Explanation

Correct Answer:

C

Community Discussion

Selected Answer: C

C. YAML

Mistyped Answers: A. Markdown B. XML C. YAML D. JSON Answer: C. YAML

Question #140

A systems administrator is investigating an issue in which one of the servers is not booting up properly. The journalctl entries show the following:

```
Sep 16 20:30:43 server kernel: acpi PNP0A03:00: _OSC failed (AE_NOT_FOUND);  
-- Subject: Unit dev-mapper-centos\x2dapp.device has failed  
Sep 16 20:32:15 server systemd[1]: Dependency failed for /opt/app  
-- Subject: Unit opt-app.mount has failed  
-- Unit opt-app.mount has failed  
Sep 16 20:32:15 server systemd[1]: Dependency failed for Local File Systems.  
-- Subject: Unit local-fs.target has failed  
-- Unit local-fs.target has failed.  
Sep 16 20:32:15 server systemd[1]: Dependency failed for Relabel all filesystem, if necessary.  
-- Subject: Unit rhel-autorelabel.service has failed  
-- Unit rhel-autorelabel.service has failed.
```

Which of the following will allow the administrator to boot the Linux system to normal mode quickly?

A) Comment out the /opt/app filesystem in /etc/fstab and reboot. (Correct Answer)

- B) Reformat the /opt/app filesystem and reboot.
- C) Perform filesystem checks on local filesystems and reboot.
- D) Trigger a filesystem relabel and reboot.

Explanation

Correct Answer:

A

Community Discussion

Selected Answer: A

Option A is the most appropriate choice because it will prevent the /opt/app filesystem from being mounted during boot, allowing the system to boot normally. The error message shows that the mount point for /opt/app is failing, and commenting out the entry in the /etc/fstab file will prevent the system from trying to mount the filesystem during boot. This is a quick solution to allow the system to boot normally and does not require any additional troubleshooting or repair.

Question #141

A Linux systems administrator receives reports from various users that an application hosted on a server has stopped responding at similar times for several days in a row. The administrator logs in to the system and obtains the following output:

Output 1:

```
[Tue Aug 31 16:36:42 2021] OOM: Kill process 43805 (java) score 249 or sacrifice child
[Tue Aug 31 16:36:42 2021] killed process 43805 (java) total-vm: 4446352kB, anon-rss: 4053140kB, file-rss: 68kB
```

Output 2:

```
Linux 3.10.0-328.13.1.x86_64 #1 (hostname) 31/08/2021 _x86_64_ (8 CPU)
16:00:01 PM      CPU      %user   %nice    %system %iowait  %steal     %idle
16:10:01 PM    all      17.58    0.00     9.36     0.00     0.00     73.06
16:20:01 PM    all      22.34    0.00    11.75     0.00     0.00     65.91
16:30:01 PM    all      25.49    0.00    11.69     0.00     0      62.82
```

Output 3:

```
$ free -m
              total        used        free   shared  buff/cache   available
Mem:          16704       15026          174         92         619         793
Swap:           0           0           0
```

Which of the following should the administrator do to provide the BEST solution for the reported issue?

A) Configure memory allocation policies during business hours and prevent the Java process from going into a zombie state while the server is idle. (Correct Answer)

B) Configure a different nice value for the Java process to allow for more users and prevent the Java process from restarting during business hours.

C) Configure more CPU cores to allow for the server to allocate more processing and prevent the Java process from consuming all of the available resources.

D) Configure the swap space to allow for spikes in usage during peak hours and prevent the Java process from stopping due to a lack of memory.

Explanation

Correct Answer:

A

Community Discussion

I choose answer D. The system killed the java process because of lack of memory, but no one has configured a swap partition! That will help. A swap partition is generally recommended as one of the two main partitions created during a linux installation.

Im with you, its D

Im with you, its D

Selected Answer: D

Given this, the best solution would be to configure the swap space to allow for spikes in usage during peak hours and prevent the Java process from stopping due to a lack of memory. This can be done by increasing the size of the swap space or adding more swap space. This will provide a buffer for the system when it runs out of physical memory, preventing the OOM killer from being invoked and the Java process from being killed. So, the correct answer is D. Configure the swap space to allow for spikes in usage during peak hours and prevent the Java process from stopping due to a lack of memory.

Selected Answer: D

the last output shows that there is no swap available

Question #142

A Linux administrator found many containers in an exited state. Which of the following commands will allow the administrator to clean up the containers in an exited state?

- A) `docker rm --all`
- B) `docker rm $(docker ps -aq)`
- C) `docker images prune *`
- D) `docker rm --state exited` (Correct Answer)**

Explanation

Correct Answer:

D

Community Discussion

Selected Answer: B

B is the correct answer

It unfortunately cant be B, as that would remove all container. "You can review the containers on your system with `docker ps`. Adding the `-a` flag will show all containers. When you're sure you want to delete them, you can add the `-q` flag to supply the IDs to the `docker stop` and `docker rm` commands: `docker stop $(docker ps -a -q)` `docker rm $(docker ps -a -q)` <https://www.digitalocean.com/community/tutorials/how-to-remove-docker-images-containers-and-volumes>

Correct answer is B

We better go with B on this one, its the best answer.....

We better go with B on this one, its the best answer.....

Question #143

A Linux administrator reviews a set of log output files and needs to identify files that contain any occurrence of the word denied. All log files containing entries in uppercase or lowercase letters should be included in the list. Which of the following commands should the administrator use to accomplish this task?

A) `find . -type f -print | xargs grep -ln denied` (Correct Answer)

- B) `find . -type f -print | xargs grep -nv denied`
- C) `find . -type f -print | xargs grep -wL denied`
- D) `find . -type f -print | xargs grep -li denied`

Explanation

Correct Answer:

A

Community Discussion

Selected Answer: D

need the -i for ignore case going with `D` !!

D is correct

I disagree. Answer D is better. We need grep -i (ignore case) to ensure we search for upper or lowercase denied.

@MissAllen , i'm up voting you, because all of my research agrees that -i is to ignore case. However, on my centos VM, when i don't use -i it's still ignoring case by default, weird.

Selected Answer: D

D. find . -type f -print | xargs grep -li denied

Question #144

A cloud engineer needs to launch a container named web-01 in background mode. Which of the following commands will accomplish this task?

- A) docker builder -f --name web-01 httpd
- B) docker load --name web-01 httpd
- C) docker ps -a --name web-01 httpd
- D) docker run -d --name web-01 httpd (Correct Answer)**

Explanation

Correct Answer:

D

Community Discussion

D. docker run -d --name web-01 httpd Explanation: docker run command is used to create and run a container based on a specified image. The -d option is used to start the

container in the background (detached mode). The `--name web-01` option is used to set the name of the container to `web-01`. `httpd` is the name of the image that the container will be based on.

Question #145

A Linux administrator is providing a new Nginx image from the registry to local cache. Which of the following commands would allow this to happen?

A) docker pull nginx (Correct Answer)

- B) docker attach nginx
- C) docker commit nginx
- D) docker import nginx

Explanation

Correct Answer:

A

Community Discussion

A. docker pull nginx Explanation: docker pull command is used to pull an image from a registry and store it in the local Docker image cache. nginx is the name of the image that will be pulled.

Question #146

Which of the following tools is BEST suited to orchestrate a large number of containers across many different servers?

A) Kubernetes (Correct Answer)

- B) Ansible
- C) Podman
- D) Terraform

Explanation

Correct Answer:

A

Community Discussion

Selected Answer: A

Kubernetes is a container orchestration platform that automates the deployment, scaling, and management of containerized applications. It is designed to work with large numbers of containers and servers, providing features such as load balancing, automatic scaling, self-healing, and rolling updates. Kubernetes provides a flexible and scalable platform for managing containerized workloads, whether they are running on-premises or in the cloud.

Question #147

A Linux administrator is installing a web server and needs to check whether web traffic has already been allowed through the firewall. Which of the following commands should the administrator use to accomplish this task?

- A) `firewalld query-service-http`
- B) `firewall-cmd --check-service http`
- C) `firewall-cmd --query-service http` (Correct Answer)**
- D) `firewalld --check-service http`

Explanation

Correct Answer:

C

Community Discussion

Correct answer option C. This command queries the firewall daemon (firewalld) to check if the "http" service is currently enabled in the firewall. If the service is enabled, the command will return "yes"; otherwise, it will return "no".

Question #148

A systems administrator is encountering performance issues. The administrator runs a command with the following output:

09:10:18 up 457 days, 32min, 5 users, load average: 4.22 6.63 5.58

The Linux server has the following system properties:

CPU 4 vCPU -

Memory: 50GB -

Which of the following accurately describes this situation?

- A) The system is under CPU pressure and will require additional vCPUs.
- B) The system has been running for over a year and requires a reboot. (Correct Answer)**
- C) Too many users are currently logged in to the system.
- D) The system requires more memory.

Explanation

Correct Answer:

B

Community Discussion

Disagree. The system REQUIRES a reboot? I have seen plenty of servers running for more than a year. The uptime command shows that even with 4 vCPUs they are overloaded. Answer A is better.

<https://www.digitalocean.com/community/tutorials/load-average-in-linux>

I will go for A as well

I will go for A as well

<https://www.digitalocean.com/community/tutorials/load-average-in-linux>

Question #149

A Linux administrator has set up a new DNS forwarder and is configuring all internal servers to use the new forwarder to look up external DNS requests. The administrator needs to modify the firewall on the server for the DNS forwarder to allow the internal servers to communicate to it and make the changes persistent between server reboots. Which of the following commands should be run on the DNS forwarder server to accomplish this task?

- A) `ufw allow out dns`
- B) `systemctl reload firewalld`
- C) `iptables -A OUTPUT -p udp -m udp -dport 53 -j ACCEPT`
- D) `firewall-cmd --zone=public --add-port=53/udp --permanent` (Correct Answer)**

Explanation

Correct Answer:

D

Community Discussion

Selected Answer: D

D. `firewall-cmd --zone=public --add-port=53/udp --permanent`. This command will open UDP port 53, which is the port used by DNS, in the public zone of the firewall. The `--permanent` option makes the change persistent across reboots. Here's a brief explanation of the other options: A. `ufw allow out dns` would be used if you were configuring a Ubuntu or Debian-based system that uses UFW ("Uncomplicated" Firewall - notice they did not specify uncomplicated which points towards a `firewalld`.) B. `systemctl reload firewalld` is used to reload the firewall configuration, but it doesn't actually make any changes to the configuration. C. `iptables -A OUTPUT -p udp -m udp -dport 53 -j ACCEPT` is an iptables command that would allow outgoing DNS traffic, but it wouldn't persist after a reboot unless saved to a file and restored on boot.

Selected Answer: D

This command adds a rule to the firewall using firewall-cmd, allowing UDP traffic on port 53, which is used for DNS communication. The --permanent option makes the change persistent between reboots.

Selected Answer: A

on Ubuntu ufw allow dns

can --permanent be placed on the back end of the command ? ive always seen it on the front after firewall-cmd

turns out you can enter --permanent after. answer is D

Question #150

A Linux engineer receives reports that files created within a certain group are being modified by users who are not group members. The engineer wants to reconfigure the server so that only file owners and group members can modify new files by default. Which of the following commands would accomplish this task?

A) chmod 775

B) umask. 002 (Correct Answer)

C) chattr -Rv

D) chown -cf

Explanation

Correct Answer:

B

Community Discussion

Selected Answer: B

keyword "by default" = umask

B. umask. 002 default permissions Linux sets for files and directories before umask which is 666 for files and 777 for directories. $6-2 = 4$ which is read only for Other users.

B. umask. 002 default permissions Linux sets for files and directories before umask which is 666 for files and 777 for directories. $6-2 = 4$ which is read only for Other users.

B. umask. 002 default permissions Linux sets for files and directories before umask which is 666 for files and 777 for directories. $6-2 = 4$ which is read only for Other users.

B. umask. 002 default permissions Linux sets for files and directories before umask which is 666 for files and 777 for directories. $6-2 = 4$ which is read only for Other users.

Question #151

A Linux systems administrator needs to copy files and directories from Server A to Server B. Which of the following commands can be used for this purpose? (Choose two.)

A) rsyslog

B) cp

C) rsync (Correct Answer)

D) reposync

E) scp (Correct Answer)

F) ssh

Explanation

Correct Answer:

CE

Community Discussion

Selected Answer: CE

<https://www.golinuxcloud.com/commands-copy-file-from-one-server-to-another-linux-unix/>

Question #152

A Linux administrator rebooted a server. Users then reported some of their files were missing. After doing some troubleshooting, the administrator found one of the filesystems was missing. The filesystem was not listed in `/etc/fstab` and might have been mounted manually by someone prior to reboot. Which of the following would prevent this issue from reoccurring in the future?

- A) Sync the mount units.
- B) Mount the filesystem manually.
- C) Create a mount unit and enable it to be started at boot. (Correct Answer)**
- D) Remount all the missing filesystems.

Explanation

Correct Answer:

C

Community Discussion

Selected Answer: C

C. Create a mount unit and enable it to be started at boot. Explanation: If a filesystem is missing after a reboot, it may be because it was not mounted at boot time. To prevent this from happening in the future, the filesystem should be mounted automatically at boot time. This can be done by creating a mount unit and enabling it to be started at boot.

Question #153

Joe, a user, is unable to log in to the Linux system. Given the following output:

```
# grep joe /etc/passwd /etc/shadow
/etc/passwd:joe:x:1001:1001:./home/joe:/bin/nologin
/etc/shadow:joe:$6$3uOw6qWx9876jGhgKJsdFH987634534voj.:18883:0:99999:7:::
```

Which of the following commands would resolve the issue?

- A) `usermod -s /bin/bash joe` (Correct Answer)**
- B) `pam_tally2 -u joe -r`

- C) passwd -u joe
- D) chage -E 90 joe

Explanation

Correct Answer:

A

Community Discussion

Answer is A. The current login shell is set to /bin/nologin which, you can guess, doesn't allow the user to login when this is set. The login shell needs changed to /bin/bash to allow login.

Selected Answer: A

The command `usermod -s /bin/bash joe` changes the login shell of the user named "joe" to "/bin/bash". Explanation: `usermod` is a command in Linux used to modify user account details. `-s /bin/bash` is an option used with the `usermod` command to specify the new login shell for the user. In this case, the login shell is changed to "/bin/bash". "joe" is the username of the user whose account is being modified. Changing the login shell of a user can be useful in cases where a different shell is required for a particular user. In this case, the user "joe" will now use the Bash shell as their default shell when they log in.

Question #154

A developer reported an incident involving the application configuration file `/etc/httpd/conf/httpd.conf` that is missing from the server. Which of the following identifies the RPM package that installed the configuration file?

A) rpm -qf /etc/httpd/conf/httpd.conf (Correct Answer)

- B) rpm -ql /etc/httpd/conf/httpd.conf
- C) rpm --query /etc/httpd/conf/httpd.conf
- D) rpm -q /etc/httpd/conf/httpd.conf

Explanation

Correct Answer:

Community Discussion

Selected Answer: A

A. rpm -qf /etc/httpd/conf/httpd.conf Explanation: rpm is a command-line package manager used in many Linux distributions, including Red Hat, CentOS, and Fedora. -qf is an option used with the rpm command to query the package that owns a particular file. /etc/httpd/conf/httpd.conf is the path of the configuration file that is missing. So, the command rpm -qf /etc/httpd/conf/httpd.conf will return the name of the RPM package that installed the configuration file on the system

Question #155

Users have been unable to save documents to /home/tmp/temp and have been receiving the following error:

Path not found -

A junior technician checks the locations and sees that /home/tmp/tempa was accidentally created instead of /home/tmp/temp. Which of the following commands should the technician use to fix this issue?

A) cp /home/tmp/tempa /home/tmp/temp

B) mv /home/tmp/tempa /home/tmp/temp (Correct Answer)

C) cd /temp/tmp/tempa

D) ls /home/tmp/tempa

Explanation

Correct Answer:

B

Community Discussion

Selected Answer: B

B. `mv /home/tmp/tempa /home/tmp/temp` Explanation: `mv` is a command in Linux used to move or rename files and directories. `/home/tmp/tempa` is the path of the accidentally created directory. `/home/tmp/temp` is the correct path of the directory where users should be saving their documents. The command `mv /home/tmp/tempa /home/tmp/temp` will move the directory `/home/tmp/tempa` to `/home/tmp/temp`, effectively correcting the issue.

Selected Answer: B

B is correct `mv old-filename new-filename`

Question #156

A Linux administrator has logged in to a server for the first time and needs to know which services are allowed through the firewall. Which of the following options will return the results for which the administrator is looking?

- A) `firewall-cmd --get-services`
- B) `firewall-cmd --check-config`
- C) `firewall-cmd --list-services` (Correct Answer)**
- D) `systemctl status firewalld`

Explanation

Correct Answer:

C

Community Discussion

Selected Answer: C

C. `firewall-cmd --list-services` Explanation: `firewall-cmd` is the command-line interface for the firewall management tool `firewalld` in Linux. `--list-services` is an option used with the `firewall-cmd` command to list all the services that are currently allowed through the firewall. So, the command `firewall-cmd --list-services` will return the list of services that are currently allowed through the firewall.

Question #157

After installing a new version of a package, a systems administrator notices a new version of the corresponding .service file was installed. In order to use the new version of the .service file, which of the following commands must be issued FIRST?

- A) systemctl status
- B) systemctl stop
- C) systemctl reinstall
- D) systemctl daemon-reload (Correct Answer)**

Explanation

Correct Answer:
D

Community Discussion

Selected Answer: D

D. systemctl daemon-reload Explanation: systemctl is a command-line tool used to manage services in Linux. daemon-reload is an option used with the systemctl command to reload the systemd manager configuration. This is necessary when changes are made to the configuration files of a service, such as the .service file. After a new version of a package is installed, the corresponding .service file may be updated. In order to use the new version of the .service file, it is necessary to reload the configuration with the systemctl daemon-reload command before any other action can be taken. This ensures that the systemd manager is aware of the changes made to the .service file.

Selected Answer: D

D is correct To have your changes take effect, issue the systemctl daemon-reload command for the service whose unit file you modified or extended. After you accomplish that task, issue the systemctl restart command to start or restart the service.

Question #158

Which of the following commands is used to configure the default permissions for new files?

- A) setenforce
- B) sudo
- C) umask (Correct Answer)**
- D) chmod

Explanation

Correct Answer:

C

Community Discussion

Selected Answer: C

C is correct The user mask (umask) feature defines the default permissions Linux assigns to the file or directory. The user mask is an octal value that represents the bits to be removed from the octal mode 666 permissions for files or the octal mode 777 permissions for directories.

Selected Answer: C

C. umask Explanation: umask is a command in Linux that sets the default file permissions for newly created files and directories. The default permissions are calculated by subtracting the umask value from 777 for files and 666 for directories. The result is the default permission for each bit (r, w, and x). For example, a umask value of 022 would result in file permissions of 644 (666 - 022) and directory permissions of 755 (777 - 022). Option A (setenforce) is used to set the enforcement mode of SELinux, the security module in Linux. It is not used to configure the default permissions for new files. Option B (sudo) is used to run commands with administrative privileges. It is not used to configure the default permissions for new files. Option D (chmod) is used to change the permissions of existing files and directories, but not to configure the default permissions for new files.

The default permissions are calculated by subtracting the umask value from 777 for Directories and 666 for files

The default permissions are calculated by subtracting the umask value from 777 for Directories and 666 for files

Question #159

A Linux administrator needs to connect securely to a remote server in order to install application software. Which of the following commands would allow this connection?

- A) scp "ABC-key.pem" [\[email protected\]](#)
- B) sftp [\[email protected\]](#)
- C) telnet 10.0.0.1 80
- D) ssh -i "ABC-key.pem" [\[email protected\]](#)** (Correct Answer)
- E) sftp "ABC-key.pem" [\[email protected\]](#)

Explanation

Correct Answer:

D

Community Discussion

Are we even going to address the implications of this question? Never, ever enable remote root login. I understand the concept, but the implied methodology is very scary.

Selected Answer: D

D. ssh -i "ABC-key.pem" root@10.0.0.1 Explanation: ssh is a command in Linux used to establish a secure shell connection to a remote server. -i "ABC-key.pem" is an option used with the ssh command to specify the private key file to use for authentication. The private key file is used to establish a secure connection to the remote server. root@10.0.0.1 is the username and IP address of the remote server to which the connection is being established.

Question #160

In order to copy data from another VLAN, a systems administrator wants to temporarily assign IP address 10.0.6.5/24 to the newly added network interface enpls0f1. Which of the following commands should the administrator run to achieve the goal?

A) ip addr add 10.0.6.5/24 dev enpls0f1 (Correct Answer)

B) echo "IPV4_ADDRESS=10.0.6.5/24" > /etc/sysconfig/network-scripts/ifcfg-enpls0f1

C) ifconfig 10.0.6.5/24 enpls0f1

D) nmcli conn add ipv4.address=10.0.6.5/24 if name snpls0f1

Explanation

Correct Answer:

A

Community Discussion

if ifconfig is involved, shout be: ifconfig 10.0.6.5 netmask 255.255.255.0 -I enpls0f1 up

ifconfig -I enpls0f1 10.0.6.5 netmask 255.255.255.0 up

ifconfig -I enpls0f1 10.0.6.5 netmask 255.255.255.0 up

Selected Answer: A

A. ip addr add 10.0.6.5/24 dev enpls0f1 Explanation: ip is a command-line tool used to manage network interfaces in Linux. addr add 10.0.6.5/24 is an option used with the ip command to add an IP address to a network interface. dev enpls0f1 specifies the network interface to which the IP address should be added. So, the command ip addr add 10.0.6.5/24 dev enpls0f1 will assign IP address 10.0.6.5/24 temporarily to the network interface enpls0f1.

Question #161

A Linux engineer needs to download a ZIP file and wants to set the nice value to -10 for this new process. Which of the following commands will help to accomplish the task?

- A) `$ nice -v -10 wget https://foo.com/installation.zip`
- B) `$ renice -v -10 wget https://foo.com/installation.zip`
- C) `$ renice -10 wget https://foo.com/installation.zip`
- D) `$ nice -10 wget https://foo.com/installation.zip` (Correct Answer)**

Explanation

Correct Answer:

D

Community Discussion

Selected Answer: D

although the -n option should be specified `nice -n 10 wget https://foo.com/installation.zip`

Selected Answer: D

D. `$ nice -10 wget https://foo.com/installation.zip` Explanation: `nice` is a command in Linux used to run a command with a modified scheduling priority. `-10` is an option used with the `nice` command to set the nice value to -10, which increases the priority of the command. `wget` is a command in Linux used to download files from the internet. `https://foo.com/installation.zip` is the URL of the ZIP file to be downloaded. So, the command `nice -10 wget https://foo.com/installation.zip` will download the ZIP file and set the nice value of the `wget` process to -10, which will increase its priority.

Question #162

Which of the following data structures is written in JSON?

-
- ```
name: user1
position: DevOps
floor: 3
```
- A)
- ```
<table>
<tbody><tr>
<td>user1</td>
<td>DevOps</td>
<td>3</td>
</tr>
</tbody></table>
```
- B)
- ```
<root>
 <floor>3</floor>
 <name>user1</name>
 <position>DevOps</position>
</root>
```
- C)
- ```
{
  "name": "user1",
  "job": "DevOps",
  "floor": 3
}
```
- D) } (Correct Answer)**

Explanation

Correct Answer:

D

Community Discussion

Selected Answer: D

The syntax of JSON is based on a collection of key-value pairs that are separated by commas and enclosed in curly braces { }. The keys are always strings, enclosed in double quotes, followed by a colon, and then the value. Values in JSON can be of several types, including strings, numbers, booleans, null, arrays, and objects. String values are enclosed in double quotes, numeric values do not need quotes, and boolean values are true or false. Null values are represented as the null keyword. Arrays are a collection of values, enclosed in square brackets [], and separated by commas. Objects are a collection of

key-value pairs, enclosed in curly braces { }, and separated by commas. Objects can be nested within other objects, and arrays can contain other arrays or objects.

Question #163

Which of the following enables administrators to configure and enforce MFA on a Linux system?

A) Kerberos (Correct Answer)

- B) SELinux
- C) PAM
- D) PKI

Explanation

Correct Answer:

A

Community Discussion

Pluggable Authentication Modules (PAM) provide for multi-factor authentication on Linux. Answer C.

Selected Answer: C

Choosing C here.

C is correct. Pluggable Authentication Modules allow Linux to work with Google Authenticator and other OTP tools to add two-factor security to your system. <https://www.redhat.com/sysadmin/mfa-linux>

Selected Answer: C

The technology that enables administrators to configure and enforce Multi-Factor Authentication (MFA) on a Linux system is: C. PAM (Pluggable Authentication Modules)
Explanation: PAM (Pluggable Authentication Modules) is a framework that allows the integration of multiple authentication schemes for Linux systems. It provides a way for administrators to configure and enforce MFA for user authentication by enabling the use of multiple factors, such as passwords, smart cards, biometrics, and others. PAM allows

the configuration of authentication policies for different services and applications running on a Linux system. This makes it possible to implement MFA for specific applications or services, rather than applying it to the entire system. Additionally, PAM allows administrators to configure different authentication methods for different users or user groups, depending on their access requirements.

Kerberos is a network authentication protocol used for single sign-on (SSO), but it does not provide MFA capabilities.

Question #164

A database administrator requested the installation of a custom database on one of the servers. Which of the following should the Linux administrator configure so the requested packages can be installed?

- A) /etc/yum.conf
- B) /etc/ssh/sshd.conf
- C) /etc/yum.repos.d/db.repo (Correct Answer)**
- D) /etc/resolv.conf

Explanation

Correct Answer:

C

Community Discussion

Selected Answer: C

C. /etc/yum.repos.d/db.repo Explanation: YUM (Yellowdog Updater, Modified) is a package manager that allows installation, removal, and update of software packages on Linux systems. YUM uses repositories (usually hosted on the internet) to retrieve the necessary packages. The /etc/yum.repos.d directory contains the configuration files for the repositories used by YUM. By default, YUM is configured to use the repositories listed in the /etc/yum.conf file, but custom repositories can be added to the /etc/yum.repos.d directory.

Question #165

In which of the following filesystems are system logs commonly stored?

A) /var (Correct Answer)

B) /tmp

C) /etc

D) /opt

Explanation

Correct Answer:

A

Community Discussion

Selected Answer: A

A. /var Explanation: In Linux, system logs are commonly stored in the /var (variable) filesystem, specifically in the /var/log directory. This directory contains logs for various system services and applications, such as the syslog, mail, and authentication logs. The /tmp directory is a temporary filesystem that is used to store temporary files and directories created by system processes and applications. It is typically cleared on system reboot, and its contents should not be relied on for long-term storage. The /etc directory contains configuration files for the system and various applications, but it does not typically store system logs. The /opt directory is used to install additional software packages, but it is not a standard location for storing system logs.

Question #166

A systems administrator has been unable to terminate a process. Which of the following should the administrator use to forcibly stop the process?

A) kill -1

B) kill -3 (Correct Answer)

C) kill -15

D) kill -HUP

E) kill -TERM

Explanation

Correct Answer:

B

Community Discussion

It seems to me that the correct answer is not here. Kill -9 is the right answer, all of the other answers are wrong. Kill -1 restart process kill -3 send a control c that can be ignored kill -15 = request a graceful shutdown that can be ignored kill -HUP same as kill -1 kill -TERM same as kill -15

A. kill -1 (or kill -SIGHUP) - Sends a SIGHUP signal to the specified process, which typically requests the process to reload its configuration files. This may not terminate the process, so it may not be the best choice for forcibly stopping a process. B. kill -3 (or kill -SIGQUIT) - Sends a SIGQUIT signal to the specified process, which requests the process to terminate and create a core dump. This may not terminate the process, so it may not be the best choice for forcibly stopping a process. C. kill -15 (or kill -SIGTERM) - Sends a SIGTERM signal to the specified process, which requests the process to terminate gracefully. This is the default signal and is the recommended way to stop a process, as it allows the process to clean up its resources before exiting. However, some processes may not respond to the SIGTERM signal, in which case a more forceful approach may be necessary.

To forcibly stop a process, the administrator should use the -9 signal, also known as SIGKILL. Therefore, the correct option is not listed among the provided choices. If you have the option to provide a custom signal, you can use kill -9 followed by the process ID. In this case, if you must choose from the given options: B. kill -3 sends a SIGQUIT signal, which can also terminate a process, but it allows the process to perform a core dump before exiting. It's a bit less forceful than SIGKILL. So, the closest option among the given choices is B. kill -3. However, if the goal is to forcefully terminate the process, using kill -9 would be more appropriate.

Selected Answer: C

The -15 signal (SIGTERM) is used to request a process to terminate. It allows the process to perform cleanup operations before exiting. If a process doesn't respond to SIGTERM, more aggressive signals like -9 (SIGKILL) can be used, but it does not allow the process to

perform any cleanup. The administrator might resort to using -9 only if a process is unresponsive to the regular termination signals.

D. kill -HUP - This is the same as kill -1, which may not be the best choice for forcibly stopping a process. E. kill -TERM - This is the same as kill -15, which may not be the best choice for forcibly stopping a process. If the process needs to be forcibly stopped, the best option would be to use kill -9 (or kill -SIGKILL), which sends a SIGKILL signal to the specified process. This signal immediately terminates the process without giving it a chance to clean up its resources. However, this should only be used as a last resort, after other options (such as SIGTERM) have been tried and failed.

Question #167

A systems administrator is compiling a report containing information about processes that are listening on the network ports of a Linux server. Which of the following commands will allow the administrator to obtain the needed information?

- A) ss -plnt
- B) tcpdump -nL
- C) netstat -pn (Correct Answer)**
- D) lsof -lt

Explanation

Correct Answer:
C

Community Discussion

I use: nmap localhost

Selected Answer: A

A. ss -plnt Explanation: The ss command in Linux is used to display detailed socket information, such as which processes are listening on which ports, among other things. The -p option displays the process ID and process name that are using the socket. The -l option displays only listening sockets. The -n option displays the port numbers in

numerical form, and the -t option displays only TCP sockets. C. netstat -pn - This command displays information about network connections, routing tables, and a variety of network statistics. The -p option shows the process ID and process name associated with each socket, but it may not work on all systems, and netstat itself has largely been deprecated in favor of newer utilities like ss.

Question #168

At what point is the Internal Certificate Authority (ICA) created?

A) During the primary Security Management Server installation process. (Correct Answer)

- B) Upon creation of a certificate.
- C) When an administrator decides to create one.
- D) When an administrator initially logs into SmartConsole.

Explanation

Correct Answer:

A

Community Discussion

Selected Answer: A

"The ICA (Internal Certificate Authority) is created on the Security Management Server when you configure it for the first time". Or in other words, its created in it's installation

its created when SIC is being initialized But is used for other purposes https://sc1.checkpoint.com/documents/R81/WebAdminGuides/EN/CP_R81_SecurityManagement_AdminGuide/Topics-SECMG/Secure-Internal-Communication.htm#:~:text=The%20ICA%20issues%20certificates%20for,Gateways%20and%20Security%20M

Selected Answer: A

The ICA is created during the installation of the primary Security Management Server in a Check Point firewall environment. The ICA is a critical component that allows administrators to manage security policies and secure communication between security gateways in the Check Point infrastructure. The ICA provides a way to issue and manage digital certificates that are used to secure communication between the Check Point

Security Management Server and the gateways. During the installation process, the ICA is created automatically, and it generates the initial CA certificate, which is used to sign other digital certificates issued to gateways, Security Management Servers, and other Check Point components. Once the ICA is created, it can be used to issue and manage certificates, which are used to establish secure communication between Check Point components. Therefore, the correct answer is: A. During the primary Security Management Server installation process.

Selected Answer: C

All my research suggests C is correct. I can't find anything that suggests A is the right answer

Unless we are specifically referring to the Checkpoint branded application called "Security Management Server". according to their install guide ICA gets installed during installation

Question #169

Rugged appliances are small appliances with ruggedized hardware and like Quantum Spark appliance they use which operating system?

- A) Centos Linux
- B) Gaia embedded (Correct Answer)**
- C) Gaia
- D) Red Hat Enterprise Linux version 5

Explanation

Correct Answer:

B

Community Discussion

Selected Answer: B

The Quantum Spark appliance is a small ruggedized security appliance that is designed for use in remote and harsh environments. It is part of Check Point's line of security appliances and is used for firewall, VPN, and other security functions. The appliance is

based on the Gaia embedded operating system, which is a stripped-down version of Check Point's flagship operating system, Gaia. Gaia is a Linux-based operating system that is used in many of Check Point's security appliances and provides a unified management platform for Check Point's security solutions. Gaia embedded is a smaller version of the operating system that is designed to run on lower-powered devices such as the Quantum Spark appliance. It provides many of the same features as the full Gaia operating system but with a smaller footprint and reduced resource requirements. Therefore, the correct answer is: B. Gaia embedded.

Selected Answer: C

Rugged appliances, like the Check Point 1200R1 and FortiGate Rugged Series2, are designed to operate reliably in harsh usage environments and conditions. They typically run on specialized versions of Linux that are tailored for their specific hardware and use cases. In the case of Check Point's Quantum Spark appliances, they run on the Gaia Operating System3. Therefore, the correct answer is C. Gaia. Please note that "Gaia embedded" is not a standard term used to describe the operating system of these appliances. Remember, the operating system used can vary between different manufacturers and models of rugged appliances. Always refer to the specific product documentation for accurate information.

Question #170

Using AD Query, the security gateway connections to the Active Directory Domain Controllers using what protocol?

A) Windows Management Instrumentation (WMI) (Correct Answer)

- B) Hypertext Transfer Protocol Secure (HTTPS)
- C) Lightweight Directory Access Protocol (LDAP)
- D) Remote Desktop Protocol (RDP)

Explanation

Correct Answer:

A

Community Discussion

When you query Active Directory you have to speak its language, which is LDAP. I would answer C.

Selected Answer: C

C. Lightweight Directory Access Protocol (LDAP)

Selected Answer: A

AD Query is a Windows tool that uses Active Directory's root\directory\ldap namespace. Each LDAP schema class and attribute is mapped to corresponding WMI classes or properties.

Selected Answer: A

It is not C because LDAP is not the protocol that AD Query uses to communicate with the Active Directory Domain Controllers. LDAP stands for Lightweight Directory Access Protocol, and it is a standard way of accessing and managing directory services over a network¹. However, AD Query is a Windows-specific tool that uses WMI (Windows Management Instrumentation) as its underlying protocol². WMI is a framework that allows applications to access and manage information about Windows-based systems³. WMI can use LDAP as one of its data sources, but it is not the same as LDAP itself

Selected Answer: C

When using AD Query, the security gateway connections to the Active Directory Domain Controllers using the Lightweight Directory Access Protocol (LDAP). Explanation: AD Query is a feature of Check Point security gateways that enables administrators to perform queries against Active Directory Domain Controllers. These queries can be used for a variety of purposes, such as user authentication and authorization, group policy enforcement, and other security-related tasks. To perform these queries, the security gateway needs to communicate with the Active Directory Domain Controllers. This communication is typically done using the Lightweight Directory Access Protocol (LDAP), which is a client-server protocol used for accessing directory services. LDAP is a widely used protocol for accessing Active Directory and is used by many different applications and services for authentication, authorization, and other directory-related functions. Therefore, the correct answer is: C. Lightweight Directory Access Protocol (LDAP).

Question #171

What is the main objective when using Application Control?

A) To filter out specific content.

B) To assist the firewall blade with handling traffic.

C) To see what users are doing. (Correct Answer)

D) Ensure security and privacy of information.

Explanation

Correct Answer:

C -

Community Discussion

right answer is D

I have a startup that utilizes some sophisticated technology. I had to learn a variety of skills in order to ensure that the management of our resources is both effective and secure.

Selected Answer: D

The main objective of using Application Control is to ensure the security and privacy of information. Explanation: Application Control is a security feature in network security appliances that is used to identify and control the applications that are allowed to access the network. The feature allows administrators to create policies that specify which applications are allowed to access the network and what actions are allowed for each application. The main objective of Application Control is to ensure the security and privacy of information by controlling the applications that are allowed to access the network. This helps to prevent unauthorized access, data leakage, and other security risks that may arise from the use of unapproved applications. By enforcing policies on allowed applications, administrators can ensure that sensitive information remains secure and confidential. Therefore, the correct answer is: D. Ensure security and privacy of information.

Selected Answer: D

<https://digitalguardian.com/blog/what-application-control>

Selected Answer: B

Application Control in the context of security on Linux systems involves managing and controlling the applications that are allowed to run on the system. It allows

administrators to define and enforce policies for application usage, helping to secure the system by controlling which applications can execute and how they interact with the network. The focus is on enhancing security and managing the traffic associated with various applications.

Question #172

As part of the requirements for installing a new application, the swappiness parameter needs to be changed to 0. This change needs to persist across reboots and be applied immediately. A Linux systems administrator is performing this change. Which of the following steps should the administrator complete to accomplish this task?

A) echo "vm.swappiness=0" >> /etc/sysctl.conf && sysctl -p (Correct Answer)

B) echo "vm.swappiness=0" >> /proc/meminfo && sysctl -a

C) sysctl -v >> /proc/meminfo && echo "vm.swapiness=0"

D) sysctl -h "vm.swapiness=0" && echo /etc/vmswapiness

Explanation

Correct Answer:

A

Community Discussion

Selected Answer: A

To persist the parameter value indefinitely -- that is, until the user wants to change it again -- they must manually update the sysctl.conf file. The command calls the nano text editor, which lets the user edit the file directly in the terminal. When the file opens, the user scrolls to the end of the text and adds the following setting. "vm.swappiness=0" >> append /etc/sysctl.conf && sysctl -p sysctl -p ((allows you to load the settings from a configuration file)

Linux swappiness is a kernel parameter that determines how aggressively the Linux virtual machine (VM) swaps pages between memory and the swap space on the system's disk. Users can fine-tune the rate at which a Linux kernel can move pages into and out of active memory using the swappable parameter, which can be set between 0 and 200. The

higher the number, the more aggressively the VM swaps data to disk. By default, the swappiness value on most Linux computers is set to 60. The user can change this level to a different setting either temporarily or indefinitely. Some database servers managed cache to use RAM, and is capable of managing what should be in RAM and what shouldn't be. Allowing the Linux operating system to have too much control over what memory pages are in RAM is likely to lower your database performance. Therefore, it is recommended that swappiness should be set to 0.

Question #173

A systems administrator wants to list all local accounts in which the UID is greater than 500. Which of the following commands will give the correct output?

- A) `find /etc/passwd -size +500`
- B) `cut -d: -f1 /etc/passwd > 500`
- C) `awk -F: '$3 > 500 {print $1}' /etc/passwd` (Correct Answer)**
- D) `sed '/UID/' /etc/passwd < 500`

Explanation

Correct Answer:

C

Community Discussion

Selected Answer: C

`awk -F: '$3 > 500 {print $1}' /etc/passwd {print $1}` The \$1 stands for the first field, in this case the first column. Option - F --field-separator I did a test and it works !!

Question #174

A systems administrator is tasked with changing the default shell of a system account in order to disable iterative logins. Which of the following is the BEST option for the administrator to use as the new shell?

- A) `/sbin/nologin` (Correct Answer)**

- B) /bin/sh
- C) /sbin/setenforce
- D) /bin/bash

Explanation

Correct Answer:

A

Community Discussion

Selected Answer: A

In order to limit the login we can add /bin/false to the end of the string (in the file /etc/passwd), thus we will have a return code = 1 (when the user tries to open the terminal it will close immediately). Another method, in the same way, we add /sbin/nologin (in this case I will have the message "This account is currently not available")

Question #175

A junior systems administrator recently installed an HBA card in one of the servers that is deployed for a production environment. Which of the following commands can the administrator use to confirm on which server the card was installed?

A) lspci | egrep 'hba|fibr' (Correct Answer)

- B) lspci | zgrep 'hba|fibr'
- C) lspci | pgrep 'hba|fibr'
- D) lspci | 'hba|fibr'

Explanation

Correct Answer:

A

Community Discussion

Selected Answer: A

A. `lspci | egrep 'hba|fibr'` Egrep is a shortcut to the `grep -E` command. The `-E` option allows the use of regular expressions for text filtering. For example: `+`, `|`, `()` etc.

Question #176

A new disk was presented to a server as `/dev/sdd`. The systems administrator needs to check if a partition table is on that disk. Which of the following commands can show this information?

- A) `lsscsi`
- B) `fdisk` (Correct Answer)**
- C) `blkid`
- D) `partprobe`

Explanation

Correct Answer:

B

Community Discussion

Selected Answer: C

'`blkid`' lists PARTUUID if the partition is available. '`fdisk`' alone is not a valid command. With `-l` it provides information about partition but '`fdisk -l`' is not an answer.

Selected Answer: B

A. `lsscsi` -> The `lsscsi` command lists information about SCSI devices in Linux B. `fdisk` -> The `fdisk` utility is a common partition table manipulator for Linux. Use `fdisk -l` to list the partition table C. `blkid` -> The `blkid` command allows you to display information about available block devices. D. `partprobe` -> `partprobe` is a program that informs the operating system kernel of partition table changes, by requesting that the operating system re-read the partition table.

Question #177

A systems administrator wants to upgrade `/bin/someapp` to a new version, but the administrator does not know the package name. Which of the following will show the RPM package name that provides that binary file?

A) `rpm -qf /bin/someapp` (Correct Answer)

B) `rpm -Vv /bin/someapp`

C) `rpm -p /bin/someapp`

D) `rpm -i /bin/someapp`

Explanation

Correct Answer:

A

Community Discussion

Selected Answer: A

Use the command '`rpm -qf`' or '`yum whatprovides`' to get the package name from a binary file or library file on the server.

Question #178

A systems administrator is installing various software packages using a package manager. Which of the following commands would the administrator use on the Linux server to install the package?

A) `winget`

B) `softwareupdate`

C) `yum-config`

D) `apt` (Correct Answer)

Explanation

Correct Answer:

Community Discussion

Selected Answer: D

Red Hat Package Managers Red Hat Package Manager (RPM) Yellow Dog Update Manager (YUM) Dandified YUM (DNF) package manager Debian Package Managers (RPM) dpkg package manager Advanced Packaging Tool (APT)

Question #179

Users are unable to create new files on the company's FTP server, and an administrator is troubleshooting the issue. The administrator runs the following commands:

```
# df -h /ftpusers/

Filesystem      Size      Used      Avail     Use%      Mounted on
/dev/sda4       150G      40G       109G      26%       /ftpusers

# df -i /ftpusers/

Filesystem      Inodes     Iused     Ifree     Iuse%      Mounted on
/dev/sda4       34567      34567      0         100%       /ftpusers
```

Which of the following is the cause of the issue based on the output above?

- A) The users do not have the correct permissions to create files on the FTP server.
- B) The ftpusers filesystem does not have enough space.
- C) The inodes is at full capacity and would affect file creation for users. (Correct Answer)**
- D) ftpusers is mounted as read only.

Explanation

Correct Answer:

C

Community Discussion

Question #180

DRAG DROP

-

You have been asked to parse a log file of logins to determine various information about who is logging in and when.

INSTRUCTIONS

-

Open and inspect the Login log file.

Drag and drop the correct commands onto the output that was generated from that command.

Tokens can be used only once and not all will be used.

If at any time you would like to bring back the initial state of the simulation, please click the Reset All button.

Command Ouput

View Login Log

Commands

tr "[a-z]" "[A-Z]" < log.txt | grep -i "mar 12"

awk '{ print \$1 }' log.txt | uniq

grep -i "mar 12" log.txt | sed 's/[a-z]/[A-Z]/g'

grep "Mar 13" log.txt

grep Mar 13 log.txt

awk '{ print \$1 }' log.txt | sort | uniq -c

awk '{ print toupper(\$0) }' log.txt

awk '{ print \$1 }' log.txt | sort | uniq

grep "Mar13" log.txt

grep log.txt "Mar 13"

awk '{ print \$2 }' log.txt | sort | uniq

tr "[A-Z]" "[a-z]" < log.txt | grep -i "mar 12"

1

2

3

[comptia@localhost exercise]\$
?

```

eric    pts/3    :0      Mon Mar 13 10:52 - 08:48 (1+12:50)
david   pts/3    :0      Mon Mar 13 10:52 - 08:48 (1+12:50)
ann     pts/3    :0      Mon Mar 13 10:52 - 08:48 (1+12:50)
chris   pts/3    :0      Mon Mar 13 10:52 - 08:48 (1+12:50)
carl    pts/3    :0      Mon Mar 13 10:52 - 08:48 (1+12:50)
joe     pts/3    :0      Mon Mar 13 10:52 - 08:48 (1+12:50)
lee     pts/3    :0      Mon Mar 13 10:52 - 08:48 (1+12:50)

```

Command Ouput

View Login Log

Commands

tr "[a-z]" "[A-Z]" < log.txt | grep -i "mar 12"

awk '{ print \$1 }' log.txt | uniq

grep -i "mar 12" log.txt | sed 's/[a-z]/[A-Z]/g'

grep "Mar 13" log.txt

grep Mar 13 log.txt

awk '{ print \$1 }' log.txt | sort | uniq -c

awk '{ print toupper(\$0) }' log.txt

awk '{ print \$1 }' log.txt | sort | uniq

grep "Mar13" log.txt

grep log.txt "Mar 13"

awk '{ print \$2 }' log.txt | sort | uniq

tr "[A-Z]" "[a-z]" < log.txt | grep -i "mar 12"

1

2

3

[comptia@localhost exercise]\$
?

```

COMPTIA PTS/1      :0      SUN MAR 12 14:20 - 09:48 (1+19:28)
CHRIS   PTS/2      :0      SUN MAR 12 14:52 - 09:48 (1+18:56)
REBOOT  SYSTEM BOOT 3.10.0-693.21.1. SUN MAR 12 11:01 - 13:26 (2+02:24)
LEE     PTS/2      :0      SUN MAR 12 14:52 - 09:48 (1+18:56)
COMPTIA PTS/0      :0      SUN MAR 12 14:14 - 14:22 (00:07)
ERIC    PTS/2      :0      SUN MAR 12 14:52 - 09:48 (1+18:56)
COMPTIA PTS/0      :0      SUN MAR 12 11:37 - 11:58 (00:20)
REBOOT  SYSTEM BOOT 3.10.0-693.21.1. SUN MAR 12 11:02 - 13:26 (2+02:23)
DAVID   PTS/2      :0      SUN MAR 12 14:52 - 09:48 (1+18:56)
COMPTIA PTS/0      :0      SUN MAR 12 09:53 - 10:59 (01:05)
COMPTIA :0          :0      SUN MAR 12 11:01 - 11:01 (00:00)
REBOOT  SYSTEM BOOT 3.10.0-693.21.1. SUN MAR 12 10:59 - 13:26 (2+02:26)
COMPTIA :0          :0      SUN MAR 12 11:03 - 13:26 (2+02:22)
COMPTIA PTS/2      :0      SUN MAR 12 14:52 - 09:48 (1+18:56)
ANN     PTS/2      :0      SUN MAR 12 14:52 - 09:48 (1+18:56)
CARL    PTS/2      :0      SUN MAR 12 14:52 - 09:48 (1+18:56)
REBOOT  SYSTEM BOOT 3.10.0-693.EL7.X SUN MAR 12 09:39 - 10:59 (01:19)
JOE     PTS/2      :0      SUN MAR 12 14:52 - 09:48 (1+18:56)
COMPTIA :0          :0      SUN MAR 12 09:43 - DOWN  (01:15)
COMPTIA PTS/0      :0      SUN MAR 12 11:04 - 11:34 (00:30)

```

Command Ouput

View Login Log

Commands

tr "[a-z]" "[A-Z]" < log.txt | grep -i "mar 12"

awk '{ print \$1 }' log.txt | uniq

grep -i "mar 12" log.txt | sed 's/[a-z]/[A-Z]/g'

grep "Mar 13" log.txt

grep Mar 13 log.txt

awk '{ print \$1 }' log.txt | sort | uniq -c

awk '{ print toupper(\$0) }' log.txt

awk '{ print \$1 }' log.txt | sort | uniq

grep "Mar13" log.txt

1

2

3

[comptia@localhost exercise]\$
?

```

ann
carl
chris
comptia
david
eric
joe
lee
reboot

```

Page 219 of 382

Explanation

Correct Answer:

Command Output

View Login Log

Commands

```
tr "[a-z]" "[A-Z]" < log.txt | grep -i "mar 12"
awk '{ print $1 }' log.txt | uniq
grep -i "mar 12" log.txt | sed 's/[a-z]/[A-Z]/g'
grep "Mar 13" log.txt
grep Mar 13 log.txt
awk '{ print $1 }' log.txt | sort | uniq -c
awk '{ print toupper($0) }' log.txt
awk '{ print $1 }' log.txt | sort | uniq
grep "Mar13" log.txt
grep log.txt "Mar 13"
awk '{ print $2 }' log.txt | sort | uniq
tr "[A-Z]" "[a-z]" < log.txt | grep -i "mar 12"
```

1

2

3

[comptia@localhost exercise]\$ grep "Mar 13" log.txt

```
eric pts/3 :0 Mon Mar 13 10:52 - 08:48 (1+12:50)
david pts/3 :0 Mon Mar 13 10:52 - 08:48 (1+12:50)
ann pts/3 :0 Mon Mar 13 10:52 - 08:48 (1+12:50)
chris pts/3 :0 Mon Mar 13 10:52 - 08:48 (1+12:50)
carl pts/3 :0 Mon Mar 13 10:52 - 08:48 (1+12:50)
joe pts/3 :0 Mon Mar 13 10:52 - 08:48 (1+12:50)
lee pts/3 :0 Mon Mar 13 10:52 - 08:48 (1+12:50)
```

Command Output

View Login Log

Commands

```
tr "[a-z]" "[A-Z]" < log.txt | grep -i "mar 12"
awk '{ print $1 }' log.txt | uniq
grep -i "mar 12" log.txt | sed 's/[a-z]/[A-Z]/g'
grep "Mar 13" log.txt
grep Mar 13 log.txt
awk '{ print $1 }' log.txt | sort | uniq -c
awk '{ print toupper($0) }' log.txt
awk '{ print $1 }' log.txt | sort | uniq
grep "Mar13" log.txt
grep log.txt "Mar 13"
awk '{ print $2 }' log.txt | sort | uniq
tr "[A-Z]" "[a-z]" < log.txt | grep -i "mar 12"
```

1

2

3

[comptia@localhost exercise]\$ tr "[a-z]" "[A-Z]" < log.txt | grep -i "mar 12"

```
COMPTIA PTS/1 :0 SUN MAR 12 14:20 - 09:48 (1+19:28)
CHRIS PTS/2 :0 SUN MAR 12 14:52 - 09:48 (1+18:56)
REBOOT SYSTEM BOOT 3.10.0-693.21.1. SUN MAR 12 11:01 - 13:26 (2+02:24)
LEE PTS/2 :0 SUN MAR 12 14:52 - 09:48 (1+18:56)
COMPTIA PTS/0 :0 SUN MAR 12 14:14 - 14:22 (00:07)
ERIC PTS/2 :0 SUN MAR 12 14:52 - 09:48 (1+18:56)
COMPTIA PTS/0 :0 SUN MAR 12 11:37 - 11:58 (00:20)
REBOOT SYSTEM BOOT 3.10.0-693.21.1. SUN MAR 12 11:02 - 13:26 (2+02:23)
DAVID PTS/2 :0 SUN MAR 12 14:52 - 09:48 (1+18:56)
COMPTIA PTS/0 :0 SUN MAR 12 09:53 - 10:59 (01:05)
COMPTIA :0 :0 SUN MAR 12 11:01 - 11:01 (00:00)
REBOOT SYSTEM BOOT 3.10.0-693.21.1. SUN MAR 12 10:59 - 13:26 (2+02:26)
COMPTIA :0 :0 SUN MAR 12 11:03 - 13:26 (2+02:22)
COMPTIA PTS/2 :0 SUN MAR 12 14:52 - 09:48 (1+18:56)
ANN PTS/2 :0 SUN MAR 12 14:52 - 09:48 (1+18:56)
CARL PTS/2 :0 SUN MAR 12 14:52 - 09:48 (1+18:56)
REBOOT SYSTEM BOOT 3.10.0-693.EL7.X SUN MAR 12 09:39 - 10:59 (01:19)
JOE PTS/2 :0 SUN MAR 12 14:52 - 09:48 (1+18:56)
COMPTIA :0 :0 SUN MAR 12 09:43 - DOWN (01:15)
COMPTIA PTS/0 :0 SUN MAR 12 11:04 - 11:34 (00:30)
```

Command Output

View Login Log

Commands

```
tr "[a-z]" "[A-Z]" < log.txt | grep -i "mar 12"
awk '{ print $1 }' log.txt | uniq
grep -i "mar 12" log.txt | sed 's/[a-z]/[A-Z]/g'
grep "Mar 13" log.txt
grep Mar 13 log.txt
awk '{ print $1 }' log.txt | sort | uniq -c
awk '{ print toupper($0) }' log.txt
awk '{ print $1 }' log.txt | sort | uniq
grep "Mar13" log.txt
grep log.txt "Mar 13"
awk '{ print $2 }' log.txt | sort | uniq
tr "[A-Z]" "[a-z]" < log.txt | grep -i "mar 12"
```

1

2

3

[comptia@localhost exercise]\$ awk '{ print \$1 }' log.txt | sort | uniq

```
ann
carl
chris
comptia
david
eric
joe
lee
reboot
```

Community Discussion

pls help me THX

```
grep "Mar 13" log.txt tr "[a-z]" "[A-Z]" < log.txt | grep -i "mar 12" awk '{print $1}' log.txt | sort  
| uniq
```

output 2 , maybe it's not better to `grep -i "mar 12" log.txt | sed 's/[a-z]/[A-Z]/g'` , the `tr` command should be with `'` and not `"` so `tr '[a-z]' '[A-Z]' < log.txt | grep -i "mar 12"`

Question #181

An administrator started a long-running process in the foreground that needs to continue without interruption. Which of the following keystrokes should the administrator use to continue running the process in the background?

A) bg (Correct Answer)

- B) bg
- C) jobs -1
- D) bg &

Explanation

Correct Answer:

A

Community Discussion

If the command is already running in your terminal, you can enter CTRL+Z to pause the process, then "bg" to resume the process in the background. Or, if your process is not running yet, add an "&" to the end of your command to run it in the background.

Selected Answer: A

bg — run jobs in the background & ampersand single ampersand & causes the command to execute in the background.

Question #182

A Linux administrator has physically added a new RAID adapter to a system. Which of the following commands should the Linux administrator run to confirm that the device has been recognized? (Choose two.)

- A) rmmod
- B) ls -ll /etc
- C) lshw -class disk (Correct Answer)**
- D) pvdisplay
- E) rmdir /dev
- F) dmesg (Correct Answer)**

Explanation

Correct Answer:
CF

Community Discussion

Selected Answer: CF

lshw -class disk (option -class shows only a certain class of hardware) dmesg - diagnostic message; dmesg is a command-line-based utility in Linux that displays kernel-related messages retrieved from the kernel ring buffer. rmmod - simple program to remove a module from the Linux Kernel ls -l. The -l option signifies the long list format. pvdisplay — Display various attributes of physical volume(s) rmdir /dev - Remove directory "dev" in this case

Selected Answer: CF

The correct answers are C. lshw -class disk. and F. dmesg

Question #183

Based on the given command:

```
find /home/user1 -type f -name "*.txt" -size +100M
```

Which of the following will the command accomplish?

- A) It finds all the files ending with .txt with a size less than 100MB in /home/user1 directory.
- B) It finds all the files ending with .txt with a size of 100MB in /home/user1 directory.
- C) It finds all the directories with .txt files with a size of 100MB in /home folder.

D) It finds all the files ending with .txt with a size of over 100MB in /home/user1 directory.
(Correct Answer)

Explanation

Correct Answer:

D

Community Discussion

Selected Answer: D

find /home/user1 -type f -name "*.txt" -size +100M find command is first. The -type field specifies whether it is a file or a directory. -type f means that it is looking for files. The -name field is means that it is looking for files that end .txt. When you see the * that is a wildcard saying anything can come before it. The -size field is denotes the file size which is greater than 100M

Question #184

While inspecting a recently compromised Linux system, the administrator identified a number of processes that should not have been running:

PID	USER	PR	NI	VIRT	RES	SHR	S	%CPU	%MEM	TIME+	COMMAND
5545	joe	30	-10	5465	56465	8254	R	0.5	1.5	00:35.3	upload.sh
2567	joe	30	-10	6433	75544	9453	R	0.7	1.8	00:25.1	upload_passwd.sh
8634	joe	30	-10	3584	74537	6435	R	0.3	1.1	00:17.6	uploadpw.sh
4846	joe	30	-10	6426	63234	9683	R	0.8	1.9	00:22.2	upload_shadow.sh

Which of the following commands should the administrator use to terminate all of the identified processes?

A) pkill -9 -f "upload*.sh" (Correct Answer)

B) kill -9 "upload*.sh"

C) killall -9 "upload*.sh"

D) skill -9 "upload*.sh"

Explanation

Correct Answer:

A

Community Discussion

Selected Answer: A

pkill An alternative command to kill is the pkill command, which also references the process by PID. Additionally, pkill can also terminate processes based on name, users, and other criteria. When -f is set, the full command line is used. (complete command line) The kill command is used to pass these signals to processes. For example, to initiate a graceful exit, type: \$ sudo kill -15 {PID} Likewise, kill -9 {PID} sends a SIGKILL message to the process. **skill** These tools are obsolete and unportable.

Question #185

A senior Linux administrator has created several scripts that will be used to install common system applications. These scripts are published to a repository to share with the systems team. A junior Linux administrator needs to retrieve the scripts and make them available on a local workstation. Which of the following Git commands should the junior Linux administrator use to accomplish this task?

- A) fetch
- B) checkout
- C) clone (Correct Answer)**
- D) branch

Explanation

Correct Answer:
C

Community Discussion

Question #186

Users are reporting that writes on a system configured with SSD drives have been taking longer than expected, but reads do not seem to be affected. A Linux systems administrator is investigating this issue and working on a solution. Which of the following should the administrator do to help solve the issue?

- A) Run the corresponding command to trim the SSD drives. (Correct Answer)**
- B) Use fsck on the filesystem hosted on the SSD drives.
- C) Migrate to high-density SSD drives for increased performance.
- D) Reduce the amount of files on the SSD drives.

Explanation

Correct Answer:
A

Community Discussion

Selected Answer: A

What is TRIM TRIM is the function that allows you to improve the long-lasting performance of SSDs by better managing the various reads and writes of the cells, so you will not only increase the life of the SSD but also speed up the read and write operations. SSD TRIM is an Advanced Technology Attachment (ATA) command that enables an operating system to inform a NAND flash solid-state drive (SSD) which data blocks it can erase because they are no longer in use. The use of TRIM can improve the performance of writing data to SSDs and contribute to longer SSD life

Question #187

A Linux engineer needs to block an incoming connection from the IP address 2.2.2.2 to a secure shell server and ensure the originating IP address receives a response that a firewall is blocking the connection. Which of the following commands can be used to accomplish this task?

- A) `iptables -A INPUT -p tcp --dport ssh -s 2.2.2.2 -j DROP`
- B) `iptables -A INPUT -p tcp --dport ssh -s 2.2.2.2 -j RETURN`
- C) `iptables -A INPUT -p tcp --dport ssh -s 2.2.2.2 -j REJECT` (Correct Answer)**
- D) `iptables -A INPUT -p tcp --dport ssh -s 2.2.2.2 -j QUEUE`

Explanation

Correct Answer:

C

Community Discussion

Selected Answer: C

In general, use REJECT if you want the other side to know that the port is unreachable, and DROP for connections to hosts you don't want to see.

both DROP & REJECT works

Selected Answer: C

I agree with Damon54

Question #188

A systems administrator detected corruption in the /data file system. Given the following output:

root@localhost ~]# lsblk -f			
NAME	FSTYPE	LABEL/UUID	MOUNTPOINT
sda			
sda1	vfat	4E7D-9539	/boot/efi
sda2	xfs	98442caf-473d-448e-aee5-561a82297314	/boot
sda3	swap	19f064e4-7c51-4b02-8219-99362a3c45ec	[SWAP]
sda4	xfs	25d96ada-4289-4def-9202-6ab11affbed3	/
sda5	xfs	61435ee9-855d-4de9-9c67-39aeb7f3edb5	/home
sdc			
sdc1	ext4	92435ff9-745e-4fg9-9c67-39aeb7f3exf5	/data

Which of the following commands can the administrator use to BEST address this issue?

- umount /data
 mkfs.xfs /dev/sdc1
 mount /data
 A)
- umount /data
 xfs_repair /dev/sdc1
 mount /data
 B)

```
umount /data  
fsck /dev/sdc1
```

c) `mount /data`

(Correct Answer)

```
umount /data  
pvs /dev/sdc1
```

D) `mount /data`

Explanation

Correct Answer:

C

Community Discussion

Selected Answer: C

mkfs ("MaKe FileSystem") format the new partitions, xfs_repair that would be fine but our partition is formatted ext4 fsck A file system's superblock contains metadata about that file system, including its size, type, and status. The superblock is critical to the function of the file system, and if it becomes corrupt, you may be unable to mount and work with the file system. You can use a tool like fsck to repair the superblock, if necessary. pvs LVM command tool , Display information about physical volumes.

Question #189

A user reported issues when trying to log into a Linux server. The following outputs were received:

Output1:

```
user1:$6SQWSeda12DeafLO09uyvbn12/fHYws1Lm45NMCnbashU:18885:0:99999:7:::
```

Output 2:

```
user1:x:39981:501:user1 account:/home/user1:/bin/bash
```

Output 3:

```
AllowGroups wheel admin desktop user
```

Output 4:

```
lp:x:7:daemon
desktopuser:x:502
user:x:501
postfix:x:32
```

Given the outputs above, which of the following is the reason the user is unable to log in to the server?

- A) User1 needs to set a long password.
- B) User1 is in the incorrect group. (Correct Answer)**
- C) The user1 shell assignment is incorrect.
- D) The user1 password is expired.

Explanation

Correct Answer:

B

Community Discussion

Selected Answer: B

A The password is set correctly (encrypted password) C bin/bash shell assignment is correct D password is not expired 18885 : 0 : 99999 : 7 : : by exclusion I would say B , but Output 4 should be like this AllowGroups wheel admin desktopuser

Question #190

A junior administrator updated the PostgreSQL service unit file per the database administrator's recommendation. The service has been restarted, but changes have not been applied. Which of the following should the administrator run for the changes to take effect?

- A) systemctl get-default
- B) systemctl daemon-reload (Correct Answer)**
- C) systemctl enable postgresql
- D) systemctl mask postgresql

Explanation

Correct Answer:
B

Community Discussion

Selected Answer: B

The “daemon-reload” reloads changes or modifies the systemd unit files. These unit files are placed in the “/etc/systemd/system” directory and require reloading if any changes are made to them.

Question #191

A systems administrator intends to use a UUID to mount a new partition permanently on a Linux system. Which of the following commands can the administrator run to obtain information about the UUIDs of all disks attached to a Linux system?

- A) fcstat

B) blkid (Correct Answer)

- C) dmsetup
- D) lscsi

Explanation

Correct Answer:

B

Community Discussion

blkid is a command-line utility in Linux that is used to display information about block devices, such as hard drives, solid-state drives (SSDs), or removable storage devices, such as USB drives. blkid can be used to display a variety of information about a block device, including its device name, type, UUID (Universally Unique Identifier), filesystem, and label. <https://www.thegeekdiary.com/blkid-command-examples-in-linux/>

Question #192

A systems administrator creates a public key for authentication. Which of the following tools is MOST suitable to use when uploading the key to the remote servers?

- A) scp
- B) ssh-copy-id (Correct Answer)**
- C) ssh-agent
- D) ssh-keyscan

Explanation

Correct Answer:

B

Community Discussion

Question #193

A systems administrator checked out the code from the repository, created a new branch, made changes to the code, and then updated the main branch. The systems administrator wants to ensure that the Terraform state files do not appear in the main branch. Which of following should the administrator use to meet this requirement?

A) clone

B) .gitignore (Correct Answer)

C) get

D) .ssh

Explanation

Correct Answer:

B

Community Discussion

Question #194

Following the migration from a disaster recovery site, a systems administrator wants a server to require a user to change credentials at initial login. Which of the following commands should be used to ensure the aging attribute?

A) chage -d 2 user

B) chage -d 0 user (Correct Answer)

C) chage -E 0 user

D) chage -d 1 user

Explanation

Correct Answer:

B

Community Discussion

Selected Answer: B

From the man page: If the LAST_DAY is set to 0 the user is forced to change his password on the next log on.

Question #195

A systems administrator wants to permit access temporarily to an application running on port 1234/TCP on a Linux server. Which of the following commands will permit this traffic?

- A) `firewall-cmd --new-service=l234/tcp`
- B) `firewall-cmd --service=l234 --protocol=tcp`
- C) `firewall-cmd --add-port=1234/tcp` (Correct Answer)**
- D) `firewall-cmd --add-whitelist-uid=l234`

Explanation

Correct Answer:

C

Community Discussion

Question #196

An engineer needs to insert a character at the end of the current line in the vi text editor. Which of the following will allow the engineer to complete this task?

- A) p
- B) r
- C) bb
- D) A (Correct Answer)**

E) i

Explanation

Correct Answer:

D

Community Discussion

Selected Answer: D

Shift+A will put the cursor at the end of the line and put vim in insert mode.

Question #197

A systems administrator is adding a Linux-based server and removing a Windows-based server from a cloud-based environment. The changes need to be validated before they are applied to the cloud-based environment. Which of the following tools should be used to meet this requirement?

- A) Ansible
- B) git clone
- C) git pull

D) terraform plan (Correct Answer)

Explanation

Correct Answer:

D

Community Discussion

Terraform is an open-source infrastructure as code (IaC) software tool which can be used to provision the infrastructure of a cloud platform. The scripts which have been used to provision infrastructure can be human-readable configuration files that can be versioned, reused, and shared. You can use a wide provision wide range of resources in the

cloud by using terraform like compute, storage, networking, and application services, across a variety of cloud providers and on-premises environments. <https://www.geeksforgeeks.org/what-is-terraform/>

I was confused of when I would use ansible vs terraform. It looks like Terraform is used with cloud infrastructure and ansible is used within a network. I believe that the question aligns more with Terraform. Correct me if I am missing something

Ansible is an open source automation platform. What that means is that it lets you manage software installations, updates, configurations and tasks within your network environment. It's especially handy in situations where you need to carry out sequential operations. It's also comparably easy to use, when put against other similar solutions. <https://www.linux.com/audience/devops/introduction-ansible-0/>

Ansible is an open source automation platform. What that means is that it lets you manage software installations, updates, configurations and tasks within your network environment. It's especially handy in situations where you need to carry out sequential operations. It's also comparably easy to use, when put against other similar solutions. <https://www.linux.com/audience/devops/introduction-ansible-0/>

I was confused of when I would use ansible vs terraform. It looks like Terraform is used with cloud infrastructure and ansible is used within a network. I believe that the question aligns more with Terraform. Correct me if I am missing something

Question #198

A systems administrator wants to check for running containers. Which of the following commands can be used to show this information?

- A) docker pull
- B) docker stats
- C) docker ps (Correct Answer)**
- D) docker list

Explanation

Correct Answer:

C

Community Discussion

Question #199

A Linux systems administrator is troubleshooting an I/O latency on a single CPU server. The administrator runs a top command and receives the following output:

```
%Cpu(s): 0.2 us, 33.1 sy, 0.0 ni, 0.0 id, 52.4 wa, 0.0 hi, 0.2 si, 0.0 st
```

Which of the following is correct based on the output received from the executed command?

- A) The server's CPU is taking too long to process users' requests.
- B) The server's CPU shows a high idle-time value.
- C) The server's CPU is spending too much time waiting for data inputs. (Correct Answer)**
- D) The server's CPU value for the time spent on system processes is low.

Explanation

Correct Answer:

C

Community Discussion

Question #200

An administrator needs to get network information from a group of statically assigned workstations before they are reconnected to the network. Which of the following should the administrator use to obtain this information?

- A) ip show
- B) ifcfg -a (Correct Answer)**
- C) ifcfg -s

D) ifname -s

Explanation

Correct Answer:

B

Community Discussion

Selected Answer: B

B ifcfg is a command used to manipulate network interfaces in Unix-like operating systems. The -a option stands for "all" and is used to display all network interfaces along with their configuration settings.

Debian12 -bash: ifcfg: command not found

Selected Answer: A

The command ifcfg -a is not a valid command in Linux for obtaining network information. The ifcfg command is used in some Linux distributions to configure network interfaces, but it does not have an -a option.

Question #201

A user created the following script file:

```
#!/bin/bash
# FILENAME: /home/user/script.sh
```

```
echo "hello world"
exit 1
```

However, when the user tried to run the script file using the command `~/script.sh`, an error returned indicating permission was denied. Which of the following should the user execute in order for the script to run properly?

A) `chmod u+x /home/user/script.sh` (Correct Answer)

- B) `chmod 600 /home/user/script.sh`
- C) `chmod 007 /home/user/script.sh`
- D) `chmod o+r /home/user/script.sh`

Explanation

Correct Answer:

A

Community Discussion

Question #202

A systems administrator is enabling LUKS on a USB storage device with an ext4 filesystem format. The administrator runs `dmesg` and notices the following output:

```
sd 8:0:0:0: [sdc] Attached SCSI disk
EXT4-fs (sdc1): mounting ext3 file system using the ext4 subsystem
EXT4-fs (sdc1): mounted filesystem with ordered data mode. Opts: (null)
```

Given this scenario, which of the following should the administrator perform to meet these requirements? (Choose three.)

- A) `gpg /dev/sdc1`
- B) `pvcreate /dev/sdc`
- C) `mkfs.ext4 /dev/mapper/LUKS0001 - L ENCRYPTED`
- D) `umount /dev/sdc` (Correct Answer)**
- E) `fdisk /dev/sdc` (Correct Answer)**
- F) `mkfs.vfat /dev/mapper/LUKS0001 - L ENCRYPTED`
- G) `wipefs -a /dev/sdb1`
- H) `cryptsetup luksFormat /dev/sdc1` (Correct Answer)**

Explanation

Correct Answer:

Community Discussion

Selected Answer: DH

Given the scenario where a systems administrator is enabling LUKS (Linux Unified Key Setup) on a USB storage device with an ext4 filesystem format, the following steps should be performed: 1. **Unmount the USB storage device** to ensure that it is not in use. This is necessary to modify the filesystem. - **D. ``umount /dev/sdc``** 2. **Initialize LUKS on the USB storage device** to set up encryption. - **H. ``cryptsetup luksFormat /dev/sdc1``** 3. **Create the filesystem on the LUKS encrypted device**. Since the requirement is to have an ext4 filesystem, use `mkfs.ext4`. - **C. ``mkfs.ext4 /dev/mapper/LUKS0001 -L ENCRYPTED``** Thus, the correct steps are: - **D. ``umount /dev/sdc``** - **H. ``cryptsetup luksFormat /dev/sdc1``** - **C. ``mkfs.ext4 /dev/mapper/LUKS0001 -L ENCRYPTED``**

Question #203

A developer wants to ensure that all files and folders created inside a shared folder named `/Group00DEV` inherit the group name of the parent folder. Which of the following commands will help achieve this goal?

- A) `chmod g+x /Group00DEV/`
- B) `chmod g+w /Group00DEV/`
- C) `chmod g+r /Group00DEV/`
- D) `chmod g+s /Group00DEV/` (Correct Answer)**

Explanation

Correct Answer:

D

Community Discussion

Question #204

Ann, a security administrator, is performing home directory audits on a Linux server. Ann issues the `su Joe` command and then issues the `ls` command. The output displays files that reside in Ann's home directory instead of Joe's. Which of the following represents the command Ann should have issued in order to list Joe's files?

A) `su - Joe` (Correct Answer)

- B) `sudo Joe`
- C) `visudo Joe`
- D) `pkexec Joe`

Explanation

Correct Answer:

A

Community Discussion

Question #205

A developer needs to launch an Nginx image container, name it `Web001`, and expose port 8080 externally while mapping to port 80 inside the container. Which of the following commands will accomplish this task?

- A) `docker exec -it -p 8080:80 --name Web001 nginx`
- B) `docker load -it -p 8080:80 --name Web001 nginx`
- C) `docker run -it -p 8080:80 --name Web001 nginx` (Correct Answer)**
- D) `docker pull -it -p 8080:80 --name Web001 nginx`

Explanation

Correct Answer:

C

Community Discussion

A developer needs to launch an Nginx image container, name it Web00l, and expose port 8080 externally while mapping to port 80 inside the container. Which of the following commands will accomplish this task? A. `docker exec -it -p 8080:80 --name Web001 nginx` B. `docker load -it -p 8080:80 --name Web001 nginx` C. `docker run -it -p 8080:80 --name Web001 nginx` D. `docker pull -it -p 8080:80 --name Web001 nginx`

Question #206

The security team has identified a web service that is running with elevated privileges. A Linux administrator is working to change the systemd service file to meet security compliance standards. Given the following output:

[Unit]

Description=CompTIA server daemon

Documentation=man:webserver(8) man:webserver_config(5)

After=network.target -

[Service]

Type=notify -

EnvironmentFile=/etc/webserver/config

ExecStart=/usr/sbin/webserver -D \$OPTIONS

ExecReload=/bin/kill -HUP \$MAINPID

KillMode=process -

Restart=on-failure -

RestartSec=42s -

[Install]

WantedBy=multi-user.target -

Which of the following remediation steps will prevent the web service from running as a privileged user?

- A) Removing the ExecStart=/usr/sbin/webserver -D \$OPTIONS from the service file
- B) Updating the EnvironmentFile line in the [Service] section to /home/websevice/config
- C) Adding the User=websevice to the [Service] section of the service file (Correct Answer)**
- D) Changing the multi-user.target in the [Install] section to basic.target

Explanation

Correct Answer:

C

Community Discussion

Selected Answer: C

Yes, you should specify a User option in the service file. But, I would also like to see this web server binary stored somewhere else, not /usr/sbin; that is meant for super-user system-wide binaries. /usr/local/bin would be more appropriate in my opinion.

Question #207

A Linux administrator provisioned a new web server with custom administrative permissions for certain users. The administrator receives a report that user1 is unable to restart the Apache web service on this server. The administrator reviews the following output:

```
[ root@server ] # id user1
UID=1011(user1) gid=1011(USER1) groups=1011(user1),101(www-data),1120(webadmin)

[ root@server ] # cat /etc/sudoers.d/custom.conf
user1 ALL=/usr/sbin/systemctl start httpd, /usr/sbin/systemctl stop httpd

webadmin ALL=NOPASSWD: /etc/init.d.httpd restart, /sbin/service httpd restart, /usr/sbin/apache2ctl
restart

#%wheel ALL=(ALL) NOPASSWD: ALL
```

Which of the following would most likely resolve the issue while maintaining a least privilege security model?

- A) User1 should be added to the wheel group to manage the service.
- B) User1 should have "NOPASSWD:" after the "ALL=" in the custom.conf.
- C) The wheel line in the custom.conf file should be uncommented.
- D) Webadmin should be listed as a group in the custom.conf file. (Correct Answer)**

Explanation

Correct Answer:

D

Community Discussion

Selected Answer: D

I choose D. A: user1 does not need to be part of the wheel group, for that typically gives them permission to run any commands anywhere. B: Removing the password prompt before running sensitive commands would weaken security. C: User1 is not part of the wheel group. Uncommenting this line would not solve their problem. D: Correct. You specify a group entry with the % symbol, like this: %webadmin ALL=(ALL) NOPASSWD: <allowed_commands>

Question #208

Which of the following would significantly help to reduce data loss if more than one drive fails at the same time?

- A) Server clustering
- B) Load balancing
- C) RAID (Correct Answer)**
- D) VDI

Explanation

Correct Answer:

C

Community Discussion

server equipped 2hard drives in RAID 1 (mirror) more than one hard drives fail how RAID can help me ?

Question #209

A systems administrator receives reports that several virtual machines in a host are responding slower than expected. Upon further investigation, the administrator obtains the following output from one of the affected systems:

16:00:01 PM	CPU	%user	%nice	%system	%iowait	%steal	%idle
16:10:01 PM	all	17.58	0.00	9.36	0.00	54.33	18.73
16:20:01 PM	all	22.34	0.00	11.75	0.00	48.69	17.22
16:30:01 PM	all	25.49	0.00	11.69	0.00	57.85	4.97
16:40:01 PM	all	25.49	0.00	11.69	0.00	53.21	9.61
16:50:01 PM	all	25.49	0.00	11.69	0.00	56.49	6.33

Which of the following best explains the reported issue?

A) The physical host is running out of CPU resources, leading to insufficient CPU time being allocated to virtual machines. (Correct Answer)

B) The physical host has enough CPU cores, leading to users running more processes to compensate for the slower response times.

C) The virtual machine has enough CPU cycles, leading to the system use percentage being higher than expected.

D) The virtual machine is running out of CPU resources, leading to users experiencing longer response times.

Explanation

Correct Answer:

A

Community Discussion

Selected Answer: A

Yes, A, because over half the time the virtual CPU is in steal mode, waiting for resources from the physical host CPU.

Question #210

A user is unable to remotely log on to a server using the server name server1 and port 22. The Linux engineer troubleshoots the issue and gathers the following information:

```
#nslookup server1
Server: 123.1.2.4
Address: 123.1.2.4#53

Name: server1
Address: 123.1.2.3

#systemctl status sshd
sshd.service - Openssh sshd daemon
Loaded: loaded (usr/lib/systemd/system/sshd.service; enabled; vendor preset: disabled)
Active: active (running) since Thu 2021-09-09 12:44:49 CDT; 6 days ago
Main PID: 1900 (sshd)
CGroup: /system.slice/sshd.service
        -1900 /usr/share/sbin/sshd -D

#netstat -anp | grep sshd
tcp    0    0 123.1.2.3:2222    0.0.0.0:*        LISTEN  2701/sshd
```

Which of the following is most likely causing the issue?

- A) server1 is not in the DNS.
- B) sshd is running on a non-standard port. (Correct Answer)**
- C) sshd is not an active service.
- D) server1 is using an incorrect IP address.

Explanation

Correct Answer:

B

Community Discussion

Selected Answer: B

B: SSHD is running on port 2222, not 22.

Question #211

A systems administrator needs to extend the logical volume `/dev/mapper/linux-root` to use all available space in its volume group. The filesystem should be extended as well. Which of the following commands will allow the administrator to achieve this goal?

- A) `lvm pvresize -r -l100% /dev/mapper/linux-root`
- B) `lvchange -L100% /dev/mapper/linux-root`
- C) `lvextend -l+100% /dev/mapper/linux-root`
- D) `lvresize -l100%FREE -r /dev/mapper/linux-root` (Correct Answer)**

Explanation

Correct Answer:
D

Community Discussion

Selected Answer: D

`lvextend` or `lvresize` could both get the job done here. The key here is `"-r"`, resize the associated filesystem.

Selected Answer: D

`lvresize -l100%FREE -r /dev/mapper/linux-root` Explanation: `-l //` `[+|-]LogicalExtentsNumber[%{VG|LV|PVS|FREE|ORIGIN}]` eg. `"-l +100%FREE" -r //` Resize underlying filesystem together with the logical volume using `fsadm`

Selected Answer: C

answer is C. <https://www.redhat.com/sysadmin/resize-lvm-simple>

Question #212

A systems administrator is gathering information about a file type and the contents of a file. Which of the following commands should the administrator use to accomplish this task?

A) file filename (Correct Answer)

- B) touch filename
- C) grep filename
- D) lsof filename

Explanation

Correct Answer:

A

Community Discussion

Question #213

A systems administrator is investigating a service that is not starting up. Given the following information:

```
root@localhost ~]# systemctl status network
network.service - LSB: Bring up/down networking
Loaded: loaded (/etc/rc.d/init.d/network; bad; vendor preset: disabled)
Active: failed (Result: exit-code) since Jan 2022-01-02 22:55:15 CST;
Docs: man:systemd-sysv-generator(8)
Process: 1083 ExecStart=/etc/rc.d/init.d/network start (code=exited, status=1/FAILURE)
Jan 02 22:55:15 localhost.localdomain network[1083]: Bringing up interface enp0s25: Error: Con...n.
Jan 02 22:55:15 localhost.localdomain network[1083]: [FAILED]
[...]
```

Which of the following systemd commands should the administrator use in order to obtain more details about the failing service?

- A) systemctl analyze network
- B) systemctl info network
- C) sysctl -a network
- D) journalctl -xu network (Correct Answer)**

Explanation

Correct Answer:

D

Community Discussion

Question #214

A DevOps engineer wants to allow the same Kubernetes container configurations to be deployed in development, testing, and production environments. A key requirement is that the containers should be configured so that developers do not have to statically configure custom, environment-specific locations. Which of the following should the engineer use to meet this requirement?

- A) Custom scheduler
- B) Node affinity
- C) Overlay network

D) Ambassador container (Correct Answer)

Explanation

Correct Answer:

D

Community Discussion

Selected Answer: C

Overlay networks in Kubernetes provide network abstraction and enable containers to communicate with each other across different environments without being concerned about the underlying network configurations. By using overlay networks, containers can be deployed in any environment without the need for developers to statically configure custom, environment-specific locations.

Selected Answer: C

Overlay networks in Kubernetes allow containers to communicate with each other across different environments (development, testing, production) without needing to statically configure custom, environment-specific locations. By using overlay networks, containers can be deployed with consistent configurations regardless of the underlying environment, ensuring that developers do not have to manage environment-specific configurations manually. This approach promotes consistency and portability across different environments, simplifying the deployment and management process for containerized applications.

Selected Answer: D

The ambassador pattern (or sidecar pattern) allows the application container to remain agnostic about its environment, and simply send and receive requests to its ambassador next door. The ambassador is responsible for network configuration, such as DB endpoints, egress endpoint, TLS, and monitoring.

Question #215

After starting an Apache web server, the administrator receives the following error:

```
Apr 23 11:49:06 localhost.localdomain httpd[4618]: (98)Address already in use: AH00072: make_sock: could not bind to address [::]:80
```

Which of the following commands should the administrator use to further troubleshoot this issue?

A) ss (Correct Answer)

- B) ip
- C) dig
- D) nc

Explanation

Correct Answer:

A

Community Discussion

Question #216

Which of the following commands should a technician use to create an administrative account for the username Joe?

A) sudo useradd -G wheel joe (Correct Answer)

B) sudo useradd joe; sudo passwd -l joe

C) sudo useradd joe; sudo cat key.pem > ~/.ssh/authorized_keys

D) sudo useradd joe; groupadd admin joe

Explanation

Correct Answer:

A

Community Discussion

Selected Answer: A

sudo useradd -G wheel joe This will create the user Joe, with his own userId and groupId, and will set his additional group attachments to "wheel". The wheel group typically allows running any command on the system, for any user associated with the group.

what "wheel" ?

Question #217

Due to performance issues on a server, a Linux administrator needs to terminate an unresponsive process. Which of the following commands should the administrator use to terminate the process immediately without waiting for a graceful shutdown?

A) kill -SIGKILL 5545 (Correct Answer)

B) kill -SIGTERM 5545

C) kill -SIGHUP 5545

D) kill -SIGINT 5545

Explanation

Correct Answer:

A

Community Discussion

Selected Answer: A

SIGINT 2 Signals when the Linux user presses 'CONTROL-C' SIGHUP 1 Hangs up signals when controlling the terminal or at the end of the controlling processes. SIGQUIT 3 Signals when the Linux user presses 'CONTROL-D' SIGFPE 8 Signals when any unexpected mathematical operation is performed. SIGKILL 9 When any of the process issues this signal, it will quit immediately. SIGALRM 14 Signals for an alarm clock SIGTERM 15 Signals to terminate the process or the software. SIGSTOP 17,19,23 Signals to stop the process in Linux. And now: The SIGKILL is used for immediate termination of a process. This signal cannot be ignored. The SIGTERM can be referred as a soft kill because the process that receives the SIGTERM signal may choose to ignore it.

Question #218

A Linux administrator generated a list of users who have root-level command-line access to the Linux server to meet an audit requirement. The administrator analyzes the following `/etc/passwd` and `/etc/sudoers` files:

```
$ cat /etc/passwd
root:x:0:0::/home/root:/bin/bash
lee:x:500:500::/home/lee:/bin/tcsh
mallory:x:501:501::/root:/bin/bash
eve:x:502:502:/home/eve:/bin/nologin
carl:x:0:503::/home/carl:/bin/sh
bob:x:504:504::/home/bob:/bin/ksh
alice:x:505:505::/home/alice:/bin/rsh

$ cat /etc/sudoers
Cmnd_Alias SHELLS = /bin/tcsh, /bin/sh, /bin/bash
Cmnd_Alias SYSADMIN = /usr/sbin/tcpdump
lee      ALL = (ALL) ALL
bob      ALL = NOPASSWD: SYSADMIN
```

Which of the following users, in addition to the root user, should be listed in the audit report as having root-level command-line access? (Choose two.)

A) Carl (Correct Answer)

B) Lee

C) Mallory (Correct Answer)

D) Eve

E) Bob

F) Alice

Explanation

Correct Answer:

AC

Community Discussion

Selected Answer: AB

Answer is Carl, as they share the same ID as the root, and Lee has the sudoers privileges. The confusion is that Mallory has the root directory listed as her home directory. However, this does not necessarily mean she has root level command line access, but she can view and manipulate files and directories within.

Selected Answer: AB

I think it is AB. Lee has an entry in sudoers, allowing him to run any command on the system lee ALL=(ALL) ALL Carl has the userId 0, the same as root. That sounds like a terrible idea to me, he should NOT share the same userId as root. That is like logging into the system as root user. Nothing here indicates to me that Mallory has admin level access.

The confusion is that Mallory has the root directory listed as her home directory. However, this does not necessarily mean she has root level command line access, but she can view and manipulate files and directories within.

The confusion is that Mallory has the root directory listed as her home directory. However, this does not necessarily mean she has root level command line access, but she can view and manipulate files and directories within.

NOT UNDERSTANDING THE PROVIDED ANSWERS ON THIS ONE... can anyone explain?

Question #219

The group named support is unable to make changes to the config file. An administrator is reviewing the permissions and sees the following:

```
$ ls -l config
-rw-rw----. 1 root app 4682 02-15 11:25 config
```

Which of the following should the administrator execute in order to give the support group access to modify the file while preserving the current ownership?

- A) chown :support config
- B) setfacl -m g:support:rw- config (Correct Answer)**
- C) chmod 664 config
- D) chmod g+s config

Explanation

Correct Answer:

B

Community Discussion

Question #220

Which of the following actions are considered good security practices when hardening a Linux server? (Choose two.)

A) Renaming the root account to something else

B) Removing unnecessary packages (Correct Answer)

C) Changing the default shell to /bin/csh

D) Disabling public key authentication

E) Disabling the SSH root login possibility (Correct Answer)

F) Changing the permissions on the root filesystem to 600

Explanation

Correct Answer:

BE

Community Discussion

Question #221

A Linux user is trying to execute commands with sudo but is receiving the following error:

```
$ sudo visudo
>>> /etc/sudoers: syntax error near line 28 <<<
sudo: parse error in /etc/sudoers near line 28
sudo: no valid sudoers sources found, quitting
```

The following output is provided:

```
# grep root /etc/shadow
root:*LOCK*:14600::::::
```

Which of the following actions will resolve this issue?

- A) Log in directly using the root account and comment out line 28 from /etc/sudoers.
- B) Boot the system in single user mode and comment out line 28 from /etc/sudoers. (Correct Answer)**
- C) Comment out line 28 from /etc/sudoers and try to use sudo again.
- D) Log in to the system using the other regular user, switch to root, and comment out line 28 from /etc/sudoers.

Explanation

Correct Answer:

B

Community Discussion

Selected Answer: B

So the root login is administratively locked. With *LOCK* in the password field, it means that nobody can log into the system as root user using a password. If the root account was configured for other authentication methods, like SSH keys or certificates, they could do so that way. That alone would not prevent regular users the ability to run "sudo", if they are configured properly in /etc/sudoers. That is because "sudo" uses the user's

password, not the root password. But, the sudoers file is misconfigured, maybe someone tried editing it directly with "vim" rather than "visudo" and broke it. Now it seems that nobody can perform sudo and nobody can log in as root (thus, no permission to fix the sudoers file). A good course of action is to boot into single user mode, which would allow root access to a simplified system without a password, and fix the syntax error.

Question #222

Users are experiencing high latency when accessing a web application served by a Linux machine. A systems administrator checks the network interface counters and sees the following:

```
# ip -s link list dev enp0s25

2: enp0s25: <BROADCAST,MULTICAST,LOWER_UP,UP> mtu 1500 qdisc fq_codel state DOWN mode DEFAULT group
default qlen 1000    link/ether ac:12:34:56:78:cd brd ff:ff:ff:ff:ff:ff

    RX: bytes  packets  errors  dropped missed  mcast

    2011664755 3579033 2394390 508      0        0

    TX: bytes  packets  errors  dropped carrier collsns

    309541780 1705408 0        0        12340    0
```

Which of the following is the most probable cause of the observed latency?

A) The network interface is disconnected.

B) A connection problem exists on the network interface. (Correct Answer)

C) No IP address is assigned to the interface.

D) The gateway is unreachable.

Explanation

Correct Answer:

B

Community Discussion

Selected Answer: B

There is some reachability on this network, we can rule out answers that would result in zero connectivity. We can see that in received packets (RX), we have lots of errors, and some dropped. In transmitted packets (TX), we have carrier failures. Carrier problems would result in a poor connection and difficulty communicating. Thus, the answer is something vaguely about connection problems on the network interface.

Question #223

A Linux administrator needs to create a symlink for `/usr/local/bin/app-a`, which was installed in `/usr/local/share/app-a`. Which of the following commands should the administrator use?

A) `ln -s /usr/local/bin/app-a /usr/local/share/app-a` (Correct Answer)

B) `mv -f /usr/local/share/app-a /usr/local/bin/app-a`

C) `cp -f /usr/local/share/app-a /usr/local/bin/app-a`

D) `rsync -a /usr/local/share/app-a /usr/local/bin/app-a`

Explanation

Correct Answer:

A

Community Discussion

Selected Answer: A

`ln -s /path/filename /linkpath/linkfilename`

Question #224

The applications team is reporting issues when trying to access the web service hosted in a Linux system. The Linux systems administrator is reviewing the following outputs:

Output 1:

```
* httpd.service = The Apache HTTPD Server
   Loaded: loaded (/usr/lib/systemd/system/httpd.service; disabled; vendor preset: disabled)
   Active: inactive (dead)
     Docs: man:httpd(8)
           man:apachectl(8)
```

Output 2:

```
16:51:16 up 28 min, 1 user, load average: 0.00, 0.00, 0.07
```

Which of the following statements best describe the root cause? (Choose two.)

- A) The httpd service is currently started.
- B) The httpd service is enabled to auto start at boot time, but it failed to start.
- C) The httpd service was manually stopped.
- D) The httpd service is not enabled to auto start at boot time. (Correct Answer)**
- E) The httpd service runs without problems.
- F) The httpd service did not start during the last server reboot. (Correct Answer)**

Explanation

Correct Answer:
DF

Community Discussion

Question #225

Which of the following specifications is used to perform disk encryption in a Linux system?

A) LUKS (Correct Answer)

B) TLS

C) SSL

D) NFS

Explanation

Correct Answer:

A

Community Discussion

Question #226

A junior developer is unable to access an application server and receives the following output:

```
[root@server1 ~]# ssh [email protected]
[email protected]'s password:
Permission denied, please try again.
[email protected]'s password:
Permission denied, please try again.
[email protected]'s password:
Account locked due to 4 failed logins
Account locked due to 5 failed logins
Last login: Mon Apr 22 21:21:06 2021 from 172.16.16.52
```

The systems administrator investigates the issue and receives the following output:

```
[root@server1 ~]# pam_tally2 --user=dev2
Login Failures Latest failure From
dev2 5 04/22/21 21:22:37 172.16.16.52
```

Which of the following commands will help unlock the account?

- A) `pam_tally2 --user=dev2 --quiet`
- B) `pam_tally2 --user=dev2`
- C) `pam_tally2 --user+dev2 --quiet`
- D) `pam_tally2 --user=dev2 --reset` (Correct Answer)**

Explanation

Correct Answer:

D

Community Discussion

Question #227

A systems technician is configuring an application server to accept packets from a trusted source with the IP address 192.168.10.22. Which of the following commands will allow communication between the two servers?

- A) `iptables -L -s 192.168.10.22 -j ACCEPT`
- B) `iptables -D INPUT -s 192.168.10.22 -j ACCEPT`
- C) `iptables -A INPUT -s 192.168.10.22 -j ACCEPT` (Correct Answer)**
- D) `iptables -A OUTPUT -s 192.168.10.22 -j ACCEPT`

Explanation

Correct Answer:

C

Community Discussion

C iptables is a firewall management tool that allows you to add or remove entries. iptables -A is the "append" command to add a rule in a firewall iptables -D is the "delete" command used to delete a rule in a firewall.

Question #228

An administrator needs to increase the system priority of a process with PID 2274. Which of the following commands should the administrator use to accomplish this task?

- A) `renice -n -15 2274` (Correct Answer)**
- B) `nice -15 2274`
- C) `echo "-15" > /proc/PID/2274/priority`
- D) `ps -ef | grep 2274`

Explanation

Correct Answer:

A

Community Discussion

B is correct also

Question #229

A Linux administrator is creating a new sudo profile for the accounting user. Which of the following should be added by the administrator to the sudo configuration file so that the accounting user can run `/opt/acc/report` as root?

- A) accounting localhost=/opt/acc/report
- B) accounting ALL=/opt/acc/report (Correct Answer)**
- C) %accounting ALL=(ALL) NOPASSWD: /opt/acc/report
- D) accounting /opt/acc/report=(ALL) NOPASSWD: ALL

Explanation

Correct Answer:
B

Community Discussion

Question #230

An administrator needs to make an application change via a script that must be run only in console mode. Which of the following best represents the sequence the administrator should execute to accomplish this task?

- A) systemctl isolate multi-user.target**
sh script.sh
systemctl isolate graphical.target (Correct Answer)
- B) systemctl isolate graphical.target
sh script.sh
systemctl isolate multi-user.target

C) sh script.sh
systemctl isolate multi-user.target
systemctl isolate graphical.target
D) systemctl isolate multi-user.target
systemctl isolate graphical.target
sh script.sh

Explanation

Correct Answer:

A

Community Discussion

Question #231

A Linux administrator logs in to a system and identifies that an important backup has been started. The backup process is consuming a considerable amount of CPU time but needs to continue. Which of the following should the administrator use to reduce the impact this process has on other services?

A) renice -n 15 -p (Correct Answer)

- B) nice -n 15 -p
- C) renice -n -15 -p
- D) nice -n -15 -p

Explanation

Correct Answer:

A

Community Discussion

1 to 20 Runs the specified processes slower than the base priority. 0 Sets priority of the specified processes to the base scheduling priority. -20 to -1 Runs the specified processes quicker than the base priority.

Question #232

The application team has reported latency issues that are causing the application to crash on the Linux server. The Linux administrator starts troubleshooting and receives the following output:

```
# netstat -s
15762 packets pruned from receive queue because of socket buffer over
690 times the listen queue of a socket overflowed
690 SYNs to LISTEN sockets ignored
2150128 packets collapsed in receive queue due to low socket buffer
TCPBacklogDrop: 844165

# ethtool -s eth0
rx_fw_discards: 4487
```

Which of the following commands will improve the latency issue?

A) # echo 'net.core.net_backlog = 5000000' >> /etc/sysctl.conf

sysctl -p

systemctl daemon-reload

B) # ifdown eth0

ip link set dev eth0 mtu 800

ifup eth0

C) # systemctl stop network

ethtool -g eth0 512

systemctl start network

D) # echo 'net.core.rmem_max = 12500000' >> /etc/sysctl.conf

echo 'net.core.wmem_max = 12500000' >> /etc/sysctl.conf

sysctl -p (Correct Answer)

Explanation

Correct Answer:

D

Community Discussion

Selected Answer: D

A: net.core.net_backlog does not exist AFAIK. There is a similar config called net.core.netdev_max_backlog. That one determines how many incoming packets can wait in queue on your network interface. D: net.core.rmem_max configures how much data your network interface can receive (or read), kind of like the allowed size of your mailbox. net.core.wmem_max configures the same for your "outbox" size. "sysctl -p" will load your new configuration from /etc/sysctl.conf.

Question #233

An administrator recently updated the BIND software package and would like to review the default configuration that shipped with this version. Which of the following files should the administrator review?

A) /etc/named.conf.rpmnew (Correct Answer)

- B) /etc/named.conf.rpmsave
- C) /etc/named.conf
- D) /etc/bind/bind.conf

Explanation

Correct Answer:

A

Community Discussion

Should this be "B" or am I not understanding this??? An .rpmnew file contains the new default configuration file and leaves your original configuration file untouched. By contrast, and .rpmsave file is a copy of your original configuration file, which has been replaced by the new default file.

Selected Answer: C

The default configuration file for BIND (Berkeley Internet Name Domain) DNS server is typically located at: C. `/etc/named.conf` `/etc/named.conf.rpmnew` and `/etc/named.conf.rpmnew` are likely created during updates to preserve the previous configuration files. They usually represent configurations that have been changed or replaced during package updates. `/etc/bind/bind.conf` is not a standard location for the default BIND configuration file. It's possible that the BIND configuration directory might be `/etc/bind`, but the main configuration file is typically named `named.conf`. Therefore, the administrator should review the file located at `/etc/named.conf` to review the default configuration that shipped with the updated BIND software package.

For Debian correct is: D. `/etc/bind/bind.conf`

Question #234

A systems administrator received a request to change a user's credentials. Which of the following commands will grant the request?

A) `sudo passwd` (Correct Answer)

- B) `sudo userdel`
- C) `sudo chage`
- D) `sudo usermod`

Explanation

Correct Answer:

A

Community Discussion

Selected Answer: A

It would depend on how you define credentials, but `sudo passwd` would be used to specifically change the password. `sudo usermod` would be used to change other attributes about the user account.

Selected Answer: D

`sudo usermod`

to change a user's credentials: D. sudo usermod to change a user's password: A. sudo passwd

Question #235

A user is unable to log on to a Linux workstation. The systems administrator executes the following command:

```
cat /etc/shadow | grep user1
```

The command results in the following output:

```
user1:!!$6$QERgAsdvojadv4asdvaarC/9dj34GdafGVaregmksfa:18875:0:99999:7:::
```

Which of the following should the systems administrator execute to fix the issue?

- A) chown -R user1:user1 /home/user1
- B) sed -i '!:/:/g' /etc/shadow
- C) chgrp user1:user1 /home/user1

D) passwd -u user1 (Correct Answer)

Explanation

Correct Answer:

D

Community Discussion

Selected Answer: D

The exclamation mark at the beginning of the user's hashed password indicates that the account is locked. The user may not log in with this password. This is typically used to temporarily disable an account. You can unlock the account by setting a new password for the user with `passwd`.

Question #236

A systems administrator is tasked with creating an Ansible playbook to automate the installation of patches on several Linux systems. In which of the following languages should the playbook be written?

- A) SQL
- B) YAML (Correct Answer)**
- C) HTML
- D) JSON

Explanation

Correct Answer:
B

Community Discussion

Question #237

A systems administrator is working on a security report from the Linux servers. Which of the following commands can the administrator use to display all the firewall rules applied to the Linux servers? (Choose two.)

- A) ufw limit
- B) iptables -F
- C) systemctl status firewalld
- D) firewall-cmd --list-all (Correct Answer)**
- E) ufw status (Correct Answer)**
- F) iptables -A

Explanation

Correct Answer:
DE

Community Discussion

Selected Answer: DE

A: `ufw limit` applies rate limiting to a given port. It is meant to prevent an IP from attacking the system, such as an overwhelming amount of SSH login attempts. Typically for Debian systems. B: `iptables -F` means to flush all the chains in the table, with no chain specified. This would delete all firewall rules. C: `systemctl status firewalld` would tell you if the firewall daemon is running, but wouldn't show you any firewall rules. D: `firewall-cmd --list-all` does as it says. It will list its firewall rules. E: `ufw status` would show you whether ufw is active or not, and also an "uncomplicated" list of its firewall rules. Typically for Debian systems. F: `iptables -A` is meant for appending a new firewall rule. It not not a complete command.

Question #238

A Linux administrator intends to start using KVM on a Linux server. Which of the following commands will allow the administrator to load the KVM module as well as any related dependencies?

A) `modprobe kvm` (Correct Answer)

- B) `insmod kvm`
- C) `depmod kvm`
- D) `hotplug kvm`

Explanation

Correct Answer:

A

Community Discussion

Question #239

The group owner of the `/home/test` directory would like to preserve all group permissions on files created in the directory. Which of the following commands should the group owner execute?

A) `chmod g+s /home/test` (Correct Answer)

B) `chgrp test /home/test`

C) `chmod 777 /home/test`

D) `chown -hR test /home/test`

Explanation

Correct Answer:

A

Community Discussion

Selected Answer: A

Commonly noted as SGID, this special permission has a couple of functions: If set on a file, it allows the file to be executed as the group that owns the file (similar to SUID). If set on a directory, any files created in the directory will have their group ownership set to that of the directory owner.

Question #240

The administrator comptia is not able to perform privileged functions on a newly deployed system. Given the following command outputs:

```
[root@newserver ~]# id comptia
uid=1000(comptia) gid=1000(comptia) groups=1000(comptia)

[root@newserver ~]# cat /etc/sudoers.d/admin
%admin ALL= (root) NOPASSWD: EXEC: /usr/bin/ps, /usr/bin/chmod, /usr/bin/yum, /usr/bin/cat, /usr/sbin/lvm, /usr/sbin/pvs

[root@newserver ~]# grep comptia /etc/passwd
comptia:x:1000:1000:comptia:/home/comptia:/bin/bash

[root@newserver ~]# chage -l comptia
Last password change : never
Password expires : never
Password inactive : never
Account expires : never
Minimum number of days between password change : 0
Maximum number of days between password change : 99999
Number of days of warning before password expires : 7
```

Which of the following is the reason that the administrator is unable to perform the assigned duties?

- A) The administrator needs a password reset.
- B) The administrator is not a part of the correct group. (Correct Answer)**
- C) The administrator did not update the sudo database.
- D) The administrator's credentials need to be more complex.

Explanation

Correct Answer:

B

Community Discussion

Selected Answer: B

Yes it is B. But, I am suspicious of this line here: %admin ALL=(root) ... This suggests to me "In the admin group, on ALL systems, only the user in the group named 'root' is allowed to perform the listed commands" What you actually want is `%admin ALL=(ALL) ...` so that any user in the admin group can run the commands.

Question #241

A Linux administrator deployed a large-scale application service as a containerized pod. The Linux administrator needs to continually send the application's log file to the company's central log store. Which of the following should the Linux administrator do to efficiently perform this task?

A) Configure a sidecar to perform log shipping. (Correct Answer)

- B) Run `docker exec` to access the logs within the container.
- C) Execute `docker inspect` and run a `log rsync` replication.
- D) Set up a custom task scheduler via `kubectrl`.

Explanation

Correct Answer:

A

Community Discussion

Question #242

A systems administrator is trying to track down a rogue process that has a TCP listener on a network interface for remote command-and-control instructions. Which of the following commands should the systems administrator use to generate a list of rogue process names? (Choose two.)

A) `netstat -antp | grep LISTEN` (Correct Answer)

B) `lsof -iTCP | grep LISTEN` (Correct Answer)

- C) `lsof -i:22 | grep TCP`
- D) `netstat -a | grep TCP`
- E) `nmap -p1-65535 | grep -i tcp`
- F) `nmap -sS 0.0.0.0/0`

Explanation

Correct Answer:

AB

Community Discussion

The answers make sense to me. I would add that ``nmap`` is also a good command to run in this situation, but the syntax would go something like: ``nmap -p1-65535 127.0.0.1 | grep -i tcp`` Scan every port between 0 and 65535 on the local host, grep for lines containing "tcp" (ignore case) in the output.

Question #243

A Linux engineer finds multiple failed login entries in the security log file for application users. The Linux engineer performs a security audit and discovers a security issue. Given the following:

```
# grep -iE '*www*|db' /etc/passwd
www-data:x:502:502:www-data:/var/www:/bin/bash
db:x:505:505:db:/opt/db:/bin/bash
```

Which of the following commands would resolve the security issue?

- A) `usermod -d /srv/www-data www-data && usermod -d /var/lib/db db`
- B) `passwd -u www-data && passwd -u db`
- C) `renice -n 1002 -u 502 && renice -n 1005 -u 505`
- D) `chsh -s /bin/false www-data && chsh -s /bin/false db` (Correct Answer)**

Explanation

Correct Answer:

D

Community Discussion

Selected Answer: D

Application users should not be allowed for shell login. Disable shell for users to stop failed logins

Question #244

A systems administrator is cloning a system partition `/dev/sda1` to `/dev/sdc1`. Which of the following commands would allow the administrator to accomplish this task most efficiently?

- A) `dd if=/dev/sda1 of=/dev/sdc1` (Correct Answer)**
- B) `cat /dev/sda1 > /dev/sdc1`

- C) `cp -R /dev/sda1 /dev/sdc1`
- D) `rsync -av /dev/sda1 /dev/sdc1`

Explanation

Correct Answer:

A

Community Discussion

Question #245

An administrator would like to securely connect to a server and forward port 8080 on a local machine to port 80 on the server. Which of the following commands should the administrator use to satisfy both requirements?

A) `ssh -L 8080:localhost:80 admin@server` (Correct Answer)

- B) `ssh -R 8080:localhost:80 admin@server`
- C) `ssh -L 80:localhost:8080 admin@server`
- D) `ssh -R 80:localhost:8080 admin@server`

Explanation

Correct Answer:

A

Community Discussion

Question #246

A Linux administrator needs to transfer a local file named `accounts.pdf` to a remote `/tmp` directory of a server with the IP address `10.10.10.80`. Which of the following commands needs to be executed to transfer this file?

- A) `rsync [email protected]:/tmp accounts.pdf`

- B) scp accounts.pdf :/tmp** (Correct Answer)
- C) cp :/tmp accounts.pdf
- D) ssh accounts.pdf :/tmp

Explanation

Correct Answer:
B

Community Discussion

all IPs are spelled incorrect: l0 instead 10

Question #247

Application code is stored in Gil. Due to security concerns, the DevOps engineer does not want to keep a sensitive configuration file, app.conf, in the repository. Which of the following should the engineer do to prevent the file from being uploaded to the repository?

- A) Run git exclude app.conf.
- B) Run git stash app.conf.
- C) Add app.conf to .exclude.
- D) Add app.conf to .gitignore. (Correct Answer)**

Explanation

Correct Answer:
D

Community Discussion

Question #248

An administrator has source code and needs to rebuild a kernel module. Which of the following command sequences is most commonly used to rebuild this type of module?

**A) ./configure
make
make install (Correct Answer)**

B) wget
gcc
cp
C) tar xvzf
build
cp
D) build
install
configure

Explanation

Correct Answer:
A

Community Discussion

Question #249

An administrator would like to measure possible packet loss between a workstation and a remote web application that is running on port 443. Which of the following would be the best command for the administrator to use to display this information?

A) ping -c 50
B) tcpdump -p 443
C) mtr -T -P 443 (Correct Answer)
D) traceroute -p 443

Explanation

Correct Answer:

C

Community Discussion

The mtr command (My Traceroute) is a network diagnostic tool that combines the functionality of ping and traceroute. The -T option specifies the use of TCP mode, and the -P 443 option sets the destination port to 443.

Selected Answer: C

ping works on ICMP/UDP. MTR allows to use specific port.

Selected Answer: A

ping: This command is commonly used to check network connectivity and measure round-trip time. The -c option allows you to specify the number of packets to send, and in this case, it's set to 50. By sending multiple packets, you can get a more accurate measure of packet loss.

Question #250

Users in the human resources department are trying to access files in a newly created directory. Which of the following commands will allow the users access to the files?

A) chattr

B) chgrp (Correct Answer)

C) chage

D) chcon

Explanation

Correct Answer:

B

Community Discussion

Question #251

A Linux administrator is troubleshooting an issue in which users are not able to access `https://portal.comptia.org` from a specific workstation. The administrator runs a few commands and receives the following output:

```
# cat /etc/hosts
```

```
10.10.10.55 portal.comptia.org
```

```
# host portal.comptia.org
```

```
portal.comptia.org has address 192.168.1.55
```

```
#cat /etc/resolv.conf
```

```
nameserver 10.10.10.5
```

Which of the following tasks should the administrator perform to resolve this issue?

A) Update the name server in `resolv.conf` to use an external DNS server.

B) Remove the entry for `portal.comptia.org` from the local hosts file. (Correct Answer)

C) Add a network route from the `10.10.10.0/24` to the `192.168.0.0/16`.

D) Clear the local DNS cache on the workstation and rerun the `host` command.

Explanation

Correct Answer:

B

Community Discussion

Selected Answer: B

The issue is likely caused by a conflicting entry for portal.comptia.org in the local /etc/hosts file, which is mapping the domain to an incorrect IP address (10.10.10.55). The correct IP address for portal.comptia.org according to DNS resolution is 192.168.1.55. Option A: Updating the name server in resolv.conf to use an external DNS server wouldn't resolve the issue since the issue is related to the local hosts file. Option C: Adding a network route wouldn't address the DNS resolution issue. Option D: Clearing the local DNS cache and rerunning the host command might help if there were changes made to DNS configurations, but the root cause here is the conflicting entry in the hosts file. Removing the entry for portal.comptia.org from the local hosts file (/etc/hosts) will allow the system to resolve the domain through DNS properly.

Selected Answer: D

I argue that B does not make sense here. We can see that there is a local DNS entry mapping portal.comptia.org -> 10.10.10.55. But, the actual DNS resolution, from the `host` command, is different, portal.comptia.org -> 192.168.1.55. Something is taking priority over the local DNS in /etc/hosts. If we delete the entry in /etc/hosts, I doubt it will resolve our connection problem, because the system is getting its DNS resolution from something else, possibly a cache. I think we should flush the local DNS cache and try again. We should also verify what is the correct IP it should resolve to. Perhaps we SHOULD also remove the entry from /etc/hosts, but for the reason that the private DNS server (at 10.10.10.5) should be responsible for resolving DNS, not workstations.

Question #252

A Linux administrator is trying to start the database service on a Linux server but is not able to run it. The administrator executes a few commands and receives the following output:

```
#systemctl status mariadb
mariadb.service
   Loaded: masked (Reason: Unit mariadb.service is masked)
   Active: inactive (dead)

#systemctl enable mariadb
Failed to enable unit: ...

#systemctl start mariadb
Failed to start mariadb.service ...
```

Which of the following should the administrator run to resolve this issue?
(Choose two.)

A) systemctl unmask mariadb (Correct Answer)

B) journalctl -g mariadb

C) dnf reinstall mariadb

D) systemctl start mariadb (Correct Answer)

E) chkconfig mariadb on

F) service mariadb reload

Explanation

Correct Answer:

AD

Community Discussion

Question #253

An administrator would like to list all current containers, regardless of their running state. Which of the following commands would allow the administrator to accomplish this task?

A) docker ps -a (Correct Answer)

B) docker list

C) docker image ls

D) docker inspect image

Explanation

Correct Answer:

A

Community Discussion

Question #254

Which of the following should be used to verify the integrity of a file?

A) sha256sum (Correct Answer)

B) fsck

C) gpg -d

D) hashcat

Explanation

Correct Answer:

A

Community Discussion

The question is a bit vague IMO. What are you comparing your file to? Where did you get this file? You can use sha526sum to generate a checksum of a given file. Then, you could compare your checksum with the checksum of the source file, assuming one exists.

Question #255

An administrator created an initial Git repository and uploaded the first files. The administrator sees the following when listing the repository:

```
__init__.py    Initial Commit Just now
main.py        Initial Commit Just now
.DS_Store      Initial Commit Just now
setup.sh       Initial Commit Just now
README.md      Initial Commit Just now
```

The administrator notices the file .DS_Store should not be included and deletes it from the online repository. Which of the following should the administrator run from the root of the local repository before the next commit to ensure the file is not uploaded again in future commits?

- A) `rm -f .DS_Store && git push`
- B) `git fetch && git checkout .DS_Store`
- C) `rm -f .DS_Store && git rebase origin main`
- D) `echo .DS_Store >> .gitignore` (Correct Answer)**

Explanation

Correct Answer:

D

Community Discussion

Question #256

An administrator runs `ping comptia.org`. The result of the command is:

```
ping: comptia.org: Name or service not known
```

Which of the following files should the administrator verify?

- A) `/etc/ethers`
- B) `/etc/services`
- C) `/etc/resolv.conf` (Correct Answer)**
- D) `/etc/sysctl.conf`

Explanation

Correct Answer:

C

Community Discussion

Question #257

A new application container was built with an incorrect version number. Which of the following commands should be used to rename the image to match the correct version 2.1.2?

- A) `docker tag comptia/app:2.1.1 comptia/app:2.1.2` (Correct Answer)**
- B) `docker push comptia/app:2.1.1 comptia/app:2.1.2`
- C) `docker rmi comptia/app:2.1.1 comptia/app:2.1.2`
- D) `docker update comptia/app:2.1.1 comptia/app:2.1.2`

Explanation

Correct Answer:

A

Community Discussion

Question #258

Which of the following is the best tool for dynamic tuning of kernel parameters?

A) tuned (Correct Answer)

B) tune2fs

C) tuned-adm

D) turbostat

Explanation

Correct Answer: A [?](#)

Community Discussion

Question #259

A non-privileged user is attempting to use commands that require elevated account permissions, but the commands are not successful. Which of the following most likely needs to be updated?

A) /etc/passwd

B) /etc/shadow

C) /etc/sudoers (Correct Answer)

D) /etc/bashrc

Explanation

Correct Answer: C [?](#)

Community Discussion

Question #260

A network administrator issues the `dig www.comptia.org` command and receives an NXDOMAIN response. Which of the following files should the administrator check first?

A) `/etc/resolv.conf` (Correct Answer)

B) `/etc/hosts`

C) `/etc/sysconfig/network-scripts`

D) `/etc/nsswitch.conf`

Explanation

Correct Answer: A [?](#)

Community Discussion

Question #261

SIMULATION

-

A new drive was recently added to a Linux system. Using the environment and tokens provided, complete the following tasks:

- Create an appropriate device label.
- Format and create an ext4 file system on the new partition.

The current working directory is `/`.

INSTRUCTIONS

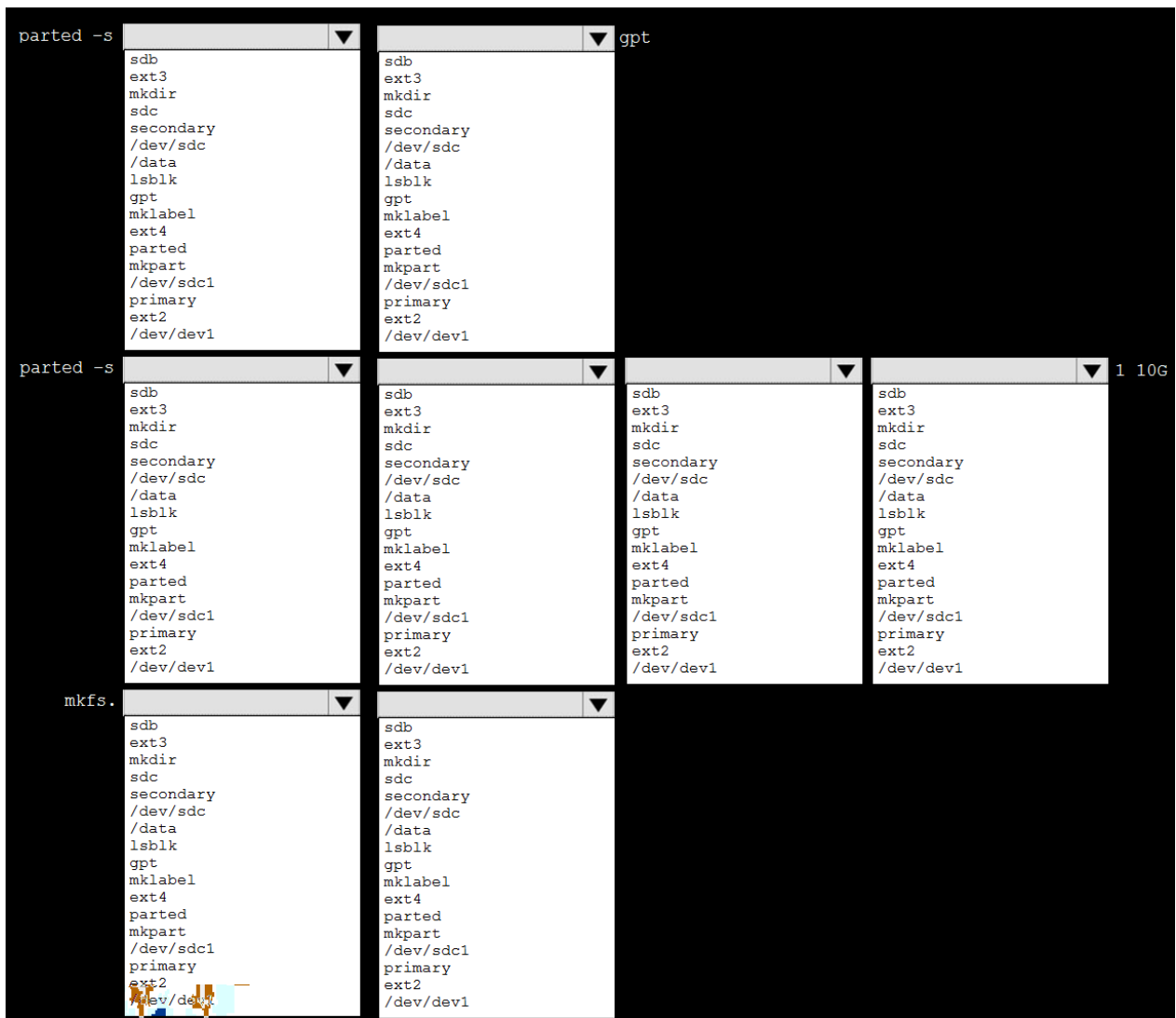
-

Use the drop-down menus to select the proper objects to complete these tasks.

Some commands are partially complete.

Not all objects will be used, and some may be used more than once.

If at any time you would like to bring back the initial state of the simulation, please click the Reset All button.



Explanation

Correct Answer:

Step 1: Create a Device Label

Command: `parted -s` - This command initiates the partitioning tool.

Choose: `/dev/sdc` - This is the new device added, likely `/dev/sdc`.

Action: `mklabel gpt` - This command will create a GPT partition table on `/dev/sdc`.

Step 2: Create a New Partition

Command: `parted -s` - Again, use `parted` to partition the drive.

Choose: `/dev/sdc` - Still working with the new drive.

Action: `mkpart primary ext4 1 10G` - This command creates a new primary partition with ext4 file system type from 1GB to 10GB on `/dev/sdc`.

Step 3: Format the New Partition to ext4

Command: `mkfs.ext4` - This will format the new partition.

Choose: `/dev/sdc1` - The partition created is `/dev/sdc1`, so use it to create the ext4 file system.

Community Discussion

Question #262

A Linux systems administrator has to write a script to be used to replace an IP address in a network configuration file.

Given the following file content:

```
# cat /etc/netplan/private.yaml
network:
  version: 2
  ethernets:
    enp0s8:
      addresses:
        - 10.10.10.2/24
      routes:
        - to: default
          via: 10.10.10.1
```

Which of the following commands will accomplish this task?

- `NEWIP=$1`
 A) `sed -i 's/10.10.10.2/$NEWIP/' /etc/netplan/private.yaml`

```
NEWIP=$1
```

B) `sed -i "s/10.10.10.2/$NEWIP/" /etc/netplan/private.yaml`

(Correct Answer)

```
NEWIP=$1
```

C) `sed "s/10.10.10.2/$NEWIP/" /etc/netplan/private.yaml`

```
NEWIP=$1
```

D) `sed -i "/10.10.10.2/$NEWIP/" /etc/netplan/private.yaml`

Explanation

Correct Answer: B ?

Community Discussion

Question #263

An administrator attempts to connect to a remote server by running the following command:

```
$ nmap 192.168.10.36
```

```
Starting Nmap 7.60 ( https://nmap.org ) at 2022-03-29 20:20 UTC
Nmap scan report for www1 (192.168.10.36)
Host is up (0.000091s latency).
Not shown: 979 closed ports
PORT STATE SERVICE
21/tcp open  ftp
22/tcp filtered ssh
631/tcp open  ipp
```

```
Nmap done: 1 IP address (1 host up) scanned in 0.06 seconds
```

Which of the following can be said about the remote server?

A) A firewall is blocking access to the SSH server. (Correct Answer)

B) The SSH server is not running on the remote server.

C) The remote SSH server is using SSH protocol version 1.

D) The SSH host key on the remote server has expired.

Explanation

Correct Answer: A [?](#)

Community Discussion

Question #264

A systems administrator just downloaded a source code package with the file extension TGZ. Which of the following commands would decompress and unarchive the package?

- A) tar -xf
- B) tar -zcvpf
- C) tar -xvf
- D) tar -zxvf (Correct Answer)**

Explanation

Correct Answer: D [?](#)

Community Discussion

Question #265

A Linux administrator would like to run the cleanup script /home/admin/script.sh at 9:00 p.m. on March 31. Which of the following commands should the administrator use to accomplish this task?

- A) at -f /home/admin/script.sh 9pm March 31
- B) echo < /home/admin/script.sh | at 9pm March 31 (Correct Answer)**
- C) at 9pm March 31 | echo > /home/admin/script.sh
- D) at 9pm March 31 -f /home/admin/script.sh

Explanation

Correct Answer: B [?](#)

Community Discussion

Question #266

Which of the following will prevent non-root SSH access to a Linux server?

A) Creating the /etc/nologin file (Correct Answer)

- B) Creating the /etc/nologin.allow file containing only a single line root
- C) Creating the /etc/nologin/login.deny file containing a single line +all
- D) Ensuring that /etc/pam.d/sshd includes account sufficient pam_nologin.so

Explanation

Correct Answer: A [?](#)

Community Discussion

Question #267

A Linux administrator attempts to log in to a server over SSH as root, and receives the following error message: Permission denied, please try again. The administrator is able to log in to the console of the server directly with root and has confirmed the password is correct. The administrator reviews the configuration of the SSH service and gets the following output:

```
Port 22
PermitRootLogin prohibit-password
PasswordAuthentication yes
PermitEmptyPassword no
Use PAM no
MaxSessions 1
MaxAuthTries 3
```

Based on the above output, which of the following will most likely allow the administrator to log in over SSH to the server?

- A) Log out other user sessions because only one is allowed at a time.
- B) Enable PAM and configure the SSH module.
- C) Modify the SSH port to use 2222.

D) Use a key to log in as root over SSH. (Correct Answer)

Explanation

Correct Answer: D [?](#)

Community Discussion

Question #268

A Linux administrator needs to determine if prerequisites are met. One of the application requirements is to install Perl on a system. Which of the following commands would accomplish this task?

- A) rpm -Uf perl

B) rpm -iv perl (Correct Answer)

- C) rpm -qa perl
- D) rpm -eh perl

Explanation

Correct Answer: B [?](#)

Community Discussion

Question #269

A server administrator has scheduled a cron job to run YUM updates on all servers without manual intervention, but kernel packages must be excluded from the updates at all times.

Which of the following configuration files need to be modified to make these changes persistent?

- A) /etc/yum.conf (Correct Answer)**
- B) /etc/dnf/dnf.conf
- C) /etc/yum/pluginconf.d/packages.conf
- D) /etc/yum.repo.d/kernel.repo

Explanation

Correct Answer: A [?](#)

Community Discussion

Question #270

SIMULATION

-

A senior administrator has placed a private key for user admin in your home directory. The server you need to remotely access is server1 and SSH is listening on port 2222.

INSTRUCTIONS

-

Part 1

-

Review the command output and build the correct command to place the private key into your SSH folder.

Part 2

-

Review the command output and build the correct command to set the file permissions.

Part 3

-

Review the command output and build the correct command to set the correct ownership.

In each part, click on objects to build a complete command. Command objects may be used more than once, but not all will be used. Click the arrow to remove any unwanted objects from your command.

Part 4

-

Select the proper file to edit for remote server access. Then, build the correct configuration output based on the server name, ports, and files.

If at any time you would like to bring back the initial state of the simulation, please click the Reset All button.

Part 1

Part 2

Part 3

Part 4

Local host command output

cat ~/.ssh/authorized_keys
cat ~/.ssh/known_hosts
ls -l .ssh
ls -l ~
ls -l ~/.ssh

ls -l /etc/ssh
ls -Z .ssh
ls -Z ~/.ssh
ssh-add ~/.ssh/id_rsa
ssh-keyscan -H server1 >> ~/.ssh/known_hosts

1777	chown	mv	server1	:
0444	comptia	root	/etc/ssh	chcon
0500	;	chage	0600	localhost
~/	~/.ssh/	admin	0777	chmod
server2	-	-R	/	~

[comptia@localhost ~] \$

Part 1

Part 2

Part 3

Part 4

Local host command output

```

cat ~/.ssh/authorized_keys      ls -l /etc/ssh
cat ~/.ssh/known_hosts          ls -Z .ssh
ls -l .ssh                      ls -Z ~/.ssh
ls -l ~                         ssh-add ~/.ssh/id_rsa
ls -l ~/.ssh                    ssh-keyscan -H server1 >> ~/.ssh/known_hosts

```

1777	chown	mv	server1	:
0444	comptia	root	/etc/ssh	chcon
0500	;	chage	0600	localhost
~/	~/.ssh/	admin	0777	chmod
server2	-	-R	/	↩

[comptia@localhost ~] \$



Part 1

Part 2

Part 3

Part 4

Local host command output

```

cat ~/.ssh/authorized_keys      ls -l /etc/ssh
cat ~/.ssh/known_hosts          ls -Z .ssh
ls -l .ssh                      ls -Z ~/.ssh
ls -l ~                         ssh-add ~/.ssh/id_rsa
ls -l ~/.ssh                    ssh-keyscan -H server1 >> ~/.ssh/known_hosts

```

1777	chown	mv	server1	:
0444	comptia	root	/etc/ssh	chcon
0500	;	chage	0600	localhost
~/	~/.ssh/	admin	0777	chmod
server2	-	-R	/	↩

[comptia@localhost ~] \$



Select the proper file to edit for remote server access:

- ☐ ~/.ssh/config
- ☐ ~/.ssh/known_hosts
- ☐ ~/.ssh/server1
- ☐ ~/.ssh/authorized_keys

Select the command that generated the output in tab 1:

Host server 1

Select output	Select output
Select output	Select output
2222	2222
server1	server1
RemoteServer	RemoteServer
admin	admin
Compression	Compression
~/.ssh/authorized_keys	~/.ssh/authorized_keys
User	User
Port	Port
Yes	Yes
22	22
comptia	comptia
~/.ssh/server1	~/.ssh/server1
INFO	INFO
Host	Host
HostName	HostName
4444	4444
IdentityFile	IdentityFile
~/.ssh/id_rsa	~/.ssh/id_rsa
LogLevel	LogLevel
Select output	Select output
Select output	Select output
2222	2222
server1	server1
RemoteServer	RemoteServer
admin	admin
Compression	Compression
~/.ssh/authorized_keys	~/.ssh/authorized_keys
User	User
Port	Port
Yes	Yes
22	22
comptia	comptia
~/.ssh/server1	~/.ssh/server1
INFO	INFO
Host	Host
HostName	HostName
4444	4444
IdentityFile	IdentityFile
~/.ssh/id_rsa	~/.ssh/id_rsa
LogLevel	LogLevel
Select output	Select output
Select output	Select output
2222	2222
server1	server1
RemoteServer	RemoteServer
admin	admin
Compression	Compression
~/.ssh/authorized_keys	~/.ssh/authorized_keys
User	User
Port	Port
Yes	Yes
22	22
comptia	comptia
~/.ssh/server1	~/.ssh/server1
INFO	INFO
Host	Host
HostName	HostName
4444	4444
IdentityFile	IdentityFile
~/.ssh/id_rsa	~/.ssh/id_rsa
LogLevel	LogLevel

Explanation

Correct Answer:

Part 1: Place the private key into the SSH folder

The goal is to move the private key for user `admin` to the `.ssh` folder and make sure it is named appropriately, using only the blocks provided in the image.

Based on the available blocks, here is how you can construct the correct command. Use the `mv` command to move the key file: `mv ~/admin ~/.ssh/`

This will move the `admin` key file from your home directory to the `.ssh` folder.

Part 2: Set the correct file permissions

To set the correct permissions for the private key file, you need to use the `chmod` command. The goal is to ensure that the private key file can only be read and written by the user, which is typically done by setting permissions to `0600`.

```
chmod 0600 ~/.ssh/admin
```

`chmod` is the command to change the permissions of a file.

`0600` sets the permission so that only the owner can read and write the file.

`~/.ssh/admin` refers to the private key file in the `.ssh` directory.

Part 3: Set the correct ownership

To set the correct ownership of the private key file, you need to use the `chown` command. The ownership should be assigned to the user `admin` to ensure that only this user can manage and use the key.

```
chown admin ~/.ssh/admin
```

`chown` is the command used to change the ownership of a file or directory.

`admin` is the username that should own the file.

`~/.ssh/admin` specifies the path to the private key file in the `.ssh` directory.

`chown` - to set the ownership.

`admin` - the username to assign as the owner.

`~/.ssh/` - the directory where the file is located.

`admin` - the private key file name.

Part 4: Edit the configuration file

For SSH configuration related to remote server access, the proper file to edit is `~/.ssh/config`.

This file allows you to set up shortcuts and configuration options for different hosts you want to connect to, simplifying the SSH connection process.

The SSH configuration file (`~/.ssh/config`) should contain details for connecting to `server1` with the specified ports and keys. Based on the output from the given image, the configuration file should look like:

```
Host server1
    HostName server1
    Port 2222
    User admin
    IdentityFile ~/.ssh/id_rsa
```

Host `server1`: This is the alias you use to connect (e.g., `ssh server1`).

HostName `server1`: The hostname or IP address of the server.

Port `2222`: The SSH port to use, which in this case is `2222` (non-default).

User `admin`: The user to connect as on the remote server.

IdentityFile `~/.ssh/id_rsa`: The private key file used for authentication.

Community Discussion

Question #271

A systems administrator is configuring a Linux system so the network traffic from the internal network 172.17.0.0/16 going out through the eth0 interface would appear as if it was sent directly from this interface. Which of the following commands will accomplish this task?

A) iptables -A POSTROUTING -s 172.17.0.0/16 -o eth0 -j MASQUERADE (Correct Answer)

B) firewallld -A OUTPUT -s 172.17.0.0/16 -o eth0 -j DIRECT

C) nmcli masq-traffic eth0 -s 172.17.0.0/16 -j MASQUERADE

D) ifconfig --nat eth0 -s 172.17.0.0/16 -j DIRECT

Explanation

Correct Answer: A [?](#)

Community Discussion

Question #272

An administrator is running a web server in a container named WEB, but none of the error output is showing. Which of the following should the administrator use to generate the errors on the container?

A) docker-compose inspect WEB

B) docker logs WEB (Correct Answer)

C) docker run --name WEB --volume/dev/stdout:/var/log/nginx/error.log

D) docker ps WEB -f

Explanation

Correct Answer: B [?](#)

Community Discussion

Question #273

A DevOps engineer pushed the updated configuration to an existing branch of a remote Git repository. Which of the following commands should the Linux administrator use to obtain these configuration changes?

A) git pull (Correct Answer)

B) git log

C) git fetch

D) git checkout main

Explanation

Correct Answer: A [?](#)

Community Discussion

Question #274

Users are reporting that a production application has slow performance. A systems administrator logs in to the server and performs some basic checks. Given the following:

```
[root@comptia ~]# ps -o pcpu; pidstat 1
Average:  UID      PID      %usr    %system    %wait    %CPU    Command
Average:           0   937242    32.39         0    68.21    32.39app_prd
Average:           0   937243    44.97        0.31    51.57    45.28app_prd
Average:           0   937244    28.62         0    68.55    28.62app_prd
Average:           0   937245    47.17         0    48.11    47.17app_prd
Average:           0   937246    27.67         0    68.24    27.67app_prd
Average:           0   937247     23.9         0    72.33     23.9app_prd
Average:           0   937248    44.97         0    49.06    44.97app_prd
Average:           0   937249    33.33         0    62.89    33.33app_prd
Average:           0   937250    43.4         0    52.52    43.4app_prd
Average:           0   937251    30.5         0    64.78    30.5app_prd
Average:           0   937271     1.26        2.83     5.35     4.09pidstat

[root@comptia ~]# vmstat 1
procs -----memory----- --swap--  -----io-----  -system--  -----cpu-----
 r b swpd free buff cache si so bi bo in cs us sy id wa st
16 0 978912 376320 50804 10296312 0 0 0 44 5503 2962 98 2 0 0 0
16 0 978912 376544 50804 10296316 0 0 0 48 5447 2933 98 2 0 0 0
15 0 978912 376432 50804 10296316 0 0 0 0 5448 2861 95 5 0 0 0

[root@comptia ~]# free -m
total used free shared buff/cache available
Mem: 15533 5063 365 164 10104 9975
Swap: 7931 955 6976
```

Which of the following is causing the performance issue?

- A) The server does not have enough memory.
- B) Too many processes are running on the server.
- C) The server CPU is receiving too much load. (Correct Answer)**
- D) The server is swapping.

Explanation

Correct Answer: C ?

Community Discussion

Question #275

A Linux administrator is updating the password for the logcollect account. The administrator issues the following commands:

```
#passwd -l logcollect  
#passwd logcollect
```

The administrator receives a message that the password cannot be modified. Which of the following is most likely the issue?

A) The logcollect account needs to be unlocked. (Correct Answer)

- B) The logcollect account is managed by an LDAP provider.
- C) The logcollect account must set the password.
- D) The logcollect account is configured for nologin.

Explanation

Correct Answer: A [?](#)

Community Discussion

Question #276

Which of the following best describes JSON?

- A) A programming language that developers use to build dynamic websites
- B) An open-source platform that facilitates the creation and administration of isolated environments
- C) A file format for exchanging information or data between systems (Correct Answer)**
- D) A software-based technique that allows a computer to simulate one or more machines so they appear to be real

Explanation

Correct Answer: C [?](#)

Community Discussion

Question #277

An administrator found a suspicious public IP address in a log file. Which of the following commands can be used with the IP address to resolve back to a domain? (Choose two).

A) dig -x (Correct Answer)

B) dig

C) whois

D) nslookup (Correct Answer)

E) traceroute -n

F) hostname

Explanation

Correct Answer: AD [?](#)

Community Discussion

Question #278

A systems administrator wants to list all local account names and their respective UIDs. Which of the following commands will provide output containing this information?

A) cut -c: -f3,1 /etc/passwd

B) cut -d: -s2,5 /etc/passwd

C) cut -d: -f1,3 /etc/passwd (Correct Answer)

D) cut -n: -f1,2 /etc/passwd

Explanation

Correct Answer: C [?](#)

Community Discussion

Question #279

The development team created a new branch with code changes that a Linux administrator needs to pull from the remote repository. When the administrator looks for the branch in Git, the branch in question is not visible. Which of the following commands should the Linux administrator run to refresh the branch information?

A) git fetch (Correct Answer)

- B) git checkout
- C) git clone
- D) git branch

Explanation

Correct Answer: A [?](#)

Community Discussion

Question #280

An organization's business office needs to collaborate on quarterly reports and asks the systems administrator to create a finance group. The administrator creates the following directory:

drwxrwxr-x	Business	Business	4096	Oct 15 16:40	Q1
------------	----------	----------	------	--------------	----

Which of the following commands will allow all members of the finance group to have group ownership of all files created in the Q1 directory while ensuring that members of the business group preserve access? (Choose two).

A) chgrp Finance Q1/ (Correct Answer)

B) chmod g+s Q1/ (Correct Answer)

C) chmod u+s Q1/

D) chmod a+x Q1/

E) chown Finance Q1/

F) chmod g+x Q1/

Explanation

Correct Answer: AB [?](#)

Community Discussion

Question #281

A user (userA) has reported issues while logging in to the system. The following output has been provided:

```
Could not chdir to home directory /home/userA: Permission denied
-bash: /home/userA/.bash_profile: Permission denied

# cat /etc/passwd | grep userA
userA:x:1234:3400:userA account:/home/userA:/bin/bash

# passwd -S userA
userA PS 2022-10-17 1 99999 7 0 (Password set, SHA512 crypt.)

# groups userA
admin dev usergrp

# ls -lrth /home/
drwx-----. 7 root      admin  9 Jan 17 2019  userA
drwx-----. 6 userB     dev   6 Jan 21 2021  userB
drwx-----. 4 userC     app   9 Jan 23 2020  userC
```

Which of the following describes the issues userA is having?

- A) The password for userA is not set.
- B) The group for userA is not assigned correctly.
- C) The account password for userA has expired.
- D) The assigned home directory is not owned by userA. (Correct Answer)**

Explanation

Correct Answer: D [?](#)

Community Discussion

Question #282

A Linux administrator needs to expose port 9000 for a container to listen during runtime. The Linux administrator creates a Dockerfile with the following entries.

```
# Test-Container
```

```
FROM node:9-alpine -
```

```
WORKDIR /usr/src/app -  
COPY package*.json ./
```

```
RUN npm install -  
copy . .
```

```
EXPOSE 9000 -  
CMD ['npm', 'start']
```

Which of the following commands should the administrator use to accomplish this task? (Choose two).

A) docker build -t Test-Container (Correct Answer)

B) docker tag Test-Container

C) docker inspect Test-Container

D) docker run -p 9000:80 Test-Container (Correct Answer)

E) docker push Test-Container

F) docker run -p 80:9000 Test-Container

Explanation

Correct Answer: AD [?](#)

Community Discussion

Question #283

After connecting to a remote host via SSH, an administrator attempts to run an application but receives the following error:

```
[user@workstation ~]$ ssh admin@srv1
Last login: Tue Mar 29 18:03:34 2022
[admin@srv1 ~] $ /usr/local/bin/config_manager
Error: cannot open display:
[admin@srv1 ~] $
```

Which of the following should the administrator do to resolve this error?

- A) Disconnect from the SSH session and reconnect using the `ssh -x` command.
- B) Add Options X11 to the `/home/admin/.ssh/authorized_keys` file.
- C) Open port 6000 on the workstation and restart the `firewalld` service.
- D) Enable X11 forwarding in `/etc/ssh/ssh_config` and restart the server. (Correct Answer)**

Explanation

Correct Answer: D [?](#)

Community Discussion

Question #284

While troubleshooting server issues, a Linux systems administrator obtains the following output:

```
[root@host ~]# free -m
```

	total	used	free	shared	buff/cache	available
Mem:	3736	3598	88	2	48	29
Swap:	2047	1824	223			

Which of the following best describes the state of the system?

A) The system has consumed the system memory and swap space. (Correct Answer)

B) The system has enough free memory space.

C) The system has swap disabled.

D) The system has allocated enough buffer space.

Explanation

Correct Answer: A ?

Community Discussion

Question #285

Users are reporting that a Linux server is responding slowly. A systems administrator troubleshooting the server issue sees the following:

```
iostat -m 2
Linux 3.10.0-1160.66.1.el7.x86_64 (myhost) 10/19/2022 _x86_64_ (4 CPU)

avg-cpu:  %user   %nice   %system %iowait  %steal   %idle
           24.30    0.01    0.29    50.38    4.00   19.01

Device: tps      MB_read/s MB_wrtn/s MB_read MB_wrtn
sdd      15.65    0.48      0.83    4936250 8513501

df -h
Filesystem      Size  Used Avail Use% Mounted on
devtmpfs        16G   0     16G   0%   /dev
tmpfs           16G   64K   16G   1%   /dev/shm
/dev/mapper/root 35G   8.8G   26G   26%  /
/dev/sda2       497M  144M   354M  29%  /boot
/dev/mapper/tmp  10G   38M   10G   1%   /tmp
/dev/mapper/var  15G   13G   2.2G  86%  /var
/dev/mapper/varlib 650G  397G  254G  62%  /var/lib
/dev/mapper/varlog 15G   9.5G   5.5G  64%  /var/log
/dev/mapper/varaudit 15G   72M   15G   1%   /var/log/audit
```

Which of the following is most likely the issue?

- A) The CPU is mostly waiting for I/O operations. (Correct Answer)**
- B) / filesystem does not have enough storage allocated.
- C) /var filesystem is almost full.
- D) The CPU capacity is inadequate.

Explanation

Correct Answer: A ?

Community Discussion

Question #286

A systems administrator identifies multiple processes in a zombie state. Which of the following signals would be best for the administrator to send to the PPID?

A) SIGTERM (Correct Answer)

- B) SIGHUP
- C) SIGQUIT
- D) SIGSTOP

Explanation

Correct Answer: A [?](#)

Community Discussion

Question #287

A Linux administrator wants a permission bit on a shared folder that allows only the owner of the file within that directory or the root user to delete or rename the file. Which of the following commands will help achieve this goal?

- A) `chmod u+w /directory/`
- B) `chmod a+r /directory/`
- C) `chmod o+t /directory/` (Correct Answer)**
- D) `chmod g+s /directory/`

Explanation

Correct Answer: C [?](#)

Community Discussion

Question #288

A Linux administrator needs to harden a system and guarantee that the Postfix service will not run even after a restart or system upgrade. Which of the following commands allows the administrator to fulfill the requirement?

A) `systemctl mask postfix.service` (Correct Answer)

B) `systemctl disable postfix.service`

C) `systemctl stop postfix.service`

D) `systemctl -n restart postfix.service`

Explanation

Correct Answer: A ?

Community Discussion

Question #289

The MySQL database process that was running on a Linux server suddenly stopped, and the process was killed. Which of the following commands can help identify whether this issue was produced by the OOM killer?

A) `grep -i /proc/oom_score`

B) `grep -i -r "out of memory" /var/log` (Correct Answer)

C) `cat /var/run/initramfs/overlayroot.log | grep "out of memory"`

D) `cat /sys/block/loop0/events`

Explanation

Correct Answer: B ?

Community Discussion

Question #290

A Linux administrator wants to prevent the httpd web service from being started both manually and automatically on a server. Which of the following should the administrator use to accomplish this task?

A) systemctl mask httpd (Correct Answer)

- B) systemctl disable httpd
- C) systemctl stop httpd
- D) systemctl reload httpd

Explanation

Correct Answer: A [?](#)

Community Discussion

Question #291

A systems administrator is receiving complaints about slow performance and system crashes. The administrator suspects memory and CPU issues. Which of the following is the first action the administrator should take to troubleshoot and resolve these issues?

A) Run resource monitoring tools like top or htop to view the current CPU and memory use. (Correct Answer)

- B) Look through the system logs and error messages to find any faults involving the CPU and memory.
- C) Remove and replace the CPU and memory components to address hardware issues.
- D) Reboot the server to clear any CPU and memory congestion.

Explanation

Correct Answer: A [?](#)

Community Discussion

Question #292

A systems administrator created a new directory with specific permissions. Given the following output:

```
# file: comptia
# owner: root
* group: root
user::rwx
group::r-x
other::---
default:user::rwx
default:group::r-x
default:group:wheel:rwx
default:mask::rwx
default:other::---
```

Which of the following permissions are enforced on /comptia?

- A) Members of the wheel group can read files in /comptia.
- B) Newly created files in /comptia will have the sticky bit set.
- C) Other users can create files in /comptia.

D) Only root can create files in /comptia. (Correct Answer)

Explanation

Correct Answer: D [?](#)

Community Discussion

Question #293

A security analyst is monitoring the network to identify latency or slowdowns during a vulnerability scan. Which of the following functions will best achieve this?

A) function x() { info=\$(dig \$(dig -x \$1 |grep PTR | tail -n 1 | awk -F ".in-addr" '{print \$1}').origin.asn.cymru.com TXT +short) && echo "\$1 | \$info" }

B) function x() { info=\$(ping -c 1 \$1 |awk -F "/" 'END{print \$5}') && echo "\$1 | \$info" } (Correct Answer)

C) function x() { info=\$(nc -m 40 \$1 |awk 'END{print \$1}') && echo "\$1 | \$info" }

D) function x() { info=\$(geoiplookup \$1) && echo "\$1 | \$info" }

Explanation

Correct Answer: B ?

Community Discussion

Question #294

A systems administrator notices the process list on a mission-critical server has a large number of processes that are in state “Z” and marked as “defunct.” Which of the following should the administrator do in an attempt to safely remove these entries from the process list?

A) Kill the process with PID 1.

B) Kill the PID of the processes.

C) Kill the parent PID of the processes. (Correct Answer)

D) Reboot the server.

Explanation

Correct Answer: C ?

Community Discussion

Question #295

A systems administrator is customizing a new Linux server. Which of the following settings for umask would ensure that new files have the default permissions: -rw-r-----?

A) 0017

B) 0027 (Correct Answer)

C) 0038

D) 0640

Explanation

Correct Answer: B [?](#)

Community Discussion

Question #296

A security learn discovers that a web server has been running with elevated privileges and identifies it as a security violation. Which of the following items needs to be added to the webserver.service file to remediate the issue?

A) In the [Service] section of the webserver.service file, add User=comptia. (Correct Answer)

B) In the [Unit] section of the webserver.service file, add AllowIsolate=true.

C) In the [Install] section of the webserver.service file, add WantedBy=single.target.

D) Add After=network.target to the [Install] section of the webserver.service file.

Explanation

Correct Answer: A [?](#)

Question #297

The users of a Linux system are unable to use one of the application filesystems.

The following outputs have been provided:

Output 1:

```
$ cd /app
$ touch file
touch: cannot touch 'file': Read-only file system
```

Output 2:

```
/dev/sdc1 on /app type ext4 (ro,relatime,seclabel,data=ordered)
```

Output 3:

```
/dev/sdc1 /app ext4 defaults 0 0
```

Output 4:

```
[ 302.048075 ] Buffer I/O error on dev sdc1, logical block 0, asyng page read
[ 302.048490 ] EXT4-fs (sdc1): Attempt to read block from filesystem resulted in short read while trying to re-open /dev/sdc1
```

Which of the following actions will resolve this issue?

A) umount /app
fsck -y /dev/sdc1
mount /app (Correct Answer)

B) xfs_repair /dev/sdc1
mount -o rw,remount /app
C) umount /app

```
xfs_repair /dev/sdc1  
mount /app  
D) fsck -y /dev/sdc1  
mount -o rw,remount /app
```

Explanation

Correct Answer: A [?](#)

Community Discussion

Question #298

A Linux administrator is implementing a stateful firewall on the Linux server. Which of the following iptables options will be required to build the stateful rules? (Choose two.)

- A) --name ESTABLISHED
- B) -m recent (Correct Answer)**
- C) -m conntrack (Correct Answer)**
- D) --cstate
- E) --remove
- F) -j DROP

Explanation

Correct Answer: BC [?](#)

Community Discussion

Question #299

A systems administrator receives the following errors via email from the system log:

XFS (lcp0): Metadata CRC error detected at xfs_agi_read_verify+0xcb/0xfe

XFS (loop0): First 12@ bytes of corrupted metadata buffer

XFS (loop0): metadata I/O error in "xfs_trans_read_buf_map" at daddr 0x2 len 1 error 74

A few minutes later, the administrator starts receiving reports that some of the images in the company's website are not loading properly.

The systems administrator runs some commands and receives the following outputs:

Output 1:

```
NAME      FSTYPE UUID                                MOUNTPPOINT
sda
|--sda1 ext1      02ae47-fe487-45bc-3f345      /
sdb
|--sdb1 xfs       34bc-45ad-80a3-347c7056      /var/www/html
```

Output 2:

DocumentRoot "/var/www/html"

Output 3:

```
httpd.service - The Apache HTTP Server
Loaded: loaded (usr/lib/systemd/system/httpd.service; enabled; vendor preset: disabled)
Active: active (running) since Sun 1991-05-24 16:12:43 UTC; 30y ago
Docs: man:httpd(8)
      man:apachectl(8)
Main PID 282 (httpd)
```

Which of the following would be the appropriate steps to take to solve this issue?

**A) systemctl stop httpd
umount /dev/sdb1
xfs_repair /dev/sdb1
mount /dev/sdb1 /var/www/html
systemctl start httpd (Correct Answer)**

B) umount /dev/sdb1
xfs_repair /dev/sdb1
xfs_metadump /dev/sdb1
mount /dev/sdb1 /var/www/html
systemctl restart httpd
C) umount /dev/sdb1
systemctl stop httpd
xfs_metadump /dev/sdb1
mount /dev/sdb1 /var/www/html
systemctl start httpd
D) systemctl stop httpd
xfs_check -L /dev/sdb
umount /var/www/html
systemctl start httpd

Explanation

Correct Answer: A ?

Community Discussion

Question #300

A systems administrator is reviewing the following output on the text editor that is being used to update the company's internal database records:

```
{  
    "Company Name"; {  
        "Address"; "street",  
        "City"; "State"  
    }  
}
```

Which of the following extensions should the systems administrator use when saving the file?

A) .json (Correct Answer)

B) .yaml

C) .tf

D) .sh

Explanation

Correct Answer: A [?](#)

Community Discussion

Question #301

Which of the following directories contains binaries that only a root user should run?

A) /sbin (Correct Answer)

B) /bin

C) /root

D) /lib

Explanation

Correct Answer: A [?](#)

Community Discussion

Question #302

An administrator made a change to a system's network configuration. Which of the following best represents what the administrator should do to have the new configuration take effect?

A) `systemctl restart networkd` (Correct Answer)

B) `systemctl enable networkd`

C) `systemctl status networkd`

D) `systemctl isolate networkd`

Explanation

Correct Answer: A [?](#)

Community Discussion

Question #303

A Linux administrator needs to run a web application on a server that requires the use of a desktop web browser to enter the final configuration information. The administrator determines the server is running in a multiuser mode. The administrator verifies that a web browser is installed on the server and that it has a display manager installed. Which of the following commands will best allow the administrator to temporarily use the web browser on the server?

A) `sudo systemctl isolate graphical.target` (Correct Answer)

B) `sudo systemctl set-default runlevel5.target`

- C) `sudo systemctl restart gdm`
- D) `sudo systemctl --no-wall rescue`

Explanation

Correct Answer: A ?

Community Discussion

Question #304

A Linux engineer wants to give read-write-execute permissions for the `cloud_users` directory to user Oliver. Which of the following commands should the engineer use?

- A) `setfacl -L oliver:rwx /cloud_users`
- B) `setfacl -x rwx:oliver /cloud_users`
- C) `setfacl -P rwx:oliver /cloud_users`
- D) `setfacl -m oliver:rwx /cloud_users` (Correct Answer)**

Explanation

Correct Answer: D ?

Community Discussion

Question #305

A Linux administrator needs to forward port 8000 on a remote server remote to port 8000 on a local server. Which of the following commands should the administrator run on the local server to achieve this goal?

- A) `firewall-cmd --add-forward-port=port=8000:proto=tcp:toport=8000:toaddr=remote`
- B) `iptables -A FORWARD -m state -p tcp -d remote --dport 8000 -j accept`
- C) `ssh -R8000:localhost:8000 remote` (Correct Answer)**

D) socat TCP4-LISTEN:8000,fork TCP4:remote:8000

Explanation

Correct Answer: C ?

Community Discussion

Question #306

An operations engineer is planning to start a container running a PostgreSQL database. The engineer wants the container to start automatically at system startup, mount /home/db directory as /var/db/postgresql inside the container, and expose the 5432 port to the OS. Which of the following commands should the engineer run to achieve this task?

A) docker run -d --restart-always -p 5432:5432 -v /home/db:/var/lib/postgresql postgresql:12 (Correct Answer)

B) docker run -d --restart -p 5432:5432 --volume /var/lib/postgresql:/home/db postgresql: 12

C) docker run -d --attach --platform 5432:5432 --volume /home/db:/var/lib/postgresql postgresql:12

D) docker run -d --init --restart --publish 5432:5432 --workdir /home/db:/var/lib/postgresql postgresql:12

Explanation

Correct Answer: A ?

Community Discussion

Question #307

A systems administrator needs to list the contents of archive.tar.xz. Which of the following commands will list the files?

- A) tar ddf archive.tar.xz
- B) tar tjf archive.tar.xz (Correct Answer)**
- C) tar xzf archive.tar.xz
- D) tar jvf archive.tar.xz

Explanation

Correct Answer: B [?](#)

Community Discussion

Question #308

A systems administrator wants to ensure all the latest patches and updates are automatically synced to the local repository every two hours on Sundays through Tuesdays in the month of May. Which of the following cron entries can the administrator use to achieve this goal? (Choose two.)

- A) 0 */2 * 5 0-3 /root/update-from-remote-to-local.sh
- B) 0 0,2,4,6,8,10,12,14 * 5 0-3 /root/update-from-remote-to-local.sh
- C) 0 0,2,4,6,8,10,12,14 * 5 0-2 /root/update-from-remote-to-local.sh (Correct Answer)**
- D) 0 0-23/2 * 5 0-3 /root/update-from-remote-to-local.sh
- E) 0 0-23/2 * 5 0-2 /root/update-from-remote-to-local.sh
- F) 0 */2 * 5 0-2 /root/update-from-remote-to-local.sh

Explanation

Correct Answer: C [?](#)

Community Discussion

Question #309

A DevOps engineer is working on a local copy of a Git repository. The engineer would like to switch from the main branch to the staging branch but notices the staging branch does not exist. Which of the following Git commands should the engineer use to perform this task?

- A) `git branch -m staging`
- B) `git commit -m staging`
- C) `git status -b staging`

D) `git checkout -b staging` (Correct Answer)

Explanation

Correct Answer: D ?

Community Discussion

Question #310

Joe, a user, is unable to log in to the system. The Linux administrator checks the `/etc/passwd` file and reviews the following output:

```
Joe:x:1002:1002::/home/Joe:/sbin/nologin
```

Which of the following should the administrator do to fix this issue?

- A) Remove the x entry in the `/etc/passwd` file for Joe.
- B) Change the `/sbin/nologin` entry to `/bin/sh` for Joe. (Correct Answer)**
- C) Issue the `usermod -L Joe` command.
- D) Ensure that a corresponding entry is in the `/etc/shadow` file for Joe.

Explanation

Correct Answer: B [?](#)

Community Discussion

Question #311

Linux system users report that over the past few months a server has stopped responding at least two times each day. Every time this happens, the users have to restart the server to recover it. A systems administrator diagnosing the issue collects the following outputs:

Output 1:

```
[Tue Aug 31 16:36:42 2021] OOM: Kill process 43805 (java) score 249 or sacrifice child
[Tue Aug 31 16:36:42 2021] killed process 43805 (java) total-vm: 4446352kB, anon-rss: 4053140kB, file-rss: 68kB
[Tue Aug 31 16:56:42 2021] java invoked oom-killer: gfp_mask=0x280da, order=0, oom_score_adj=0
[Tue Aug 31 16:56:42 2021] java cpuset=/ mems_allowed=0
[Tue Aug 31 16:56:42 2021] CPU: 2 PID# 88196 Comm: java Not tainted 3.10.0-328.13.1.x86_64 #1
```

Output 2:

```
Linux 3.10.0-328.13.1.x86_64 #1 (hostname) 31/08/2021 _x86_64_ (8 CPU)
```

	CPU	%user	%nice	%system	%iowait	%steal	%idle
16:00:01 PM	all	8.04	0.00	1.11	0.00	0.00	90.85
16:20:01 PM	all	1.60	0.00	1.20	0.00	0.00	99.29
16:30:01 PM	all	1.69	0.00	0.89	0.00	0.00	98.42
16:40:01 PM	all	1.08	0.00	0.88	0.00	0.00	98.04
16:50:01 PM	all	0.05	0.00	0.87	0.00	0.00	99.04

Output 3:

```
$ free -m
```

	total	used	free	shared	buff/cache	available
Mem:	15819	15026	174	92	619	134
Swap:	0	0	0			

Which of the following is the best permanent resolution for the system?

A) More memory resources (Correct Answer)

- B) More CPU resources
- C) More network bandwidth
- D) Extended root filesystem

Explanation

Correct Answer: A [?](#)

Community Discussion

Question #312

A diagnostic tool reports a “host seems down” event for a server with an IP address of 192.168.47.44. Which of the following commands should the administrator use to confirm the host down event?

- A) nmap 192.168.47.0/32
- B) netcat 192.168.47.44
- C) nmap 192.168.47.44 -Pn
- D) ping -c 1 192.168.47.44 (Correct Answer)**

Explanation

Correct Answer: D [?](#)

Community Discussion

Question #313

An administrator changed the default port of an SSH server to 2222 on myhost, and clients are not able to connect. The administrator runs some commands and receives the following output:

```
$ ssh -p 2222 myhost
```

```
ssh:connect to host myhost on port 2222: No route to host
```

```
$ ping myhost
```

```
PING myhost (192.168.0.10) 56(84) bytes of data
```

```
64 bytes from myhost (192.168.0.10): icmp_seq=1 time=0.876 ms
```

```
64 bytes from myhost (192.168.0.10): icmp_seq=2 time=0.732 ms
```

Which of the following commands should be run on myhost?

A) `firewall-cmd --zone=public --add-service=ssh`

B) `firewall-cmd --zone=public --add-port=2222` (Correct Answer)

C) `iptables --zone=public --add-service=ssh`

D) `iptables --zone=public --add-port=2222`

Explanation

Correct Answer: B ?

Community Discussion

Question #314

A Linux administrator created a virtual clone of a physical server and would like to remove any existing entries related to SSH keys from outside entities on the virtual clone. Which of the following files should the administrator remove? (Choose two.)

A) `~/.ssh/authorized_keys` (Correct Answer)

B) `~/.ssh/known_hosts` (Correct Answer)

- C) /etc/ssh/ssh_config
- D) ~/.ssh/config
- E) /etc/ssh/sshd_config
- F) /etc/ssh/ssh_hcst_rsa_key.pub

Explanation

Correct Answer: AB [?](#)

Community Discussion

Question #315

An administrator thinks that a package was installed using a snap. Which of the following commands can the administrator use to verify this information?

A) snap list (Correct Answer)

- B) snap find
- C) snap install
- D) snap try

Explanation

Correct Answer: A [?](#)

Community Discussion

Question #316

A Linux administrator is implementing a CI/CD process for the company's internal accounting web application. Which of the following best defines the purpose of this process?

A) To automate the process of building, testing, and deploying application components (Correct Answer)

- B) To perform security penetration tests on deployed applications to identify vulnerabilities
- C) To formalize the approval process of application releases and configuration changes
- D) To leverage code to document the infrastructure, configurations, and dependencies

Explanation

Correct Answer: A ?

Community Discussion

Question #317

Users report that they cannot access some files located in the `/opt/finapp` directory after a power outage caused an unexpected server restart. A Linux administrator examines the following filesystem details and system logs:

```
# mount | grep finapp
/dev/mapper/rhel-opt_finapp on /opt/finapp type xfs (rw,relatime,seclabel,attr2,inode64,logbufs=8,logbsize=32k,noquota)

#lsblk
NAME                                MAJ:MIN RM SIZE  RO TYPE MOUNTPOINT
sda                                  8:0      0 931.5G  0 disk
|-sda1                              8:1      0  1G     0 part/boot
|-sda2                              8:2      0 930.5G  0 part
  |--rhel-
root                                253:0    0 30.5G   0 lvm /
  |--rhel-
opt_finapp                          253:1    0 900G    0      /opt/finapp

# grep opt_finapp /var/log/messages
XFS (opt_finapp): Corruption detected in inode 3645, extent tree
```

Which of the following commands should the administrator run in an attempt to fix the filesystem?

- A) `fdisk /dev/mapper/rhel-opt_finapp`
- B) `xfs_repair /dev/mapper/rhel-opt_finapp` (Correct Answer)**
- C) `lvcreate -L900G -n opt_finapp rhel`
- D) `fsck.ext4 /dev/mapper/rhel-opt_finapp`

Explanation

Correct Answer: B ?

Community Discussion

Question #318

An administrator accidentally installed the httpd RPM package along with several dependencies. Which of the following options is the best way for the administrator to revert the package installation?

- A) dnf clean all
- B) rpm -e httpd
- C) apt-get clean
- D) yum history undo last (Correct Answer)**

Explanation

Correct Answer: D [?](#)

Community Discussion

Question #319

A systems administrator needs to know the mail exchange record for the foo.com domain. Which of the following commands will accomplish this task?

- A) telnet mx foo.com
- B) dig mx foo.com (Correct Answer)**
- C) nslookup mx foo.com
- D) host mx foo.com

Explanation

Correct Answer: B [?](#)

Community Discussion

Question #320

A Linux administrator encounters the following error in the Apache log files:

SQLSTATE[HY000] [2002] Permission denied

Which of the following commands should the administrator run to safely fix this issue?

- A) `chcon -t httpd_sys_content_t /home/apache`
- B) `setsebool -P httpd_can_network_connect_db on` (Correct Answer)**
- C) `setenforce 0`
- D) `touch /.autorelabel`

Explanation

Correct Answer: B [?](#)

Community Discussion

Question #321

A systems administrator received a report about issues with a Linux system. Given the following output:

File system	Size	Used	Avail	Use%	Mounted on
devtmpfs	856M	0	856M	0%	/dev
tmpfs	875M	0	875M	0%	/dev/shm
tmpfs	856M	8.7M	866M	1%	/run
tmpfs	875M	0	875M	0%	/sys/fs/cgroup
/dev/mapper/rootvg-rootlv	2.0G	310M	1.7G	16%	/
/dev/mapper/rootvg-usrlv	10G	1.7G	8.4G	17%	/usr
/dev/mapper/rootvg-homelv	1014M	40M	975M	4%	/home
/dev/mapper/rootvg-varlv	8.0G	7.9G	10M	100%	/var
/dev/mapper/rootvg-tmplv	2.0G	47M	2.0G	3%	/tmp

Which of the following is the best command to use to fix this issue?

- A) mkfs.xfs /var
- B) rm -rf /var/*
- C) lvextend -L +5G -r /dev/mapper/rootvg-varlv (Correct Answer)**
- D) lvresize -L +2G -r /dev/mapper/rootvg-rootlv

Explanation

Correct Answer: C ?

Community Discussion

Question #322

A Linux administrator is configuring a new internal web server fleet. The web servers are up and running but can only be reached by users directly via IP address. The administrator is attempting to fix this inconvenience by requesting appropriate records from the DNS team. The details are:

Hostname: devel.comptia.org -

IP address: 5.5.5.1, 5.5.5.2, 5.5.5.3, 5.5.5.4

Name server: 5.5.5.254 -

Additional names: dev.comptia.org, development.comptia.org

Which of the following types of DNS records should the Linux administrator request from the DNS team? (Choose three).

A) MX

B) NS (Correct Answer)

C) PTR

D) A (Correct Answer)

E) CNAME (Correct Answer)

Explanation

Correct Answer: BDE ?

Community Discussion

Question #323

A system engineer deploys a new application server, but the server cannot communicate with the back-end database hostname. The engineer confirms that the application server can ping the database server's IP address. Which of the following is the most likely cause of the issue?

A) Incorrect DNS servers (Correct Answer)

- B) Unreachable default gateway
- C) Missing route configuration
- D) Misconfigured subnet mask

Explanation

Correct Answer: A [?](#)

Community Discussion

Question #324

A systems administrator created a user cron to run a scheduled database backup cronjob on the server at 1 a.m. everyday. The following is the cronjob syntax:

```
01 * * * /bin/sh backup.sh
```

The backups, however, are not being created. When checking the crontab file, the administrator sees the following error:

```
$ cron@localhost:$ crontab -e
/var/spool/cron/crontab/cron: Permission denied
```

Which of the following will permit the cronjob to execute?

A) Creating the file cron.allow and adding user cron to it (Correct Answer)

- B) Giving cron ownership of the file /usr/bin/crontab

- C) Running `sudo cron restart` to activate the cronjob
- D) Running `sudo crontab -u cron -e` to edit the file with `sudo`

Explanation

Correct Answer: A [?](#)

Community Discussion

Question #325

Which of the following command sequences is used to install Linux software directly from source code?

- A) `./config`
`./compile`
`./copy`
- B) `unzip`
`gcc`
`install`
- C) `./unpack`
`make exec`
`make copy`
- D) `./configure`
`make`
`make install` (Correct Answer)**

Explanation

Correct Answer: D [?](#)

Community Discussion

Question #326

A newly hired Linux administrator needs to grant access to the Linux servers by generating a password-protected RSA key of 4096 bytes. Which of the following commands will accomplish this task?

- A) `ssh-keygen -RPb 4096`
- B) `ssh-keygen -t rsa -b 4096` (Correct Answer)**
- C) `ssh-keygen -k rsa -l 4096 -p`
- D) `ssh-keygen -t dsa -B 4096`

Explanation

Correct Answer: B [?](#)

Community Discussion

Question #327

A Linux administrator updated the configuration file for the httpd web service. The administrator needs to apply the updated changes to the service without interrupting any running sessions. Which of the following commands should the administrator run to accomplish this task?

- A) `systemctl reload httpd` (Correct Answer)**
- B) `systemctl restart httpd`
- C) `systemctl status httpd`
- D) `systemctl mask httpd`

Explanation

Correct Answer: A [?](#)

Community Discussion

Question #328

A systems administrator is tasked with configuring a repository on an RPM-based Linux system. Which of the following need to be reviewed and modified? (Choose two.)

A) /etc/yum.conf (Correct Answer)

B) /etc/apt/sources.list.d

C) /etc/pam.so.d

D) /etc/apt.conf

E) /etc/yum.repos.d (Correct Answer)

F) /etc/ssh/ssh_config

Explanation

Correct Answer: AE [?](#)

Community Discussion

Question #329

A Linux administrator needs to check the content of a log file that is appending data as the file grows. Which of the following commands should the administrator use to accomplish this task?

A) tail -f log.txt (Correct Answer)

B) tail -v log.txt

C) tail -c log.txt

D) tail log.txt

Explanation

Correct Answer: A [?](#)

Community Discussion

Question #330

Which of the following paths stores the configuration files in a Linux filesystem?

- A) /proc
- B) /home
- C) /root

D) /etc (Correct Answer)

Explanation

Correct Answer: D [?](#)

Community Discussion

Question #331

Which of the following commands is used to tune kernel parameters?

A) sysctl (Correct Answer)

- B) ss
- C) mkinitrd
- D) lsof

Explanation

Correct Answer: A [?](#)

Community Discussion

Question #332

A Linux administrator needs to remove all local firewall rules on a Linux system. Which of the following commands should the administrator run?

- A) iptables -D
- B) iptables -L
- C) iptables -F (Correct Answer)**
- D) iptables -A

Explanation

Correct Answer: C [?](#)

Community Discussion

Question #333

A Linux administrator has defined a systemd script docker-repository.mount to mount a volume for use by the Docker service. The administrator wants to ensure that Docker service does not start until the volume is mounted. Which of the following configurations needs to be added to the Docker service definition to best accomplish this task?

- A) After=docker-repository.mount
- B) ExecStart=/usr/bin/mount -a
- C) Requires=docker-repository.mount (Correct Answer)**
- D) RequiresMountsFor=docker-repository.mount

Explanation

Correct Answer: C [?](#)

Community Discussion

Question #334

A systems administrator needs to stop a running systemd service tor and prevent it from running when the system starts up. Which of the following is the best command for the administrator to use to accomplish this task?

- A) `pidof -o %PPID -x tor && apt -y purge tor`
- B) `kill tor`
- C) `service stop|disable tor`
- D) `systemctl stop tor && systemctl disable tor` (Correct Answer)**

Explanation

Correct Answer: D ?

Community Discussion

Question #335

An administrator is troubleshooting a database service outage that was reported by a monitoring system. Given the following output:

```
$ systemctl status mariadb
Oct 20 16:40:45 comptia systemd[1]: mariadb.service: Main process exited, code=killed, status=9/KILL
Oct 20 16:40:45 comptia systemd[1]: mariadb.service: Failed with result 'signal'.
Oct 20 16:40:50 comptia systemd[1]: Stopped MariaDB 10.3 database server.

$ dmesg
[ 1075.811435] Node 0 DMA32 free:8220kB min:8224kB low:8856kB high:9488kB active_anon:1456kB inactive_anon:319024kB active_file:8kB
inactive_file:2952kB unevictable:0kB writepending:4kB pre
sent:507712kB managed:456368kB mlocked:0kB bounce:0kB free_pcp:276kB local_pcp:276kB free_cma:0kB
[ 1075.811441] lowmem_reserve[]: 0 0 0 0
[ 1061.491433] oom-kill:constraint=CONSTRAINT_NONE,nodemask=
(null),cpuset=/,mems_allowed=0,global_oom,task_memcg=/system.slice/mariadb.service,task=mysql, pid=1981,uid=27
[ 1061.491453] Out of memory: Killed process 1981 (mysql) total-vm:330668kB, anon-rss:31316kB, file-rss:0kB, shmem-rss:0kB, UID:27 pgtables:324kB
oom_score_adj:0
[ 1061.525468] oom_reaper: reaped process 1981 (mysql), now anon-rss:0kB, file-rss:0kB, shmem-rss:0kB
```

Which of the following is the reason for the outage?

- A) The administrator sent a kill signal to the database.
- B) The server is missing the DMA bus.
- C) The database cannot write anything else to the storage.
- D) The server does not have enough physical memory. (Correct Answer)**

Explanation

Correct Answer: D [?](#)

Community Discussion

Question #336

Which of the following is a benefit of a service mesh?

A) Encrypted communication between two services in a Kubernetes environment (Correct Answer)

- B) Direct access to the Kubernetes API services through the use of tokens
- C) Elevated privileges in a Kubernetes pod to allow root access in a hardened cluster
- D) Creating PVCs in a Kubernetes cluster to store and manage persistent data

Explanation

Correct Answer: A [?](#)

Community Discussion

Question #337

A Linux systems administrator is working to obtain the installed kernel version of several hundred systems. Which of the following utilities should the administrator use for this task?

A) Ansible (Correct Answer)

- B) Git
- C) Docker
- D) Bash

Explanation

Correct Answer: A [?](#)

Community Discussion

Question #338

A systems administrator is creating new user accounts in several Linux machines and wants to automate the process from a Linux system used for operations. In this operations system, the list of servers is located in the /home/user/serverslist file, and the list of user accounts is located in the /home/user/userslist file. Which of the following scripts will help accomplish this task?

```
for server in $(cat /home/user/serverslist)
do
    for user in $(cat /home/user/userslist)
    do
        sudo useradd $user
    done
done
```

A) done

```
for server in $(cat /home/user/serverslist)
do
    ssh -i user@$server '
        for user in $(cat /home/user/userslist)
        do
            sudo useradd $user
        done
    exit '
done
```

B) done

```
for server in $(cat /home/user/serverslist)
do
    scp /home/user/userslist user@$server:/tmp
    ssh -i user@$server '
        for user in $(cat /tmp/userslist)
        do
            sudo useradd $user
        done
    exit '
```

c) done

(Correct Answer)

```
ssh user@$(cat /home/user/serverslist) '
    sudo useradd $(cat /home/user/userslist)
D) exit '
```

Explanation

Correct Answer: C ?

Community Discussion

Question #339

A Linux engineer receives the following notification from a system:

cp: cannot create regular file '/app/appdata.tar': No space left on device

The engineer obtains the following output:

```
[root@host ~]# df -i /app
```

Filesystem	Inodes	IUsed	IFree	IUse%	Mounted on
/dev/vdb	2048	2048	0	100%	/app

```
[root@host ~]# df -h /app
```

Filesystem	Size	Used	Avail	Use%	Mounted on
/dev/vdb	2.0G	6.1M	1.9G	1%	/app

Which of the following describes the state of the filesystem based on this output?

- A) The filesystem was mounted in read-only mode.
- B) The filesystem has depleted the available inodes. (Correct Answer)**
- C) The filesystem is not mounted in the correct location.
- D) The filesystem has consumed the available space.

Explanation

Correct Answer: B [?](#)

Community Discussion

Question #340

A DevOps engineer wants to work on a snapshot of the infrastructure as code configuration. Which of the following commands should the engineer use to accomplish this task?

- A) git pull
- B) git push
- C) git branch
- D) git tag (Correct Answer)**

Explanation

Correct Answer: D [?](#)

Community Discussion

Question #341

A Linux administrator is creating a user that can run the FTP service but cannot log in to the system. The administrator sets `/bin/false` as a log-in shell for the user. When the user tries to run the FTP service, it is rejected with an Invalid shell: `/bin/false` message. Which of the following is the best way to resolve the issue?

- A) Change ownership of `/bin/false` to the FTP user.
- B) Add `/bin/false` entry to the `/etc/shells` file. (Correct Answer)**
- C) Make `/bin/false` an executable file.
- D) Change the user's default shell to `/bin/bash`.

Explanation

Correct Answer: B [?](#)

Question #342

A Linux administrator is configuring a log collector agent to monitor logs in /var/log/collector. The agent is reporting that it cannot write in the directory. The agent runs as the logger user account. The administrator runs a few commands and reviews the following output:

Output 1:

```
drwxr-xr-x. 1 root root 0 Oct 01 20:13 collector
drwxr-xr-x. 1 root root 680 Oct 01 20:13 .
```

Output 2:

```
* file: /var/log/collector
* owner: root
* group: root
user::rwx
group::r-x
mask::rwx
other::r-x
default:user:rwx
default:group::r-x
default:mask::rwx
default:other::r-x
```

Output 3:

```
uid=1010(logger) gid=1010(monitor) groups=1010(monitor)
```

Which of the following is the best way to resolve the issue?

A) setfacl -Rm u:logger:rw /var/log/collector (Correct Answer)

B) usermod -aG root logger

C) chmod 644 /var/log/collector

D) `chown -R logger:monitor /var/log`

Explanation

Correct Answer: A [?](#)

Community Discussion

Question #343

A systems administrator receives reports of issues in one of the Linux systems. The following output has been provided:

```
# dig node.mycomptia.org
```

```
; <<>> DiG 9.11.36.x.x. <<>> node.mycomptia.org
;; global options: +cmd
;; connection timed out; no servers could be reached
```

```
# cat /etc/resolv.conf
search mycomptia.org
```

Which of the following actions will best resolve this issue? (Choose two.)

- A) `echo "8.8.8.8 node.mycomptia.org" >> /etc/hosts`
- B) `echo "nameserver 8.8.8.8" >> /run/systemd/resolve/resolv.conf` (Correct Answer)**
- C) `systemctl restart networkd`
- D) `echo "8.8.8.8" >> /etc/resolv.conf`
- E) `echo "DNS1=8.8.8.8" >> /etc/sysconfig/network-scripts/ifcfg-eth0`
- F) `systemctl restart NetworkManager` (Correct Answer)**
- G) `systemctl restart named`

Explanation

Correct Answer: BF [?](#)

Community Discussion

Question #344

The journald entries have filled a Linux machine's /var volume. Which of the following is the best command for a systems administrator to use to free up the disk space occupied by these entries?

A) journalctl --rotate

journalctl --vacuum-time=1s (Correct Answer)

B) systemctl stop systemd-journald

systemctl start systemd-journald

C) rm -rf /var/log/journal

systemctl restart systemd-journald

D) pkill -HUP systemd-journald

systemctl restart systemd-journald

Explanation

Correct Answer: A [?](#)

Community Discussion

Question #345

A Linux administrator edits a file that a system uses during the startup process to mount all the required partitions and volumes. The administrator wants to ensure no errors are in the file and the system will restart properly. Which of the following commands should the administrator use?

A) fsadm -v

B) mount -a (Correct Answer)

C) fsck -N

D) mountpoint -d

Explanation

Correct Answer: B ?

Community Discussion

Question #346

A technician just fixed a few issues in some code and is ready to deploy the code into production. Which of the following steps should the technician take next?

- A) Create a new branch using git checkout.
- B) Perform a git clone to pull main down.
- C) Create a git pull request to merge into main. (Correct Answer)**
- D) Perform a git pull to update the local copy of the code.

Explanation

Correct Answer: C ?

Community Discussion

Question #347

An administrator would like to mirror the website files on the primary web server www1, to the backup web server www2. Which of the following commands should the administrator use to most efficiently accomplish this task?

- A) [www1] \$ rsync -a -e ssh /var/www/html user1@www2:/var/www/html (Correct Answer)**
- B) [www1] \$ scp -r /var/www/html user1@www2:/var/www/html
- C) [www2] \$ cd /var/www/html; wget -m http://www1/
- D) [www1] \$ cd /var/www/html && tar cvf - . | ssh user1@www2 ' (cd /var/www/html; tar xvf -) '

Explanation

Correct Answer: A [?](#)

Community Discussion

Question #348

A systems administrator is reconfiguring existing user accounts in a Linux system. Which of the following commands should the administrator use to include “myuser” in the finance group?

- A) groupadd finance myuser
- B) groupmcd finance myuser
- C) useradd -g finance myuser
- D) usermod -aG finance myuser (Correct Answer)**

Explanation

Correct Answer: D [?](#)

Community Discussion

Question #349

Users are unable to run applications stored on the /app1 mount point. A Linux systems administrator is investigating the issue and obtains the following output from the /etc/fstab file:

Permissions have been verified and should be correct. Which of the following options should the systems administrator remove from the mount point?

A) noexec (Correct Answer)

B) nobarrier

C) noatime

D) noadirtime

Explanation

Correct Answer: A [?](#)

Community Discussion

Question #350

The development team wants to prevent a file from being modified by all users in a Linux system, including the root account. Which of the following commands can be used to accomplish this objective?

A) `chmod +t /app/conf/file`

B) `setenforce /app/conf/file`

C) `chattr +i /app/conf/file` (Correct Answer)

D) `chmod 0000 /app/conf/file`

Explanation

Correct Answer: C [?](#)

Community Discussion

Question #351

A Linux administrator wants to change the default SSH server port. Which of the following files should the administrator edit to accomplish this task?

A) /etc/ssh/sshd_config (Correct Answer)

B) ~/.ssh/config

C) /etc/ssh/ssh_config

D) /etc/ssh/ssh_host_rsa_key

Explanation

Correct Answer: A [?](#)

Community Discussion

Question #352

A systems administrator received a request to enable secure web services and SSH at the firewall level. Which of the following should the administrator use to accomplish this task? (Choose two.)

A) sudo ufw allow https (Correct Answer)

B) sudo ufw allow 22/udp

C) sudo ufw allow 8080/tcp

D) sudo ufw allow 22/tcp (Correct Answer)

E) sudo ufw allow http

F) sudo ufw allow 80/tcp

Explanation

Correct Answer: AD [?](#)

Community Discussion

Question #353

A Linux systems administrator is preparing to install updates using APT. The administrator notices the version of the packages that will be installed differs from the latest version on the Ubuntu website. Which of the following commands should the administrator use to obtain the latest version of the package?

A) apt update (Correct Answer)

- B) apt-get refresh
- C) apt purge
- D) apt-get cache update

Explanation

Correct Answer: A [?](#)

Community Discussion

Question #354

A Linux administrator is configuring a two-node cluster and needs to be able to connect the nodes to each other using SSH keys from the root account. Which of the following commands will accomplish this task?

- A) [root@nodea ~]# ssh -i ~/.ssh/id_rsa root@nodeb
- B) [root@nodea ~]# scp -i ~/.ssh/id_rsa root@nodeb
- C) [root@nodea ~]# ssh-copy-id -i ~/.ssh/id_rsa root@nodeb (Correct Answer)**
- D) [root@nodea ~]# ssh-agent -c ~/.ssh/id_rsa root@nodeb
- E) [root@nodea ~]# ssh-add -c ~/.ssh/id_rsa root@nodeb

Explanation

Correct Answer: C ?

Community Discussion

Question #355

A systems administrator issued the dm-crypt command with no parameters. Which of the following cryptographic features is the systems administrator trying to use?

A) LUKS (Correct Answer)

- B) HMAC
- C) SHA-256
- D) MD5

Explanation

Correct Answer: A ?

Community Discussion

Question #356

A Linux administrator is troubleshooting poor CPU performance on a virtual machine. The administrator thinks that the underlying hypervisor may be starved for resources. The administrator uses the top command to check the current CPU usage. Which of the following CPU metrics supports the administrator's theory?

- A) user
- B) nice
- C) steal (Correct Answer)**

D) system

Explanation

Correct Answer: C ?

Community Discussion

Question #357

A Linux engineer needs to determine the oom_score value for PID 7991. Which of the following files should the engineer verify to find this value?

A) /var/7991/oom_score

B) /proc/7991/oom_score (Correct Answer)

C) /opt/7991/oom_score

D) /usr/config/7991/oom_score

Explanation

Correct Answer: B ?

Community Discussion

Question #358

A Linux administrator is tasked with moving files in a database server. The administrator must not overwrite any existing files. Which of the following commands would indicate that the file already exists?

A) mv -i filename /tmp/backup

B) mv -b filename /tmp/backup

C) mv -n filename /tmp/backup (Correct Answer)

D) mv -f filename /tmp/backup

Explanation

Correct Answer: C ?

Community Discussion

Question #359

A Linux systems administrator is trying to execute a particular shell script on a server. The administrator reviews the following outputs:

```
$ ./startup.sh
bash: ./startup.sh Permission denied

$ ls -l startup.sh
-rw-rw-r-- 1 companyabc companyabc 18 October 15:35 startup.sh
```

Which of the following commands should the administrator use to allow the script to run?

- A) `chown root:root startup.sh`
- B) `chmod 750 startup.sh`
- C) `chmod +x startup.sh` (Correct Answer)**
- D) `chmod 400 startup.sh`

Explanation

Correct Answer: C ?

Community Discussion

Question #360

A web server will not start on a Linux server. A Linux administrator troubleshooting the issue runs a few commands, which produce the following output:

Output 1: systemctl units -

UNIT	FILE	STATE	VENDOR PRESET
httpd.service		masked	disabled
httpd@.service		disabled	disabled
firewalld.service		enabled	enabled

Output 2: service status -

```
- httpd.service
   Loaded: masked (Reason: Unit httpd.service is masked.)
   Active: inactive (dead) Since Thu 2021-09-01
   ...
```

Output 3: firewall configuration

```
ssh http mdns samba-client
```

Which of the following commands will allow the administrator to start the web server?

- A) systemctl stop firewalld.service
- B) systemctl daemon-reload
- C) systemctl unmask httpd.service (Correct Answer)**
- D) systemctl enable [\[email protected\]](#)

Explanation

Correct Answer: C ?

Community Discussion

Question #361

An administrator has received multiple tickets relating to a lack of a disk space on a few servers, but the percentage of disk space usage is below the threshold. The following shows further analysis and findings:

```
# df
Filesystem      1K-blocks  Used Available Use% Mounted on
/dev/xvda       33030016 10407780   22622236  32% /
tmpfs           368748      0     368748  0% /lib/init/rw
varrun          368748      56     368692  1% /var/run
tmpfs           368748      0     368748  0% /dev/shm

# df -i
Filesystem      Inodes          IUsed IFree   IUse% Mounted on
/dev/xvda       2080768         2080768      0   100% /
tmpfs           92187            3      92184   1% /lib/init/rw
varrun          92187           38     92149   1% /var/run
varlock         92187            4     92183   1% /var/lock
```

Which of the following commands should the administrator use to help identify the issue?

A) `find . -xdev -type f | cut -d "/" -f 2 | sort | uniq -c | sort -n` (Correct Answer)

B) `find . -xdev -type p | cut -d "/" -f 2 | sort | uniq -c | sort -n`

C) `find . -xdev -type s | cut -d "/" -f 2 | sort | uniq -c | sort -n`

D) `find . -xdev -type d | cut -d "/" -f 2 | sort | uniq -c | sort -n`

Explanation

Correct Answer: A ?

Community Discussion

Question #362

A cloud engineer wants to delete all unused networks that are not referenced by any container. Which of the following commands will achieve this goal?

- A) docker network erase
- B) docker network clear
- C) docker network prune (Correct Answer)**
- D) docker network rm

Explanation

Correct Answer: C [?](#)

Community Discussion

Question #363

An application requires a configuration file be placed in each user's home directory when an account is created. In which of the following locations should the administrator place the configuration file?

- A) /etc/skel (Correct Answer)**
- B) /etc/sysconfig
- C) /etc/ssh
- D) /etc/rc.local

Explanation

Correct Answer: A [?](#)

Community Discussion

Question #364

A Linux system is having issues. Given the following outputs:

```
# dig @192.168.2.2 mycomptiahost
; <<>> DiG 9.9.4-RedHat-9.9.4-74.el7_6.1 <<>> @192.168.2.2 mycomptiahost
; (1 server found)
;; global options: +cmd
;; connection timed out; no servers could be reached

# nc -v 192.168.2.2 53
Ncat: Version 7.70 (https://nmap.org/ncat)
Ncat : Connection timed out.

# ping 192.168.2.2
PING 192.168.2.2 (192.168.2.2) 56(84) bytes of data.
64 bytes from 192.168.2.2: icmp_seq=1 ttl=117 time=4.94 ms
64 bytes from 192.168.2.2: icmp_seq=2 ttl=117 time=10.5 ms
```

Which of the following best describes this issue?

- A) The DNS host is down.
- B) The name mycomptiahost does not exist in the DNS.
- C) The Linux engineer is using the wrong DNS port.
- D) The DNS service is currently not available or the corresponding port is blocked. (Correct Answer)**

Explanation

Correct Answer: D [?](#)

Community Discussion

Question #365

A Linux administrator is removing non-permanent rules from the system firewall. Which of the following commands will allow the administrator to achieve this goal in the safest way possible?

- A) `firewall-cmd --set-default-zone`
- B) `firewall-cmd --runtime-to-permanent` (Correct Answer)**

- C) firewall-cmd --reload
- D) firewall-cmd --complete-reload

Explanation

Correct Answer: B [?](#)

Community Discussion

Question #366

A technician wants to temporarily use a Linux virtual machine as a router for the network segment 10.10.204.0/24. Which of the following commands should the technician issue? (Choose three.)

- A) echo "1" > /proc/sys/net/ipv4/ip_forward (Correct Answer)**
- B) iptables -A FORWARD -j ACCEPT (Correct Answer)**
- C) iptables -A PREROUTING -j ACCEPT
- D) iptables -t nat -s 10.10.204.0/24 -p tcp -A PREROUTING -j MASQUERADE
- E) echo "0" > /proc/sys/net/ipv4/ip_forward
- F) echo "1" > /proc/net/tcp
- G) iptables -t nat -s 10.10.204.0/24 -A POSTROUTING -j MASQUERADE (Correct Answer)**
- H) iptables -t nat -A PREROUTING -j MASQUERADE

Explanation

Correct Answer: ABG [?](#)

Community Discussion

Question #367

A Linux administrator is troubleshooting a system with low performance. The administrator runs a few commands and receives the following output:

```
top - 14:06:18 up 27 days, 20:40, 1 user, load average: 2.43, 1.80, 5.32
Tasks: 2490 total, 3 running, 2485 sleeping, 0 stopped, 4 zombie
%Cpu(s): 55.5 us, 3.5 sy, 0.0 ni, 41.0 id, 9.1 wa, 0.0 hi, 0.4 si, 0.0 st
KiB Mem : 26388633+total, 19876084 free, 6732524 used, 23727772+buff/cache
KiB Swap : 33554428 total, 33223932 free, 330496 used. 25625606+avail Mem

PID USER PR NI VIRT RES SHR S %CPU %MEM T TIME+ COMMAND
27973 root 20 0 47848 47848 47848 R 9180 3384 R 97.9 0.0 1 :16.20 java
27969 root 20 0 47832 47832 47832 R 9064 3384 R 37.8 0.0 1 :15 :85 make_list
16698 root 20 0 3039748 93380 7544 S 20.9 0.0 1092:28 metricbeat
7238 root 20 0 175332 5188 1904 R 13.1 0.0 0:01.03 top 1921 root 20 0 103476 892
892 s 5.2 0.0
358:52.99 sam-sharefsd 25644 root 20 0 767128 5484 944 S 4.9 0.0 137:47.24 sam-
sharefsd
...
vmstat
procs -----memory-----swap-----io-----system-- -----cpu-
r b swpd free buff cache si so bi bo in cs us sy id wa st
2 1 330496 19863532 63831676 173491584 0 0 1574 366 0 0 2 5 86 7 0
```

Which of the following is the most likely cause of the issue?

- A) The disk I/O is high.
- B) The swap is low.
- C) The memory usage is high.
- D) The CPU utilization is high. (Correct Answer)**

Explanation

Correct Answer: D ?

Community Discussion

Question #368

A user on a Linux workstation needs to remotely start an application on a Linux server and then forward the graphical display of that application back to the Linux workstation. Which of the following would enable the user to perform this action?

- A) `ssh -X user@server application` (Correct Answer)**

- B) ssh -y user@server application
- C) ssh user@server application
- D) ssh -D user@server application

Explanation

Correct Answer: A [?](#)

Community Discussion

Question #369

A Linux administrator is attempting to create a KVM on a remote Linux server. After failing to create the VM, the administrator runs some commands and reviews the following output:

Output1:

```
#lscpu
Architecture: x86_64
CPU op-mode(s): 32-bit, 64-bit
...

Virtualization features:
Hypervisor vendor: Microsoft
Virtualization type: full
```

Output2:

```
# cat /etc/modprobe.d/kvm.conf
options kvm_intel nested=0
```

Which of the following is the most likely cause of the issue?

- A) Virtualization does not work on 32-bit processors.
- B) The virtual server does not allow VMs running inside the VM. (Correct Answer)**
- C) The memory capacity and swap space are both inadequate.

D) The boot partition does not have enough available space.

Explanation

Correct Answer: B [?](#)

Community Discussion

Question #370

Which of the following describes how a user's public key is used during SSH authentication?

A) The user's public key is used to hash the password during SSH authentication.

B) The user's public key is verified against a list of authorized keys. If it is found, the user is allowed to log in. (Correct Answer)

C) The user's public key is used instead of a password to allow server access.

D) The user's public key is used to encrypt the communication between the client and the server.

Explanation

Correct Answer: B [?](#)

Community Discussion

Question #371

An administrator needs to restrict a file's permission set in the following ways:

- The file needs to be read and write, but it should not be executable to the file owner.
- The file needs to be read only and executable to the group owner.
- The file needs to be read only to all other users.

Which of the following represents the next command the administrator should execute?

- A) `chmod 210 file.txt`
- B) `chmod 654 file.txt` (Correct Answer)**
- C) `chmod 753 file.txt`
- D) `chmod 777 file.txt`

Explanation

Correct Answer: B [?](#)

Community Discussion

Question #372

A Linux administrator is updating the file that contains the addresses of the software and update repositories on an Ubuntu system. Which of the following files needs to be edited to accomplish this task?

- A) `/etc/apt/sources.list` (Correct Answer)**
- B) `/etc/apt/listchanges.conf`
- C) `/etc/yum/yum.conf`
- D) `/etc/dnf/dnf.conf`

Explanation

Correct Answer: A [?](#)

Community Discussion

Question #373

A developer is unable to access a Linux server via SSH. Given the following output:

```
SSH server configuration
PermitRootLogin no
PubkeyAuthentication yes
PasswordAuthentication yes
GSSAPIAuthentication yes
X11Forwarding no

authorized_keys
ssh-rsa
AAAAB3NzaC1yc2EAAAADAQABAAQACCu+EdwagTk1XBR9rcys8cIT50/uZ5vvqfqtOgVr
9GDRVmZMyM4kNt2H1TGbgrHMct0nzRWzCTZUhx90s7u10HcyXQRidnRoH/CZzX0
YQ1tUmWcRL4Gu3Q50cOkod/spL3ovHx1LmK6kMIHfol5WvT59VF5snIFnWkRcP9U/
developer@localhost.localdomain

User information
/etc/passwd
developer:x:1000:1000:comptia:/home/developer:/bin/bash

/etc/shadow
developer:!!:0:99999:7:::

Last password change : never
Password expires : never
Password inactive : never
Account expires : never
Minimum number of days between password change : 0
Maximum number of days between password change : 99999
Number of days of warning before password expires : 7
```

Which of the following explains why the developer is unable to log in to the server?

- A) The developer's private key has been deleted from the server
- B) The developer's account has been locked out (Correct Answer)**

- C) The developer's public key is in the wrong location
- D) SSH has been disabled for user log-in

Explanation

Correct Answer: B [?](#)

Community Discussion

Question #374

After making a configuration change to an Apache web server, an administrator needs to reload the service. However, sudo access was only granted to the killall command.

Which of the following commands should the administrator use to load the new configuration?

A) sudo killall -HUP httpd (Correct Answer)

- B) sudo killall -reload httpd
- C) sudo killall -9 httpd
- D) sudo killall -TERM httpd

Explanation

Correct Answer: A [?](#)

Community Discussion

Question #375

An administrator needs to allow remote administrative access to a Linux server only to employees who are using the authorized private key. Which of the following options should the administrator set in the SSHD configuration file to achieve the goal?

- A) AuthorizedKeysCommand sudo
- B) PermitRootLogin prohibit-password
- C) AuthorizedKeysConmandUser root
- D) PasswordAuthentication no (Correct Answer)**

Explanation

Correct Answer: D ?

Community Discussion

Question #376

A systems administrator has changed the permissions on the /etc/passwd file. When a user attempts to log in, the following is displayed on the command prompt:

```
I have no name!
```

The following output is provided:

```
$ ls -l /etc(passwd,shadow)
-rw----- 1 root 7868 Oct 18 10:45 /etc/passwd
-rw-r----- 1 shadow 1054 Oct 18 10:45 /etc/shadow
```

Which of the following commands will fix the file permissions?

- A) sudo 755 /etc/passwd
- B) sudo chmod 644 /etc/passwd (Correct Answer)**

- C) sudo chmod 644 /etc/shadow
- D) sudo chmod 755 /etc/shadow

Explanation

Correct Answer: B ?

Community Discussion

Question #377

An administrator added the port 2222 for the SSH server on myhost and restarted the SSH server. The administrator noticed issues during the startup of the service. Given the following outputs:

```
$ ssh -p 2222 myhost
ssh:connect to host myhost on port 2222: Connection refused

$ nmap -p 2222 myhost
Starting Nmap 7.70 ( https://nmap.org ) at 2022-10-17 21:12 FET
Nmap scan report for myhost (10.7.3.26)
Host is up (0.00027s latency).
rDNS record for 10.7.3.26: myhost
PORT      STATE      SERVICE
2222/tcp  closed    EtherNetIP-1
MAC Address: 52:54:00:F5:DF:F8 (QEMU virtual NIC)
Nmap done: 1 IP address (1 host up) scanned in 0.57 seconds

$ systemctl status sshd
sshd.service - OpenSSH server daemon
Loaded: loaded (/usr/lib/systemd/system/sshd.service; enabled; vendor preset: enabled)
Active: active (running) since Mon 2022-10-17 19:40:07 CEST; 36min ago
Docs: man:sshd(8)
      man:sshd_config(5)
Main PID: 13186 (sshd)
Tasks: 1 (limit: 12373)
Memory: 1.1M
CGroup: /system.slice/sshd.service
        13186 /usr/sbin/sshd -D -oCiphers=aes256-gcm@openssh.com

Oct 17 19:40:07 myhost systemd[1]: Starting OpenSSH server daemon...
Oct 17 19:40:07 myhost sshd[13186]: error: Bind to port 2222 on 0.0.0.0 failed: Permission denied.
Oct 17 19:40:07 myhost system[1]: Started OpenSSH server daemon.
Oct 17 19:40:07 myhost sshd[13186]: Server listening on 0.0.0.0 port 22.
```

Which of the following commands will fix the issue?

A) semanage port -a -t ssh_port_t -p tcp 2222 (Correct Answer)

- B) chcon system_u:object_r:ssh_home_t /etc/ssh/*
- C) iptables -A INPUT -p tcp --dport 2222 -j ACCEPT
- D) firewall-cmd --zone=public --add-port=2222/tcp

Explanation

Correct Answer: A ?

Community Discussion

Question #378

After trying to install an RPM package unsuccessfully, a systems administrator verifies the integrity of the package's database. The administrator discovers that the database is corrupted and needs to be recreated. Which of the following commands will help accomplish this task?

- A) `rpmdb -D rebuild`
- B) `rpmdb --initdb`
- C) `rpmdb --rebuilddb` (Correct Answer)**
- D) `rpmdb --exportdb`

Explanation

Correct Answer: C ?

Community Discussion

Question #379

Which of the following is a simple option to deter attackers who have physical access to a Linux system from performing a root account takeover?

- A) Configuring a GRUB bootloader password (Correct Answer)**
- B) Removing the root user from the system
- C) Purchasing a hardware security module
- D) Disabling the runlevel for single-user mode

Explanation

Correct Answer: A [?](#)

Community Discussion

Question #380

A Linux administrator needs to give the correct permission on a file called exam.txt. The owner needs to have full permissions to the file, the group only needs read and execute permissions, and others will not have any permissions. Which of the following commands should the administrator use to accomplish this task?

- A) `chmod 644 exam.txt`
- B) `chmod 650 exam.txt`
- C) `chmod 750 exam.txt` (Correct Answer)**
- D) `chmod 760 exam.txt`

Explanation

Correct Answer: C [?](#)

Community Discussion